

NIKHIL

Real Analysis

Section - 02

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CHAPTER 5

FUNCTION AND THEIR PROPERTIES

5.1. Definitions

Function: Let $A, B \subset \mathbb{R}$ be two subsets of $\mathbb{R}$.

A function from A to B denoted by $f: A \rightarrow B$ is a rule which assign every element of A to unique element of B . A is defined as a domain of f & B is defined as co-domain of f and $f(A)$ is defined as range of f , where $f(A) = \{f(x) \in B : x \in A\}$.

1. Equality of two functions: Two functions f_1 & f_2 are said to be equal if and only if

(i) f_1 & f_2 have same domain D (say)

(ii) $f_1(x) = f_2(x) \quad \forall x \in D$

2. Composition of functions: Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be two functions such that $f(x) = y$ and $g(y) = z$, where $x \in X, y \in Y, z \in Z$. Then the function $h: X \rightarrow Z$ such that $h(x) = z = g(y) = g(f(x)) \quad \forall x \in X$ is known as the composition of f and g and is denoted by $g \circ f$.

3. Inverse function: Let $f: X \rightarrow Y$ be a one-one onto function. Then the function $g: Y \rightarrow X$ which associates to each element $y \in Y$ the unique element $x \in X$ such that $f(x) = y$ is known as inverse function of f . The inverse function g of f is denoted by f^{-1} . Then, we have $f^{-1}: Y \rightarrow X$ such that $f^{-1}(y) = x$, where $f(x) = y$.

4. Domain of Definition: let $y = f(x)$ be a rule, $S \subseteq \mathbb{R}$ on which f becomes real valued function i.e., if S be a subset of $\mathbb{R}$ & $f: S \rightarrow \mathbb{R}$ be real valued function then S is called domain of definition. We sometimes denote domain of definition as 'Dod'

Examples:

(a) $f(x) = \sin x, x \neq 1, \text{ Dod} = \mathbb{R} - \{1\}$

(b) $f(x) = \frac{1}{x}, \text{ Dod} = \mathbb{R} - \{0\}$

(c) $f(x) = \log_e x, \text{ Dod} = \mathbb{R}^+$

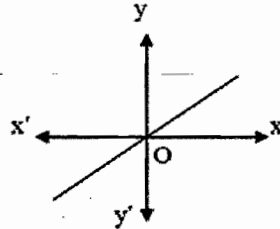
(d) $f(x) = \log_e \log_e x, \text{ Dod} = (1, \infty)$

(e) $f(x) = \log_e \log_e \log_e \log_e x, \text{ Dod} = (e^e, \infty)$

Some well known functions

1. **Identity Function:** The function defined by $f(x) = x$

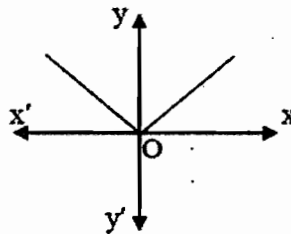
for all $x \in \mathbb{R}$ is called the identity function on $\mathbb{R}$. Clearly, the domain and range of the identity function are both equal to $\mathbb{R}$. The graph of the identity function is a straight line passing through the origin and inclined at an angle of 45° to x -axis.



2. **Polynomial Function:** A function of the form $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$, where $n \in \mathbb{N}$ and $a_0, a_1, a_2, \dots, a_n \in \mathbb{R}$. Then, f is called a polynomial function.

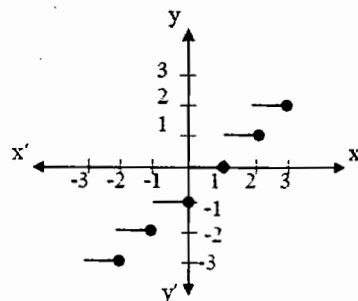
Example: $f(x) = x^3 + 2x^2 + 9$

3. **Modulus Function:** The function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$ is called the modulus function. The range is the set of all non-negative real numbers.



Properties:

- (i) $|x| \leq a \Rightarrow a \leq x \leq a; (a \geq 0)$
 - (ii) $|x| \geq a \Rightarrow x \leq -a$ or $x \geq a; (a \geq 0)$
 - (iii) $|x \pm y| \leq |x| + |y|$
 - (iv) $|x \pm y| \geq ||x| - |y||$
4. **The Greatest Integer Function:** Let $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = [x] = n \forall n \leq x < n+1, n \in \mathbb{N}$, $[x]$ indicates the integral part of x which is nearest and smallest integer to x . It is also known as floor of x . Domain of f is the set of real number and range of f is the set of integers.



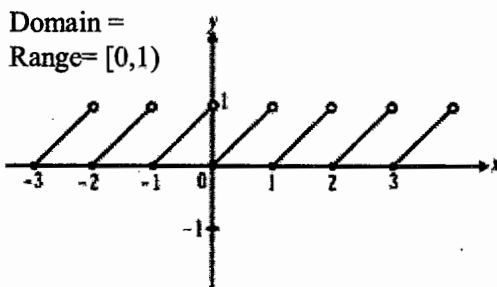
Put Your Own Notes

Example: $[2.45] = 2, [-2.1] = -3$

Properties:

- (i) $[x] = x$ holds, if $x \in \mathbb{Z}$
- (ii) $[x + I] = [x] + I$, if $I \in \mathbb{Z}$
- (iii) $[x + y] \geq [x] + [y]$
- (iv) If $[\phi(x)] \geq I$, then $\phi(x) \geq I$, where $I \in \mathbb{Z}$
- (v) If $[\phi(x)] \leq I$, then $\phi(x) < I + 1$, where $I \in \mathbb{Z}$
- (vi) $[-x] = -[x]$, if $x \in \mathbb{Z}$
- (vii) $[-x] = -[x] - 1$, if $x \notin \mathbb{Z}$

5. **Fractional Part of Function:** The function $f: \mathbb{R} \rightarrow [0, 1]$ defined by $f(x) = \{x\} = x - [x]$, $[x]$ is greatest integer function is called fractional part of x .

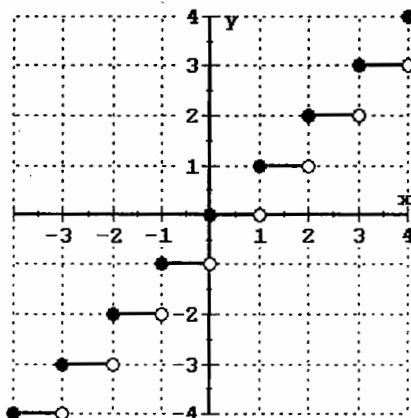


Example: $x = 1.3$, then $\{x\} = 0.3$.

Properties:

- (i) $\{x\} = x$; if $0 \leq x < 1$
- (ii) $\{x\} = 0$; if $x \in \mathbb{Z}$
- (iii) $\{-x\} = 1 - \{x\}$; if $x \in \mathbb{Z}$

6. **Least Integer Function:** A function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = \lceil x \rceil =$ the integral part of x which is nearest and greatest integer to x . It is also known as ceiling of x .



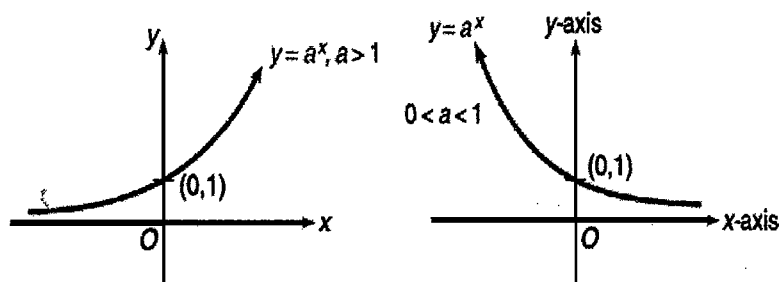
Example: $\lceil 2.3023 \rceil = 3$, $\lceil -8.0725 \rceil = -8$

Properties:

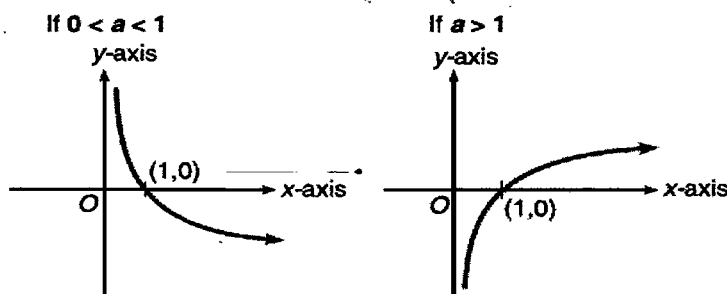
(i) $\lceil x \rceil = x$, if $x \in \mathbb{Z}$.

(ii) $\lceil x + I \rceil = \lceil x \rceil + I$; if $I \in \mathbb{Z}$

7. **Exponential Function:** Let a be positive real number. Then the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = a^x$ is called the exponential function. The range is the set of all positive real numbers.



8. **Logarithmic Function:** Let ' a ' be a positive real number, then the function $f: \mathbb{R}^+ \rightarrow \mathbb{R}$ defined as $f(x) = \log_a x$ is called the logarithmic function. The range is the set $\mathbb{R}$ of all real numbers.



Properties: Let a, b, c be positive real numbers

(i) $\log_e(ab) = \log_e a + \log_e b$

(ii) $\log_e \left(\frac{a}{b} \right) = \log_e a - \log_e b$

(iii) $\log_e a^m = m \log_e a$ $m \in \mathbb{R}$

(iv) $\log_a a = 1$ $a \neq 1$

(v) $\log_{b^m} a = \frac{1}{m} \log_b a$ $b \neq 1$ & $m \in \mathbb{R}$

(vi) $\log_b a = \frac{1}{\log_a b}$ $a, b \neq 1$

(vii) $\log_b a = \frac{\log_m a}{\log_m b}$ $a, b \neq 1$ & $m > 0$

(viii) $a^{\log_a m} = m$ $m > 0$ & $a \neq 1$

(ix) $a^{\log_c b} = b^{\log_c a}$ $c \neq 1$

Put Your Own Notes

- 9. Rational Function:** Let $p(x), q(x)$ be two polynomials. Then a rational function is a function $f: D \rightarrow \mathbb{R}$ defined as $f(x) = \frac{p(x)}{q(x)}$, where $D = \mathbb{R} - \{x \in \mathbb{R} | q(x) = 0\}$.

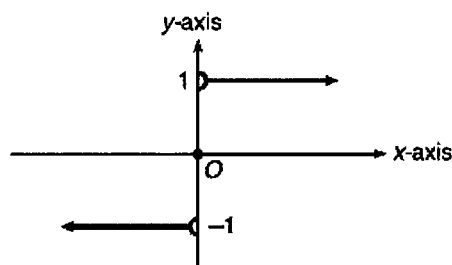
Example:

- (a) Let $f: \mathbb{R} - \{0\} \rightarrow \mathbb{R}$ defined as $f(x) = \frac{1}{x}$
 (b) Let $f: \mathbb{R} - \{0\} \rightarrow \mathbb{R}$ defined as $f(x) = \frac{1}{x^2}$.

- 10. Signum Function:** Let $f: \mathbb{R} \rightarrow \{-1, 0, 1\}$ defined as

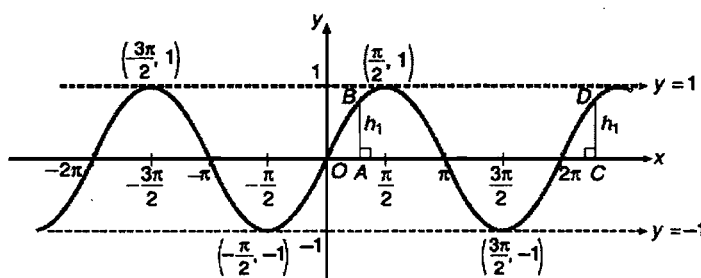
$$f(x) = \begin{cases} 1, & \text{if } x > 0 \\ -1, & \text{if } x < 0 \\ 0, & \text{if } x = 0 \end{cases}$$

is called signum function.

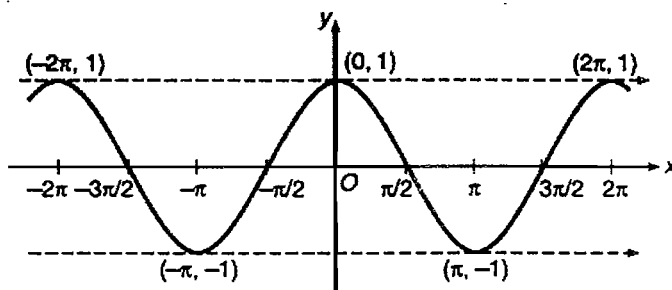


- 11. Trigonometric Functions:**

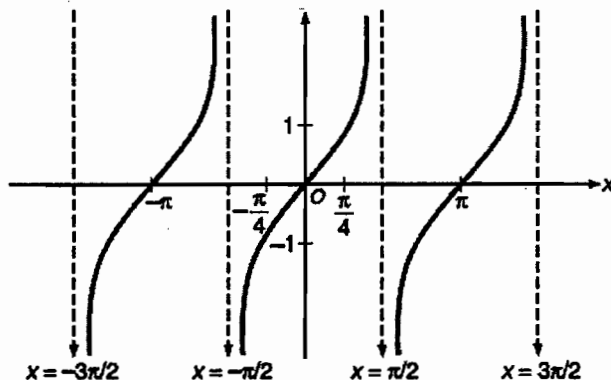
- (i) **Sine Function:** The function which associates each real number x to $\sin x$ is called the sine function. Here x is the radian measure of the angle. The domain of the sine function is $\mathbb{R}$ and the range is $[-1, 1]$.



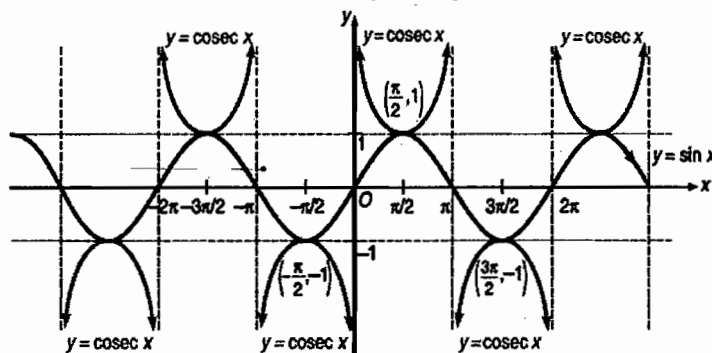
- (ii) **Cosine Function:** The function that associates each real number x to $\cos x$ is called the cosine function. Here x is the radian measure of the angle. The domain of the cosine function is $\mathbb{R}$ and the range is $[-1, 1]$.



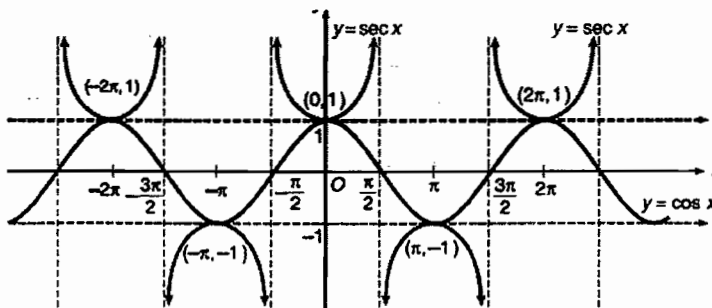
- (iii) **Tangent Function:** The function that associates each real number x to $\tan x$ is called the tangent function. Clearly, the tangent function is not defined at odd multiple of $\pi/2$, i.e. $\pm\pi/2, \pm3\pi/2$ etc. So, the domain of the tangent function is $\mathbb{R} - \{(2n+1)\pi/2 \mid n \in \mathbb{Z}\}$. Since, it takes every value between $-\infty$ and ∞ . So, the range is $\mathbb{R}$.



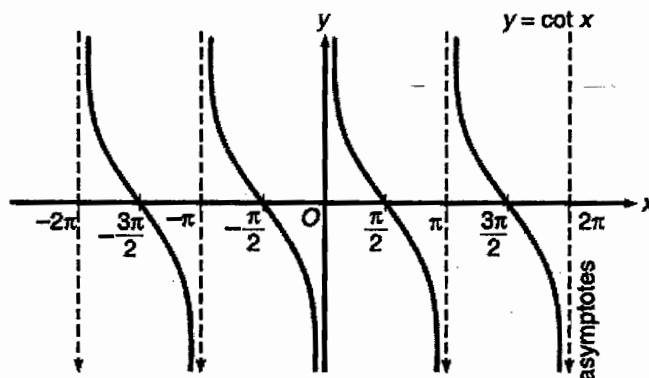
- (iv) **Cosecant Function:** The function that associates each real number x to $\csc x$ is called the cosecant function. Clearly, $\csc x$ is not defined at $x = n\pi$, $n \in \mathbb{Z}$ i.e., $0, \pm\pi, \pm2\pi, \pm3\pi$ etc. So its domain is $\mathbb{R} - \{n\pi \mid n \in \mathbb{Z}\}$. Since $\csc x \geq 1$ or $\csc x \leq -1$. Therefore, range is $(-\infty, -1] \cup [1, \infty)$.



- (v) **Secant Function:** The function that associates each real number x to $\sec x$ is called the secant function. Clearly set is not defined at odd multiples of $\pi/2$ i.e., $(2n+1)\pi/2$, $n \in \mathbb{Z}$. So, its domain is $\mathbb{R} - \{(2n+1)\pi/2 \mid n \in \mathbb{Z}\}$. Since $|\sec x| \geq 1$, therefore its range is $(-\infty, -1] \cup [1, \infty)$.



(vi) **Cotangent Function:** The function that associates a real number x to $\cot x$ is called the cotangent function. Clearly, $\cot x$ is not defined at $x = n\pi$, $n \in \mathbb{Z}$ i.e., at $n=0, \pm\pi, \pm2\pi$ etc. So domain of $\cot x$ is $\mathbb{R} - \{n\pi | n \in \mathbb{Z}\}$. The range of $f(x) = \cot x$ is $\mathbb{R}$ as is evident from its graph in figure.



12. Inverse Trigonometric Function: The inverse trigonometric function $\sin^{-1} x$, $\cos^{-1} x$, $\tan^{-1} x$ etc. are defined as the inverse of the corresponding trigonometric functions. For example, $\sin^{-1} x$ is defined as the angle whose sine is x . But since for each value of $\sin^{-1} x$, there are unlimited number of values of the angles whose sine is x and for values of x for which $|x| > 1$, $\sin^{-1} x$ does not exist. Thus, we cannot define the inverse function of $\sin x$ unless we modify the definition of $\sin x$ in such a way that it becomes a bijection. If we consider $\sin x$ as a function from $[-\pi/2, \pi/2]$ to $[-1, 1]$, then it is a bijection. Consequently $\sin^{-1} x$ exists and it is a function from $[-1, 1]$ to $[-\pi/2, \pi/2]$.

Similarly, the other inverse trigonometric functions can be defined.

The domain and range of the inverse trigonometric functions are stated below:

Function	Domain	Range	Definition of the function
$\sin^{-1} x$	$[-1, 1]$	$[-\pi/2, \pi/2]$	$y = \sin^{-1} x \Leftrightarrow x = \sin y$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$	$y = \cos^{-1} x \Leftrightarrow x = \cos y$
$\tan^{-1} x$	$(-\infty, \infty)$ or $\mathbb{R}$	$(-\pi/2, \pi/2)$	$y = \tan^{-1} x \Leftrightarrow x = \tan y$
$\cot^{-1} x$	$(-\infty, \infty)$ or $\mathbb{R}$	$(0, \pi)$	$y = \cot^{-1} x \Leftrightarrow x = \cot y$
$\operatorname{cosec}^{-1} x$	$\mathbb{R} - (-1, 1)$	$[-\pi/2, \pi/2] - \{0\}$	$y = \operatorname{cosec}^{-1} x \Leftrightarrow x = \operatorname{cosec} y$
$\sec^{-1} x$	$\mathbb{R} - (-1, 1)$	$[0, \pi] - \{\pi/2\}$	$y = \sec^{-1} x \Leftrightarrow x = \sec y$

5.2. Graphical Transformation

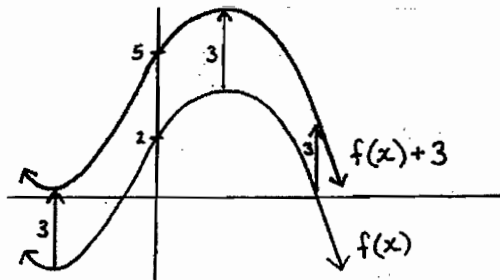
1. When $f(x)$, transforms to $f(x) \pm a$. (where a is +ve)

i.e., $f(x) \rightarrow f(x) + a$

shift the given graph of $f(x)$ upward through ' a ' units again,

$f(x) \rightarrow f(x) - a$ shift the given graph of $f(x)$ downward through ' a ' units

Example:

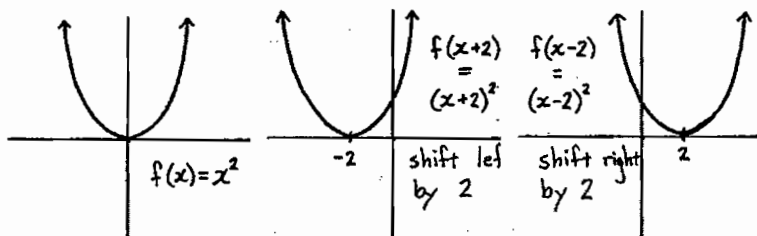


2. $f(x)$ transforms to $f(x \pm a)$

i.e., $f(x) \rightarrow f(x - a)$; ' a ' is positive. Shift the graph of $f(x)$ through ' a ' unit towards right $f(x)$ transformations to $f(x + a)$.

i.e., $f(x) \rightarrow f(x + a)$; ' a ' is positive. Shift the graph of $f(x)$ through ' a ' unit towards left.

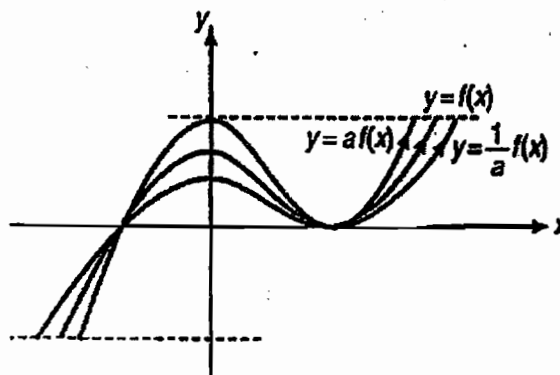
Example:



3. $f(x)$ transforms to $af(x)$

i.e., $f(x) \rightarrow af(x)$; $a > 1$ Stretch the graph of $f(x)$ ' a ' times along y-axis.

$f(x) \rightarrow \frac{1}{a}f(x)$; $a > 1$. Shrink the graph of $f(x)$ ' a ' times along y-axis.



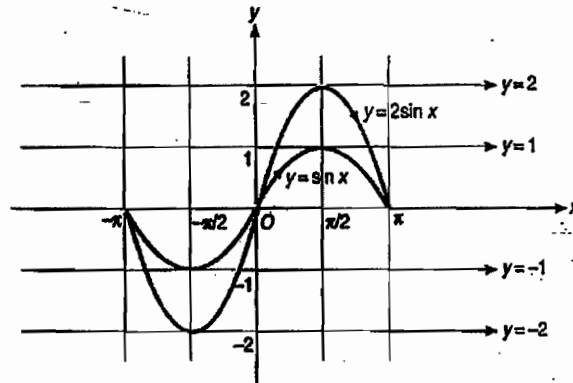
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Example: Plot $y = \sin x$ and $y = 2\sin x$.

Solution: We know; $y = \sin x$ and $f(x) \rightarrow af(x)$

$\Rightarrow$ Stretch the graph of $f(x)$ 'a' times along y-axis.

$\therefore y = 2\sin x \Rightarrow$ stretch the graph of $\sin x$ '2' times along y-axis.



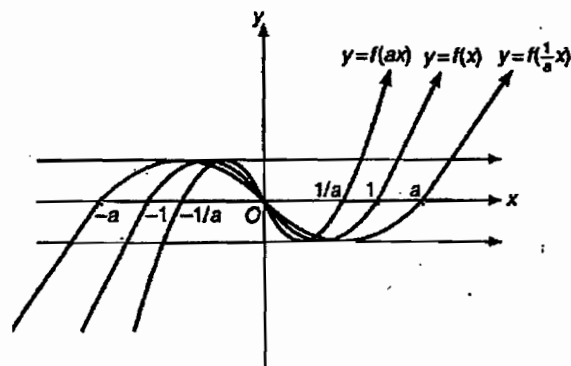
4. $f(x)$ transforms to $f(ax)$

i.e., $f(x) \rightarrow f(ax); a > 1$

Stretch (or contract) the graph of $f(x)$ 'a' times along y-axis.

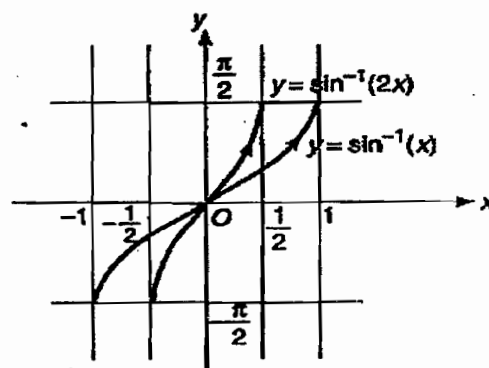
again $f(x) \rightarrow f\left(\frac{1}{a}x\right); a > 1$.

Shrink (or expand) the graph of $f(x)$ 'a' times along y-axis.



Example: Plot $y = \sin^{-1} x$ and $y = \sin^{-1}(2x)$.

Solution: Here; $y = \sin^{-1}(2x)$, is to shrink (or contract) the graph of $\sin^{-1} x$ '2' times along x-axis.



5. $f(x)$ transforms to $f(-x)$

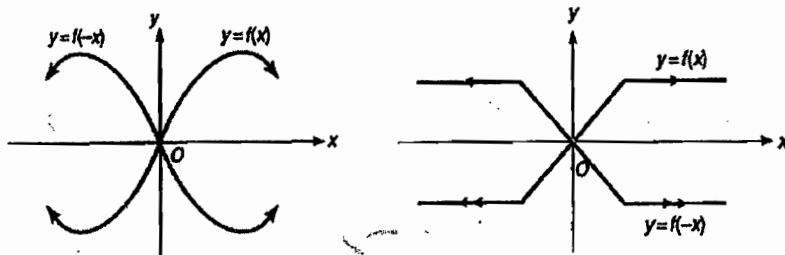
i.e., $f(x) \rightarrow f(-x)$

To draw $y = f(-x)$, take the image of the curve $y = f(x)$ in y -axis as plane mirror.

OR

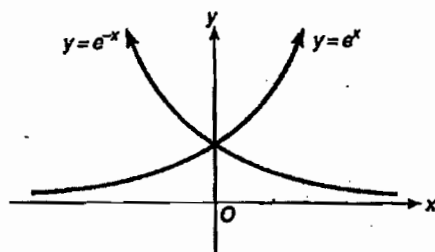
"Turn the graph of $f(x)$ by 180° about y -axis."

OR



Example: Plot $y = e^{-x}$.

Solution: As $y = e^x$ is known; then $y = e^{-x}$ is the image in y -axis as plane mirror for $y = e^x$; shown as;



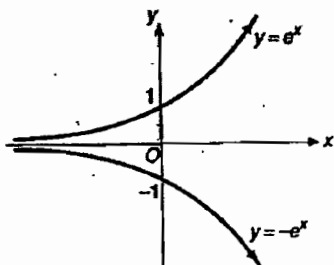
6. $f(x)$ transforms to $-f(x)$

i.e., $f(x) \rightarrow -f(x)$;

To draw $y = -f(x)$, take image of $y = f(x)$ in the x -axis as plane mirror.

OR

"Turn the graph of $f(x)$ by 180° about x -axis."



Put Your Own Notes

Example: Plot the curve $y = -e^x$.

Solution: As $y = e^x$ is known;

$\therefore y = -e^x$ take image of $y = e^x$ in the x -axis as plane mirror.

7. $f(x)$ transforms to $-f(-x)$

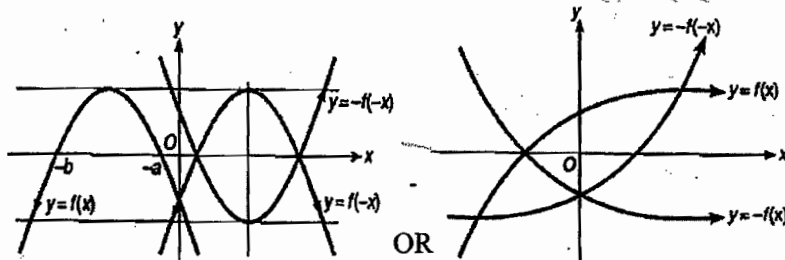
i.e., $f(x) \rightarrow -f(-x)$;

To draw $y = -f(-x)$ take image of $f(x)$ about y -axis to obtain $f(-x)$ and then take image of $f(-x)$ about x -axis to obtain $-f(-x)$.

$\therefore f(x) \rightarrow -f(-x)$

(i) Image about y -axis

(ii) Image about x -axis

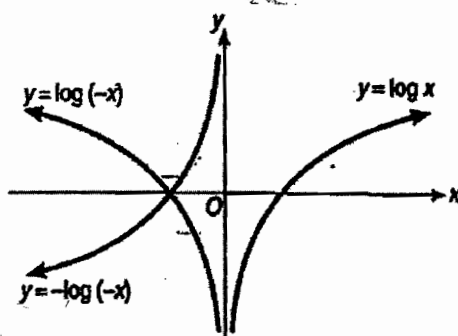


Example: Plot the curve $y = -\log(-x)$.

Solution: As $y = \log x$ is known;

(i) Take image about y -axis, for $y = \log(-x)$.

(ii) Take image of $y = \log(-x)$ about x -axis, for $y = -\log(-x)$.



8. $f(x)$ transforms to $|f(x)|$

i.e., $f(x) \rightarrow |f(x)|$

Here; $y = |f(x)|$ is drawn in two steps.

(i) In the first step, leave the positive part of $f(x)$, {i.e., the part of $f(x)$ above x -axis} as it is.

(ii) In the second step, take the mirror image of negative part of $f(x)$. {i.e., the part of $f(x)$ below x -axis} in the x -axis as plane mirror.

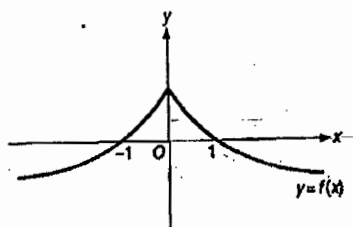
OR

Take the mirror image (in x -axis) of the portion of the graph of $f(x)$ which lies below x -axis.

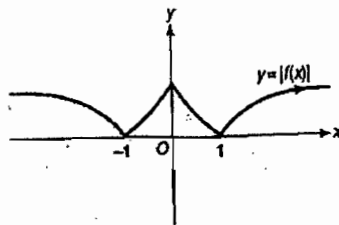
OR

Turn the portion of the graph of $f(x)$ lying below x -axis by 180° about x -axis.

Graph of $f(x)$:



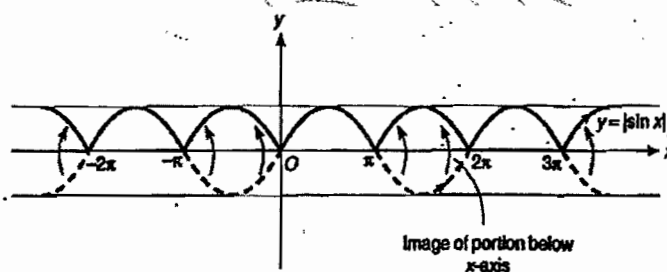
Graph for $|f(x)|$:



Example: Sketch the graph for $y = |\sin x|$.

Solution: Here, $y = \sin x$ is known.

$\therefore$ To draw $y = |\sin x|$, we take the mirror image (in x -axis) of the portion of the graph of $\sin x$ which lies below x -axis.



9. $f(x)$ transforms to $f(|x|)$

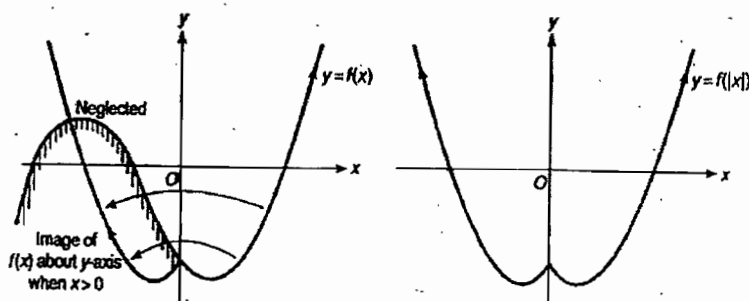
i.e., $f(x) \rightarrow f(|x|)$

If we know $y = f(x)$, then to plot $y = f(|x|)$, we should follow two steps:

- (i) Leave the graph lying right side of the y -axis as it is.
- (ii) Take the image of $f(x)$ in the right of y -axis as the plane mirror and the graph of $f(x)$ lying leftward of the y -axis (if it exists) is omitted.

OR

Neglect the curve for $x < 0$ and take the images of curve for $x \geq 0$ about y -axis.

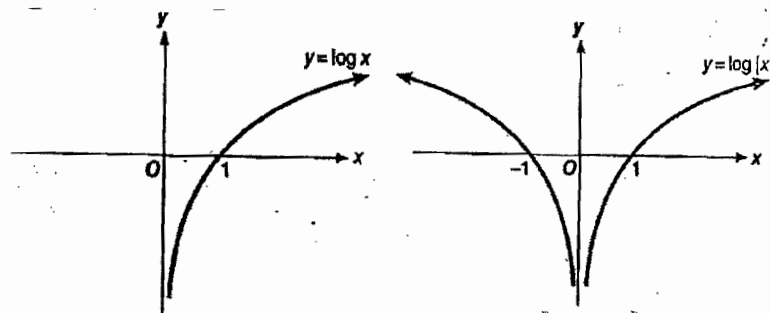


Example: Sketch the graph for $y = \log|x|$.

Solution: As we know, the curve $y = \log x$.

$\therefore y = \log|x|$ could be drawn in two steps:

- (i) Leave the graph lying right side of the y -axis as it is.
- (ii) Take the image of $f(x)$ in the right of y -axis as the plane mirror.



10. Sketching $h(x) = \text{Minimum} \{f(x), g(x)\}$ and

$$h(x) = \text{Minimum} \{f(x), g(x)\}$$

$$(i) \quad h(x) = \text{Maximum} \{f(x), g(x)\}$$

$$\Rightarrow h(x) = \begin{cases} f(x); & \text{when } f(x) > g(x) \\ g(x); & \text{when } g(x) > f(x) \end{cases}$$

$\therefore$ Sketch $f(x)$ when its graph is above the graph of $g(x)$ and sketch $g(x)$ when its graph is above the of $f(x)$.

$$(ii) \quad h(x) = \text{Minimum} \{f(x), g(x)\}$$

$$\Rightarrow h(x) = \begin{cases} f(x), & \text{when } f(x) < g(x) \\ g(x), & \text{when } g(x) < f(x) \end{cases}$$

$\therefore$ Sketch $f(x)$ when its graph is lower and otherwise sketch $g(x)$.

OR

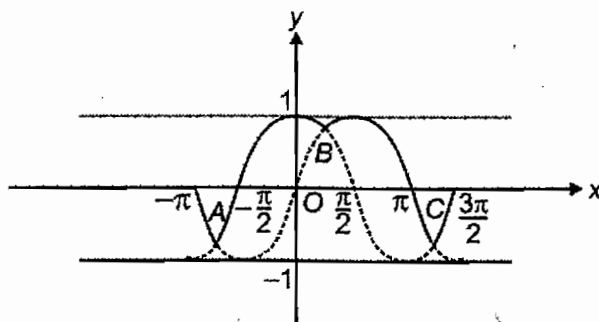
“To draw the graph of functions of the form $y = \max \{f(x), g(x)\}$ or $y = \min \{f(x), g(x)\}$.”

We first draw the graphs of both the functions $f(x)$ and $g(x)$ and their points of intersections. Then we find any two consecutive points of intersection. In between these points either $f(x) > g(x)$ or $f(x) < g(x)$, then, in order to $\max \{f(x), g(x)\}$ we take those segments of $f(x)$ for which $f(x) > g(x)$, between any two consecutive points of intersection of $f(x)$ and $g(x)$.

Similarly, in order to $\min \{f(x), g(x)\}$, we take those segments of $f(x)$ for which $f(x) < g(x)$, between any two consecutive points of intersection of $f(x)$ and $g(x)$.

Example: Sketch the graph of $y = \max\{\sin x, \cos x\}$, $\forall x \in \left(-\pi, \frac{3\pi}{2}\right)$.

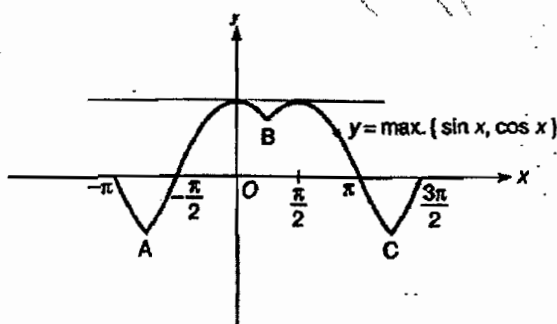
Solution:



First plot both $y = \sin x$ and $y = \cos x$ by a dotted curve as can be seen from the graph in the interval $\left(-\pi, \frac{3\pi}{2}\right)$ and then darken those dotted lines for which $f(x) > g(x)$ or $g(x) > f(x)$.

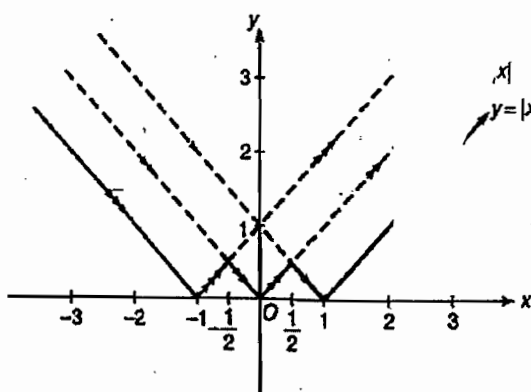
From adjacent figure the point of intersections are A, B, C.

$\therefore$ Graph of $\max\{\sin x, \cos x\}$



Example: Sketch the graph for $y = \min\{|x|, |x-1|, |x+1|\}$.

Solution:



First plot the graph for:

$y = |x|$, $y = |x-1|$, $y = |x+1|$ by a dotted curve as can be seen from the graph and then darken those dotted lines for which

$$|x| < \{|x-1|, |x+1|\}:$$

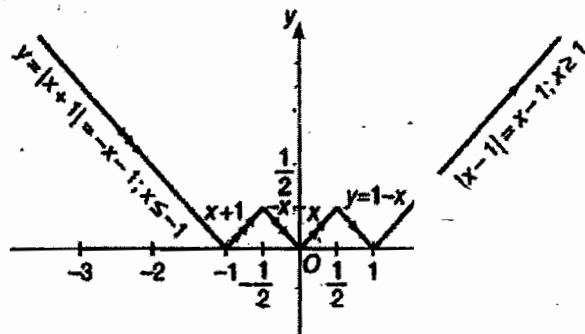
$$|x-1| < \{|x|, |x+1|\}$$

$$\text{and } |x+1| < \{|x|, |x-1|\}.$$

Graph for $y = |x|$, $y = |x-1|$, $y = |x+1|$.

Put Your Own Notes

As from the above curve graph for $y = \min\{|x-1|, |x|, |x+1|\}$ is plotted as;



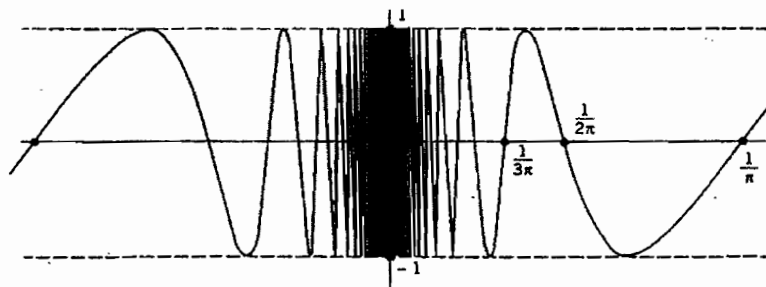
From above figure : $\min\{|x-1|, |x|, |x+1|\} =$

$$\begin{cases} -(x+1) & ; x \leq -1 \\ |x+1| & ; -1 \leq x \leq -\frac{1}{2} \\ -x & ; -\frac{1}{2} \leq x \leq 0 \\ |x| & ; 0 \leq x \leq \frac{1}{2} \\ 1-x & ; \frac{1}{2} \leq x \leq 1 \\ |x-1| & ; x \geq 1 \end{cases}$$

5.3. Some Important Functions and Their Graphs

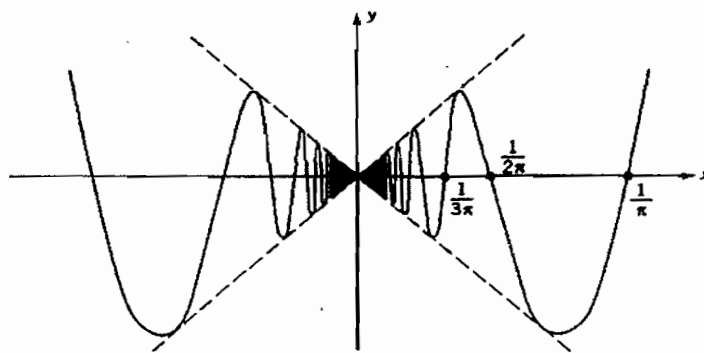
1. $f: A \rightarrow \mathbb{R}$

$$f(x) = \sin\left(\frac{1}{x}\right) \quad \forall x \in A$$



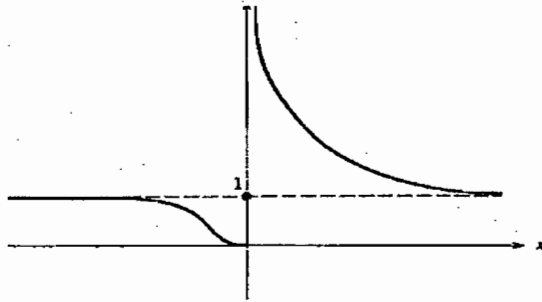
2. $f: A \rightarrow \mathbb{R}$

$$f(x) = x \sin\left(\frac{1}{x}\right) \quad \forall x \in A$$



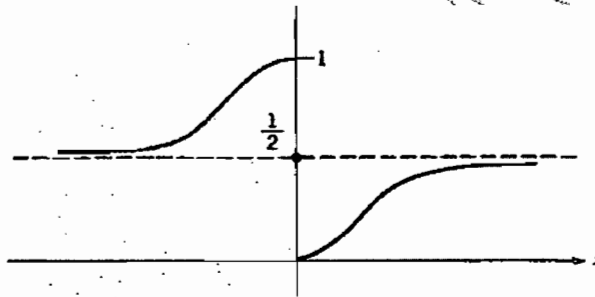
3. $f: A \rightarrow \mathbb{R}$

$$f(x) = e^{1/x} \quad \forall x \in A$$



4. $f: A \rightarrow \mathbb{R}$

$$f(x) = \left(\frac{1}{e^{1/x} + 1} \right) \quad \forall x \in A$$



5.4. Classification of Functions

1. **Algebraic Function:** A function $y = f(x)$ is said to be an algebraic function and set $S \subset \mathbb{R}$, if it is a root of the equation of the form

$$p_0(x)y^n + p_1(x)y^{n-1} + \dots + p_n(x) = 0$$

Where $p_i(x)$ are polynomial in $x \in S$ $0 \leq i \leq n$

Example:

(a) $p(y) = -3xy^2 - (1+x^2)y + 2x = 0$

$$\text{Then } y = \frac{(1+x^2) \pm \sqrt{(1+x^2)^2 + 4(3x)(2x)}}{-6x}$$

- (b) Constant functions are algebraic function even if $f(x) = \pi$.

2. **Transcendental Function:** If a function $y = f(x)$ defined on a set $S \subset \mathbb{R}$ is not an algebraic function then it is said to be a transcendental function on S .

Examples:

(a) $f(x) = x^\pi$

(b) $f(x) = c^x, c \neq 0, 1$

(c) $f(x) = x^x$

(d) $f(x) = x^{\frac{1}{x}}$

(e) $f(x) = \log_c x, c \neq 0, 1$

(f) $f(x) = \sin x$

Put Your Own Notes

3. **Periodic Function:** A function $y = f(x)$ is said to be periodic if $\exists l \in \mathbb{R}$ such that $f(x+l) = f(x) \quad \forall x \in D.o.d$ and $x+l \in D.o.d$ and such l is called a period of f .

Fundamental Period: The smallest positive $\alpha \in \mathbb{R}$ for a periodic function f is defined as the fundamental period of f .

Examples:

- (a) $f(x) = \sin x$ is a periodic function with set of periods $\{2n\pi : n \in \mathbb{Z}\}$
fundamental period of $f(x) = \sin x$ is 2π
- (b) $f(x) = \tan x$ is a periodic function with fundamental period π .

Remark:

- (a) Periodicity is a domain based property.
- (b) A periodic function may or may not have fundamental period.

Example: $f: \mathbb{R} \rightarrow \mathbb{R}$

$$f(x) = c \quad \forall x \in \mathbb{R}, \quad c \in \mathbb{R}$$

Then for any $\alpha \in \mathbb{R}$, $f(x+\alpha) = f(x) = c \quad \forall x \in \mathbb{R}$

set of periods of f is $\mathbb{R}$

but fundamental period does not exist.

- (c) If a function f has a fundamental period T . Then $af(bx+c)+d$ has period $\frac{T}{|b|}$, $a, b, c, d \in \mathbb{R}$.
- (d) There exist a periodic function whose fundamental period does not exist but set of period is dense in $\mathbb{R}$

Example: $f: \mathbb{R} \rightarrow \mathbb{R}$

$$f(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ -1 & x \in \mathbb{Q}^c \end{cases}$$

Then $f(x+\alpha) = f(x) \quad \forall x \in \mathbb{R}, \quad \forall \alpha \in \mathbb{Q}$.

$\therefore$ Set of periods is $\mathbb{Q}$ (set of rationals)

But fundamental period does not exist.

- (e) If $f_1, f_2, \dots, f_n$ are periodic functions with periods $T_1, T_2, \dots, T_n$ respectively and L.C.M. of $T_1, T_2, \dots, T_n$ exist. Then their linear combination of f_i 's is periodic function and their L.C.M. decide the period of f_i 's.

$$(f) \quad L.C.M \left\{ \frac{p_1}{q_1}, \frac{p_2}{q_2}, \dots, \frac{p_n}{q_n} \right\} p_i, q_i \in \mathbb{N} = \frac{L.C.M \{p_1, p_2, \dots, p_n\}}{H.C.F \{q_1, q_2, \dots, q_n\}}$$

- (g) If $\alpha_1, \alpha_2, \dots, \alpha_n$ are irrational numbers, obtained by multiplying same irrational number by different rational (non zero) then

$$L.C.M (\alpha_1, \alpha_2, \dots, \alpha_n) = L.C.M (\text{Rational multiplied by that irrational})$$

- (h) L.C.M of rational and irrational numbers is not defined.

Examples of Periodic Functions:

- (a) $f(x) = \sin\left(\frac{x}{3}\right)$ is a periodic function with period 6π .

Reason: Since period of $\sin x = 2\pi$

$$\therefore \text{period of } \sin\left(\frac{x}{3}\right) = 2\pi \times 3 = 6\pi$$

- (b) $f(x) = \sin^2 x$ has period π .

Reason: Since $\sin^2 x = \frac{1 - \cos 2x}{2} = \frac{1}{2} - \frac{1}{2} \cos(2x)$

as $\cos 2x$ has period π

$$\therefore \sin^2(x) \text{ has period } \frac{2\pi}{2} = \pi$$

- (c) $f(x) = \sin^3 x$ has a period 2π

Reason: $\sin^3 x = \frac{1}{4}(3\sin x - \sin 3x)$

$$= \frac{3}{4}\sin x - \frac{1}{4}\sin 3x$$

since $\frac{3}{4}\sin x$ has period 2π

$$\frac{1}{4}\sin 3x \text{ has period } \frac{2\pi}{3}$$

$$\text{and } L.C.M\left\{2\pi, \frac{2\pi}{3}\right\} = L.C.M\left\{2, \frac{2}{3}\right\}$$

$$= \frac{L.C.M\{2, 2\}}{H.C.F\{1, 3\}}$$

$$= 2$$

$\therefore \sin^3 x$ has period 2π .

- (d) $f(x) = \sin \pi x + x - [x]$ is periodic function with period 2.

Reason: Since $\sin(\pi x)$ has period $\frac{2\pi}{\pi} = 2$

and $x - [x]$ has period 1.

also $L.C.M\{2, 1\}$ is 2.

$\therefore f(x)$ is periodic function with period 2.

Notes:

- (a) Sum or difference of two periodic function may not be periodic

Example: Take $f(x) = \sin x$, $g(x) = x - [x]$

Then f and g are periodic function with period 2π and 1 resp.

But $f(x) \pm g(x)$ is not periodic.

Put Your Own Notes

- (b) Sum of two non periodic function may be periodic function.

Example: $f(x) = x^2, g(x) = -x^2 \quad \forall x \in \mathbb{R}$

Then f and g both are non periodic functions

But $(f+g)(x) = f(x) + g(x) = 0 \quad \forall x \in \mathbb{R}$

is a periodic function.

- (c) Sum of two periodic functions (having their fundamental periods) may be periodic but fundamental period may not exist

Example: $f(x) = \sin^2 x, g(x) = \cos^2 x$ both are periodic functions with fundamental period π . But $f(x) + g(x) = \sin^2 x + \cos^2 x = 1$ does not have fundamental period.

- (d) If f and g are two functions defined on $\mathbb{R}$ such that g is periodic. Then $f \circ g$ is periodic on $\mathbb{R}$.

Reason: Since g is periodic, so $\exists \alpha \in \mathbb{R}$ such that $g(x+\alpha) = g(x) \quad \forall x \in D, x+\alpha \in D$.

So $(f \circ g)(x+\alpha) = f(g(x+\alpha))$

$= f(g(x))$

$= (f \circ g)(x) \quad \forall x \in D.o.d, x+\alpha \in D.o.d$

- (e) Algebraic functions cannot be periodic unless they are constant function.

4. **Bounded Function:** Let $S \subseteq \mathbb{R}$ and $f: S \rightarrow \mathbb{R}$ be a function. Then f is bounded if and only if $f(S)$ is bounded in $\mathbb{R}$

i.e., $\exists \alpha, \beta \in \mathbb{R}$ such that

$\alpha \leq f(x) \leq \beta \quad \forall x \in S$

Notes:

- (a) If S is finite $\Rightarrow$ Range is finite $\Rightarrow f$ is bounded.
 (b) For a function to be unbounded, range must be infinite set.
 (c) f is unbounded on $S \Leftrightarrow \forall n \in \mathbb{N} \exists x_n \in S$ such that

$|f(x_n)| > n$.

i.e., $\langle f(x_n) \rangle$ is unbounded.

- (d) Sum, difference and composition of two bounded functions is bounded function.

Example: $f: [0,1] \rightarrow \mathbb{R}$

$$f(x) = \begin{cases} \frac{1}{x} & x \in (0,1] \\ 0 & x = 0 \end{cases}$$

Then f is unbounded.

Reason: For all $n \in \mathbb{N}$, $\exists \frac{1}{n} \in [0,1]$ such that

$$f\left(\frac{1}{n}\right) = n \in f([0,1])$$

$$\Rightarrow \mathbb{N} \subseteq f([0,1])$$

$\Rightarrow$ Unbounded set contained in $f([0,1])$

$\Rightarrow f([0,1])$ is unbounded.

5. Monotonic Function: Let $S \subseteq \mathbb{R}$ and $f: S \rightarrow \mathbb{R}$ then

(i) f is said to be monotonically increasing if for $x_1 < x_2$

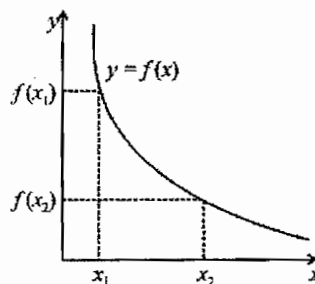
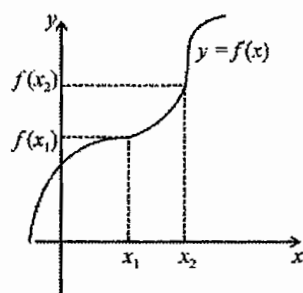
$$\Rightarrow f(x_1) \leq f(x_2) \quad \forall x_1, x_2 \in S$$

(If strict inequality hold then f is called strictly increasing function)

(ii) f is said to be monotonically decreasing if for $x_1 < x_2$

$$\Rightarrow f(x_2) \geq f(x_1) \quad \forall x_1, x_2 \in S$$

If strict inequality hold then f is called strictly decreasing function.



Monotonic Functions

Notes:

- If f is monotonically increasing (decreasing) function. Then $-f$ is monotonically decreasing (increasing) function.
- If $f_1: X_1 \rightarrow X_2$ is monotonic on X_1 and $f_2: X_2 \rightarrow X_3$ is monotonic on X_2 then $f_2 \circ f_1: X_1 \rightarrow X_3$ is monotonic on X_1 .
- Sum of two monotonically increasing (decreasing) function is monotonically increasing (decreasing).
- Difference of two monotonically increasing (decreasing) function may not be monotonically increasing (decreasing).

Example: $f: \left[0, \frac{\pi}{2}\right] \rightarrow \mathbb{R}$ $g: \left[0, \frac{\pi}{2}\right] \rightarrow \mathbb{R}$

$$f(x) = \sin x, \quad g(x) = x \quad \forall x \in \left[0, \frac{\pi}{2}\right]$$

Then f and g are monotonically increasing functions but $h(x) = \sin x - x$ is not monotonically increasing function.

- (e) Product of two monotonically increasing (decreasing) functions may not be monotonically increasing (decreasing).

Example: $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 0 & x < 0 \\ 3 & x \geq 0 \end{cases}$$

and $g: \mathbb{R} \rightarrow \mathbb{R}$

$$g(x) = \begin{cases} -7 & x < 0 \\ -2 & x \geq 0 \end{cases}$$

$$\text{But } f(x) \cdot g(x) = \begin{cases} 0 & x \leq 0 \\ -6 & x > 0 \end{cases}$$

is not monotonically increasing.

Examples of some well non monotonic functions:

- (a) $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = a^x \quad \forall x \in \mathbb{R}, a > 1 \text{ is monotonically increasing function.}$$

- (b) $f: (0, \infty) \rightarrow \mathbb{R}$ defined by — — .

$$f(x) = \log_e x \quad \forall x \in (0, \infty) \text{ is monotonically increasing function.}$$

- (c) $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = -x \quad \forall x \in \mathbb{R} \text{ is monotonically decreasing function.}$$

- 6. Even Function and Odd Function:** A function $f: S \rightarrow \mathbb{R}$ is said to be an even function if $f(-x) = f(x) \forall x \in S$ and f is called an odd function if $f(-x) = -f(x) \forall x \in S$.

Examples of even functions:

- (a) $f(x) = x^2$ on $\mathbb{R}$

- (b) $f(x) = \cos x$ on $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

- (c) $f(x) = |x|$ on $\mathbb{R}$ are even functions.

Examples of odd functions:

- (a) $f(x) = \sin x$ on $\mathbb{R}$

- (b) $f(x) = x^3$ on $\mathbb{R}$

- (c) $f(x) = x^2 \sin(x)$ on $\mathbb{R}$ all are odd functions.

Remark:

- The graph of an even function is always symmetrical about y -axis. i.e., the curve left hand side of y -axis is the mirror image of the curve on its right hand side.
- The graph of an odd function is always symmetric in opposite quadrant. i.e., the curve in the first quadrant is identical to the curve in third quadrant and the curve in second quadrant is identical to the curve in fourth quadrant.
- If f is a even and differentiable function on $\mathbb{R}$. Then $f'(x)$ is an odd function.
- If $f(x)$ is an odd differentiable function on $\mathbb{R}$, then $f'(x)$ is an even function.

5.5. Limit of a Function

5.5.1 General Principle For Existence of Limit (GPEL)

Let $S \subseteq \mathbb{R}$ be a subset of $\mathbb{R}$ and $x = \alpha$ is a limit point of S . (may or may not be member of S)

Let $f: S \rightarrow \mathbb{R}$

We say $l \in \mathbb{R}$ is the limit of $f \Leftrightarrow$ for any $\varepsilon > 0 \exists \delta > 0$

Such that

$$x_1, x_2 \in \{x: 0 < |x - \alpha| < \delta\} \Rightarrow |f(x_1) - f(x_2)| < \varepsilon$$

Note: This method is rather useful to disclaim that limit does not exist.

Examples:

- $\lim_{x \rightarrow a} \sin \frac{1}{x-a}$ does not exist.

Solution: For any $\delta > 0 \exists n \in \mathbb{N}$ such that

$$-\delta < x_1 - a = \frac{1}{-2n\pi - \frac{\pi}{2}} < x_2 - a = \frac{1}{2n\pi + \frac{\pi}{2}} < \delta, \text{ and so,}$$

$$x_1, x_2 \in \{x: 0 < |x - a| < \delta\} \Rightarrow \left| \sin \frac{1}{x_1 - a} - \sin \frac{1}{x_2 - a} \right| = |-1 - 1| = 2 \not< \varepsilon = 1.$$

Hence, by the general principle for the existence of limits, $\lim_{x \rightarrow a} \sin \frac{1}{x-a}$ does not exist.

- $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ -1 & x \in \mathbb{Q}^c \end{cases}$$

Then limit does not exist at any real number.

Solution: For any $\alpha \in \mathbb{R}$ and for any $\delta > 0$, $\exists x_1 \in \mathbb{Q}$ and $x_2 \in \mathbb{Q}^c$ and $x_1, x_2 \in (\alpha - \delta, \alpha + \delta)$ such that

$$|f(x_1) - f(x_2)| = |1 - (-1)| = 2$$

Put Your Own Notes

3. Prove that $\lim_{x \rightarrow a} \sqrt{x} = \sqrt{a}$ for $a > 0$.

Solution: The function $f(x) = \sqrt{x}$ is defined for all non-negative real numbers. Since we consider the interval $(a - \delta, a + \delta)$, we can choose $0 < \delta < a$. Now $\sqrt{x} + \sqrt{a} > \sqrt{a}$.

$$\text{Now } |\sqrt{x} - \sqrt{a}| = \left| \frac{x-a}{\sqrt{x} + \sqrt{a}} \right| < \frac{1}{\sqrt{a}} |x-a| < \varepsilon, \text{ if } |(x-a)| < \varepsilon \sqrt{a}.$$

Hence, we choose $\delta = \min(a, \varepsilon \sqrt{a})$ or any other smaller positive real number. Hence if $0 < |x-a| < \delta$, then we have

$$|\sqrt{x} - \sqrt{a}| < \frac{1}{\sqrt{a}} |x-a| < \frac{\delta}{\sqrt{a}} < \frac{\varepsilon \sqrt{a}}{\sqrt{a}} = \varepsilon.$$

4. Let $f(x) = \begin{cases} 4x+3 & \text{if } x \neq 2 \\ 12 & \text{if } x = 2 \end{cases}$

Prove that $\lim_{x \rightarrow 2} f(x) = 11$

Solution: Let $\varepsilon > 0$ be given; we want to find a $\delta > 0$ such that

$$|f(x) - 11| < \varepsilon \text{ whenever } |x-2| < \delta.$$

$$\text{Now } |f(x) - 11| = |4x+3-11| = |4x-8| = 4|x-2| \text{ we can choose } \delta = \frac{\varepsilon}{4}.$$

Then we have $|f(x) - 11| < \varepsilon$ whenever $0 < |x-2| < \frac{\varepsilon}{4}$. Hence

$\lim_{x \rightarrow 2} f(x) = 11$. But by definition, at $x = 2$, $f(x) = 12$ so that

$$\lim_{x \rightarrow 2} f(x) \neq f(2).$$

5. $\lim_{x \rightarrow a} \frac{x^2 - a^2}{x - a} = 2a$

Solution: Here $f(x) = (x^2 - a^2)/(x - a), x \neq a$.

Let $\varepsilon > 0$ be given. Then we want to find a $\delta > 0$ such that

$$|f(x) - 2a| < \varepsilon \text{ whenever } 0 < |x-a| < \delta.$$

$$\text{Now, } |f(x) - 2a| = \left| \frac{x^2 - a^2}{x - a} - 2a \right| = \left| \frac{(x-a)^2}{x-a} \right| = |x-a|$$

We can choose $\delta = \varepsilon$. Then we have

$$|f(x) - 2a| < \varepsilon \text{ whenever } 0 < |x-a| < \varepsilon$$

$$\text{Hence } \lim_{x \rightarrow a} \frac{x^2 - a^2}{x - a} = 2a.$$

6. $\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$

Solution: Here $f(x) = x \sin(1/x)$.

Let $\varepsilon > 0$ be given. Then we want to find a $\delta > 0$ such that

$$|f(x) - 0| < \varepsilon \text{ whenever } 0 < |x - 0| < \delta.$$

Now, $|f(x) - 0| = \left| x \sin \frac{1}{x} \right| = |x| \left| \sin \frac{1}{x} \right| \leq x$, as $\left| \sin \frac{1}{x} \right| \leq 1$

We can choose $\delta = \varepsilon$. Then we have

$$|f(x) - 0| < \varepsilon \text{ whenever } 0 < |x| < \varepsilon$$

Hence, $\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$

Put Your Own Notes

5.5.2 Second Definition

Let $S \subseteq \mathbb{R}$ be a subset of $\mathbb{R}$ and $x = \alpha$ be a limit point of S (may or may not belongs to S)

Let $f: S \rightarrow \mathbb{R}$.

We say $l \in \mathbb{R}$ is the limit of $f \Leftrightarrow$ for any sequence $\langle a_n \rangle$, such that $\langle a_n \rangle \rightarrow \alpha \Rightarrow f(a_n) \rightarrow l$. ($a_n \in S \forall n \in \mathbb{N}$)

Note:

- (a) If we can find two sequences $\langle a_n \rangle$ and $\langle b_n \rangle$ in S converging to α but $f(a_n)$ and $f(b_n)$ converging to different limit. Then we say the function has no limit at $x = \alpha$ or limit does not exist at $x = \alpha$.
- (b) We can say limit does not exist at $x = \alpha$ if we can find a sequence $\langle a_n \rangle$ in S such that $\langle a_n \rangle \rightarrow \alpha$ and $f(a_n)$ does not converge at all.

Examples:

- (a) $f: \mathbb{R} - \{0\} \rightarrow \mathbb{R}$ defined by

$$f(x) = \sin\left(\frac{1}{x}\right) \forall x \in \mathbb{R} - \{0\}$$

Then limit of f does not exist at $x = 0$.

Reason: $\langle a_n \rangle = \left\langle \frac{1}{n} \right\rangle$ is a sequence in $S = \mathbb{R} - \{0\}$ such that

$$\langle a_n \rangle \rightarrow 0 \text{ but } f(a_n) = f\left(\frac{1}{n}\right) = \sin\left(\frac{1}{\frac{1}{n}}\right) = \sin(n)$$

does not converge anywhere.

- (b) $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$$

Then limit of f does not exist at $x = 0$.

Reason: Take $\langle a_n \rangle = \left\langle \frac{1}{n} \right\rangle$ and $\langle b_n \rangle = \left\langle -\frac{1}{n} \right\rangle$

Then $a_n \rightarrow 0$ and $b_n \rightarrow 0$

But $f(a_n) \rightarrow 1$ and $f(b_n) \rightarrow -1$

(c) $f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ defined by

$$f(x) = \frac{1}{x} \quad \forall x \in \mathbb{R} \setminus \{0\}$$

Reason: Take $\langle a_n \rangle = \left\langle \frac{1}{n} \right\rangle$, then $\lim_{x \rightarrow 0} f(x)$ does not exist.

Then $\langle a_n \rangle \rightarrow 0$ but $f(a_n) = \left\langle \frac{1}{\frac{1}{n}} \right\rangle = \langle n \rangle$ which is not convergent.

5.5.3 One Side Limit

Definition 1: A function $f(x)$ is said to approach L as $x \rightarrow a$ from the right or from above if given $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \text{ whenever } a < x < a + \delta.$$

The number L is called the right hand limit of f at a . We write this as $\lim_{x \rightarrow a+} f(x) = L$. Note that for the existence of the right hand limit $f(x)$ should be defined in $(a, a + \delta)$.

Definition 2: A function $f(x)$ is said to approach M as x approaches a from the left or from below if given $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|f(x) - M| < \varepsilon \text{ whenever } (a - \delta < x < a).$$

The number M is called the left hand limit of f at $x = a$. We write this as $\lim_{x \rightarrow a-} f(x) = M$. Note that for the existence of the left hand limit $f(x)$ should be defined in $(a - \delta, a)$.

Note: We say limit exists if right hand limit & left hand limit are equal.

Example: Evaluate $\lim_{x \rightarrow 0-} \frac{|x|}{x}$ and $\lim_{x \rightarrow 0+} \frac{|x|}{x}$.

Solution: By definition,

$$|x| = x, \text{ if } x \geq 0 \text{ and } |x| = -x, \text{ if } x < 0.$$

$$\text{Now } \lim_{x \rightarrow 0-} f(x) = \lim_{\substack{x \rightarrow 0 \\ (x < 0)}} \frac{|x|}{x} = \lim_{x \rightarrow 0} \frac{-x}{x} = -1,$$

$$\lim_{x \rightarrow 0+} f(x) = \lim_{\substack{x \rightarrow 0 \\ (x > 0)}} \frac{|x|}{x} = \lim_{x \rightarrow 0} \frac{x}{x} = 1.$$

Hence $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist, since $-1 \neq 1$.

Example: Show that $\lim_{x \rightarrow 0} \frac{x-|x|}{x}$ does not exist.

Solution: We have $\lim_{x \rightarrow 0^-} \frac{x-|x|}{x} = \lim_{x \rightarrow 0^-} \frac{x-(-x)}{x} = \lim_{x \rightarrow 0^-} \frac{2x}{x} = 2$,

and $\lim_{x \rightarrow 0^+} \frac{x-|x|}{x} = \lim_{x \rightarrow 0^+} \frac{x-x}{x} = 0$.

Hence $\lim_{x \rightarrow 0} \frac{x-|x|}{x}$ does not exist.

Example: Evaluate $\lim_{x \rightarrow 0} [x+|x|]$.

Solution: $\lim_{x \rightarrow 0^-} [x+|x|] = \lim_{x \rightarrow 0^-} (x-x) = 0$,

$\lim_{x \rightarrow 0^+} [x+|x|] = \lim_{x \rightarrow 0^+} (x+x) = \lim_{x \rightarrow 0^+} 2x = 0$.

Hence $\lim_{x \rightarrow 0} (x+|x|) = 0$.

Example: A function f is defined by

$$f(x) = \begin{cases} \frac{1}{2} - x & \text{if } 0 < x < \frac{1}{2} \\ \frac{3}{2} - x & \text{if } \frac{1}{2} < x < 1 \end{cases}$$

Evaluate $\lim_{x \rightarrow \frac{1}{2}} f(x)$

Solution: We have $\lim_{x \rightarrow \frac{1}{2}^-} f(x) = \lim_{x \rightarrow \frac{1}{2}^-} f(x) = \lim_{x \rightarrow \frac{1}{2}^-} \left\{ \frac{1}{2} - x \right\} = 0$.

$\lim_{x \rightarrow \frac{1}{2}^+} f(x) = \lim_{x \rightarrow \frac{1}{2}^+} f(x) = \lim_{x \rightarrow \frac{1}{2}^+} \left\{ \frac{3}{2} - x \right\} = 1$.

Hence $\lim_{x \rightarrow \frac{1}{2}} f(x)$ does not exist.

5.5.4 Theorems on Limits

1. (Uniqueness Theorem)

Let $S \subseteq \mathbb{R}$ be a subset of $\mathbb{R}$ and f is defined on S .

Then $\lim_{x \rightarrow a} f(x)$ if exist then it is unique.

2. If $\lim_{x \rightarrow a} f(x) = L$, then f is bounded on some deleted neighbourhood of $x = a$.

3. $\lim_{x \rightarrow a} f(x)$ and equal to a real number $l \Leftrightarrow$ both left hand limit $\lim_{x \rightarrow a^-} f(x)$ and right limit $\lim_{x \rightarrow a^+} f(x)$ exist and are equal to l .

4. Algebra of limits

Let $\lim_{x \rightarrow a} f(x) = l$ and $\lim_{x \rightarrow a} g(x) = m$

Then

(i) $\lim_{x \rightarrow a} (f(x) \pm g(x)) = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x) = l \pm m$

(ii) $\lim_{x \rightarrow a} (f(x) \cdot g(x)) = \lim_{x \rightarrow a} (f(x)) \cdot \lim_{x \rightarrow a} (g(x)) = l \cdot m$

$$(iii) \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{l}{m} \text{ provided } m \neq 0.$$

Examples:

$$(a) \lim_{x \rightarrow -1} \frac{(x+2)(3x-1)}{x^2+3x-x}$$

$$(b) \lim_{x \rightarrow 0} \frac{\sqrt{4+x}-2}{x}$$

$$(c) \lim_{x \rightarrow 0^+} \frac{\sin x}{\sqrt{x}}$$

Solutions:

$$(a) \lim_{x \rightarrow -1} \frac{(x+2)(3x-1)}{x^2+3x-x} = \frac{\lim_{x \rightarrow -1} (x+2) \cdot \lim_{x \rightarrow -1} (3x-1)}{\lim_{x \rightarrow -1} (x^2+3x-x)}$$

$$= \frac{1 \cdot (-4)}{-4} = 1.$$

$$(b) \lim_{x \rightarrow 0} \frac{\sqrt{4+x}-2}{x} = \lim_{x \rightarrow 0} \frac{\sqrt{4+x}-2}{x} \cdot \frac{\sqrt{4+x}+2}{\sqrt{4+x}+2}$$

$$= \lim_{x \rightarrow 0} \frac{1}{\sqrt{4+x}+2} = \frac{1}{4}.$$

$$(c) \lim_{x \rightarrow 0^+} \frac{\sin x}{\sqrt{x}} = \left(\lim_{x \rightarrow 0^+} \frac{\sin x}{x} \right) \cdot \left(\lim_{x \rightarrow 0^+} \sqrt{x} \right) = 1 \cdot 0 = 0$$

Example: Evaluate $\lim_{x \rightarrow 0} \frac{e^{1/x}}{e^{1/x}+1}$.

Solution: Now when $x \rightarrow 0^+$, $1/x \rightarrow \infty$, $e^{1/x} \rightarrow \infty$ and when $x \rightarrow 0^-$, $1/x \rightarrow -\infty$, $e^{1/x} \rightarrow 0$.

$$\therefore \lim_{x \rightarrow 0^+} \frac{e^{1/x}}{e^{1/x}+1} = \lim_{x \rightarrow 0^+} \frac{1}{1+e^{-1/x}} = 1$$

$$\text{and } \lim_{x \rightarrow 0^-} \frac{e^{1/x}}{e^{1/x}+1} = \frac{0}{1} = 0$$

so that the left hand limit is not equal to the right hand limit.

Hence $\lim_{x \rightarrow 0} \frac{e^{1/x}}{e^{1/x}+1}$ does not exist.

Example: Find $\lim_{x \rightarrow 0} e^x \operatorname{sgn}(x + [x])$, where the signum function is defined

$$\text{as } \operatorname{sgn}(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \text{ and } [x] \text{ means the greatest integer } \leq x. \\ -1 & \text{if } x < 0 \end{cases}$$

Solution: Left for the reader.

(Ans. $\lim_{x \rightarrow 0} e^x \operatorname{sgn}(x + [x])$ does not exist.)

5. If f and g are defined on a deleted neighbourhood of a point a and $f(x) \geq g(x) \forall x \in D$ then $\lim_{x \rightarrow a} f(x) \geq \lim_{x \rightarrow a} g(x)$ provided limit exist.

6. **Sandwich Theorem for Functions:** If function f, g and h are defined on a deleted neighbourhood D of a point a such that $f(x) \geq g(x) \geq h(x) \forall x \in D$ and $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = l$, then $\lim_{x \rightarrow a} g(x)$ exists and equals to l .

7. Let f be strictly increasing on $I \subset \mathbb{R}$ then f^{-1} exist and is strictly increasing $f(I)$

Reason: Since f is strictly increasing on I , it is one-to-one on I and so f^{-1} exists. To prove that f^{-1} is strictly increasing, let $y_1 < y_2$ be two points of $f(I)$ and let $x_1 = f^{-1}(y_1)$ and $x_2 = f^{-1}(y_2)$. We cannot have $x_1 > x_2$ for this will imply $y_1 > y_2$ contradicting our assumption. Hence the only alternative is $x_1 < x_2$ which means $f^{-1}(y_1) < f^{-1}(y_2)$. Hence f^{-1} is strictly increasing.

8. If f is monotonic on (a, b) , then for each c in (a, b) $\lim_{x \rightarrow c^+} f(x)$ and $\lim_{x \rightarrow c^-} f(x)$ both exist may not be equal.

9. If f is monotonic increasing on (a, b) , then for each $c \in (a, b)$

$$\lim_{x \rightarrow c^-} f(x) = \sup_{x \in (a, c)} f(x) < f(c) < \inf_{x \in (c, b)} f(x) = \lim_{x \rightarrow c^+} f(x)$$

5.5.5 Some Important Limits

- (i) $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = \lim_{x \rightarrow \infty} \left(1 - \frac{1}{x}\right)^{-x} = e;$
- (ii) $\lim_{x \rightarrow 0} (1+x)^{1/x} = e;$
- (iii) $\lim_{x \rightarrow \infty} \left(1 + \frac{y}{x}\right)^x = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^{yx} = e^y;$
- (iv) $\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1;$
- (v) $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a, \forall a > 0;$
- (vi) $\lim_{x \rightarrow a} \frac{x^p - a^p}{x - a} = p a^{p-1}, \forall p \neq 0 \text{ and } a \neq 0 \text{ if } p = 0;$
- (vii) $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1;$

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(viii) $\lim_{x \rightarrow 0} \cos x = 1$.

The above limits are of basic significance.

5.5.6 Infinite Limits and Limits at Infinity

- (i) A function f is said to tend to l as $x \rightarrow \infty$ if given $\varepsilon > 0$, there exists a positive number k such that $|f(x) - l| < \varepsilon$ whenever $x > k$

Also then we write $\lim_{x \rightarrow \infty} f(x) = l$ or $f(x) \rightarrow l$ as $x \rightarrow \infty$.

- (ii) A function f is said to tend to l as $x \rightarrow -\infty$ if given $\varepsilon > 0$, there exists a positive number k such that $|f(x) - l| < \varepsilon$ whenever $x < -k$

Also then we write $\lim_{x \rightarrow -\infty} f(x) = l$ or $f(x) \rightarrow l$ as $x \rightarrow -\infty$.

- (iii) A function f is said to tend to ∞ as x tends to a , if given $k > 0$, however large, there exists a positive number δ such that $f(x) > k$ whenever $0 < |x - a| < \delta$.

Also then we write $\lim_{x \rightarrow a} f(x) = \infty$ or $f(x) \rightarrow \infty$ as $x \rightarrow a$.

- (iv) A function f is said to tend to $-\infty$ as x tends to a , if given $k > 0$, however large, there exists a positive number δ such that $f(x) < -k$ whenever $0 < |x - a| < \delta$.

Also then we write $\lim_{x \rightarrow a} f(x) = -\infty$ or $f(x) \rightarrow -\infty$ as $x \rightarrow a$.

- (v) A function f is said to tend to ∞ as $x \rightarrow \infty$ if given $k > 0$, however large, there exists a positive number K such that $f(x) > k$ whenever $x > K$

Also then we write $\lim_{x \rightarrow \infty} f(x) = \infty$ or $f(x) \rightarrow \infty$ as $x \rightarrow \infty$.

- (vi) A function f is said to tend to $-\infty$ as $x \rightarrow \infty$ if given $k > 0$, however large, there exists a positive number K such that $f(x) < -k$ whenever $x > K$

Also then we write $\lim_{x \rightarrow \infty} f(x) = -\infty$ or $f(x) \rightarrow -\infty$ as $x \rightarrow \infty$.

- (vii) A function f is said to tend to ∞ as $x \rightarrow -\infty$, if given $k > 0$, however large, there exists a positive number K such that $f(x) > k$ whenever $x < -K$.

Also then we write $\lim_{x \rightarrow -\infty} f(x) = \infty$ or $f(x) \rightarrow \infty$ as $x \rightarrow -\infty$.

- (viii) A function f is said to tend to $-\infty$ as $x \rightarrow -\infty$, if given $k > 0$, however large, there exists a positive number K such that $f(x) < -k$ whenever $x < -K$.

Example: Show that $\lim_{x \rightarrow +\infty} \frac{1}{x} = \lim_{x \rightarrow -\infty} \frac{1}{x} = 0$.

Solution: Let $\varepsilon > 0$ be given. Choose $M = 1/\varepsilon$. Then

$$x > M \Rightarrow \left| \frac{1}{x} \right| < \frac{1}{M} = \varepsilon \Rightarrow \left| \frac{1}{x} - 0 \right| < \varepsilon.$$

Hence $\lim_{x \rightarrow +\infty} \frac{1}{x} = 0$,

Now $x < -M \Rightarrow |x| > M \Rightarrow \left| \frac{1}{x} \right| < \frac{1}{M} = \varepsilon$

or $x < -M \Rightarrow \left| \frac{1}{x} - 0 \right| < \varepsilon$. Hence $\lim_{x \rightarrow -\infty} \frac{1}{x} = 0$.

Examples: Show that

(i) $\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty$

(ii) $\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$

(iii) $\lim_{x \rightarrow 0} \frac{1}{x^2} = +\infty$

Solutions:

(i) Let $M > 0$ be any large number. Let $\delta = 1/M$.

Then $0 < x < \delta \Rightarrow \frac{1}{x} > \frac{1}{\delta} \Rightarrow \frac{1}{x} > M$.

Hence $\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty$,

(ii) Again $-\delta < x < 0 \Rightarrow -\frac{1}{\delta} > \frac{1}{x} \Rightarrow -M > \frac{1}{x} \Rightarrow \frac{1}{x} < -M$.

Hence $\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$,

Note : $\lim_{x \rightarrow 0} \frac{1}{x}$ does not exist.

(iii) Now $-\delta < x < \delta \Rightarrow \frac{1}{x^2} > \delta \Rightarrow \frac{1}{x^2} > M$.

Hence $\lim_{x \rightarrow 0} \frac{1}{x^2} = +\infty$,

Example: Find $\lim_{x \rightarrow \infty} \frac{x^2 - 4}{x^2 + 7x + 12}$.

Solution: Dividing numerator and denominator by x^2 , we get

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{x^2 - 4}{x^2 + 7x + 12} &= \lim_{x \rightarrow \infty} \frac{1 - 4/x^2}{1 + 7/x + 12/x^2} \\ &= \frac{1 - 4 \lim_{x \rightarrow \infty} (1/x^2)}{1 + 7 \lim_{x \rightarrow \infty} (1/x) + 12 \lim_{x \rightarrow \infty} (1/x^2)} \\ &= \frac{1 - 0}{1 + 0 + 0} = 1. \end{aligned}$$

Example: Evaluate $\lim_{x \rightarrow 0} \frac{e^{1/x} - 1}{e^{1/x} + 1}$.

Solution: We know $e^t = 1 + t + \frac{t^2}{2!} + \dots$

It follows that $e^t \rightarrow \infty$ as $t \rightarrow +\infty$ (1)

By previous example, $x \rightarrow 0+ \Rightarrow 1/x \rightarrow +\infty \Rightarrow e^{1/x} \rightarrow +\infty$, by (1)

$$\Rightarrow \frac{1}{e^{1/x}} \rightarrow 0 \Rightarrow e^{-1/x} \rightarrow 0$$

Again $x \rightarrow 0- \Rightarrow -\frac{1}{x} \rightarrow +\infty \Rightarrow e^{-1/x} \rightarrow +\infty$, by (1)

$$\Rightarrow \frac{1}{e^{1/x}} \rightarrow +\infty \Rightarrow e^{1/x} \rightarrow 0.$$

Hence $x \rightarrow 0+ \Rightarrow e^{-1/x} \rightarrow 0$ and $x \rightarrow 0- \Rightarrow e^{1/x} \rightarrow 0$. (A)

We have $\lim_{x \rightarrow 0-} \frac{e^{1/x} - 1}{e^{1/x} + 1} = \frac{0-1}{0+1} = -1$, using (A)

and $\lim_{x \rightarrow 0+} \frac{e^{1/x} - 1}{e^{1/x} + 1} = \lim_{x \rightarrow 0+} \frac{1 - e^{-1/x}}{1 + e^{-1/x}} = \frac{1-0}{1+0} = 1$, by (A).

Hence $\lim_{x \rightarrow 0} \frac{e^{1/x} - 1}{e^{1/x} + 1}$ does not exist. ($\because -1 \neq 1$)

Example: Show that $\lim_{x \rightarrow 0} \frac{xe^{1/x}}{1 + e^{1/x}} = 0$.

Solution: We know that $x \rightarrow 0+ \Rightarrow e^{-1/x} \rightarrow 0$,

and $x \rightarrow 0- \Rightarrow e^{1/x} \rightarrow 0$.

We have $\lim_{x \rightarrow 0-} \frac{xe^{1/x}}{1 + e^{1/x}} = \frac{0 \cdot 0}{1+0} = 0$,

and $\lim_{x \rightarrow 0+} \frac{xe^{1/x}}{1 + e^{1/x}} = \lim_{x \rightarrow 0+} \frac{x}{e^{-1/x} + 1} = \frac{0}{0+1} = 0$,

Hence $\lim_{x \rightarrow 0} \frac{xe^{1/x}}{1 + e^{1/x}} = 0$.

Example: Prove that $\lim_{x \rightarrow \infty} \frac{1}{x+1} = 0$.

Solution: Let $f(x) = 1/(x+1)$ and let $\varepsilon > 0$ be given. Then

$$|f(x) - 0| = \frac{1}{x+1} < \frac{1}{x} < \varepsilon \text{ if } x > \frac{1}{\varepsilon}$$

Taking $k = 1/\varepsilon$. Then, we see that

$$\left| \frac{1}{x+1} - 0 \right| < \varepsilon \text{ whenever } x > k \Rightarrow \lim_{x \rightarrow \infty} \frac{1}{x+1} = 0.$$

Example: Evaluate $\lim_{x \rightarrow 0} \frac{1}{|x|}$

Solution: Let $f(x) = 1/|x|$. Let $k > 0$ be given and let $\delta = 1/k$. Then

$$0 < |x - 0| < \delta \Rightarrow |x| < \delta \Rightarrow |x| < 1/k \Rightarrow \frac{1}{|x|} > k$$

Thus, $f(x) > k$ whenever $0 < |x - 0| < \delta \Rightarrow \lim_{x \rightarrow 0} \frac{1}{|x|} = \infty$

Example: Find $\lim_{x \rightarrow 0} [x]$, where $[x]$ denotes the greatest integer not greater than x .

Solution: Right-hand limit = $\lim_{x \rightarrow 0+0} [x] = \lim_{h \rightarrow 0+} [0+h] = 0$

and Left-hand limit = $\lim_{x \rightarrow 0-0} [x] = \lim_{h \rightarrow 0+} [0-h] = -1$

Since right-hand limit $\neq$ left-hand limit, so $\lim_{x \rightarrow 0} [x]$ does not exist

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CHAPTER 6

CONTINUITY

6.1. Definitions:

Let $S \subseteq \mathbb{R}$ be a subset of $\mathbb{R}$ and $f: S \rightarrow \mathbb{R}$ be a real valued function. Then we say f is continuous at $\alpha \in S$ if any of the following condition is satisfied:

- (i) $\alpha \in S - S'$ (i.e. α is an isolated point of S)
- (ii) $\alpha \in S'$ and $\lim_{x \rightarrow \alpha} f(x) = f(\alpha)$ (i.e. limit of f exist at $x = \alpha$ and equal to the value of the function)

Remarks:

- (a) If any of the condition fails at a point then the function f is said to be discontinuous at that point and the point is called a point of discontinuity of f .
- (b) If limit point of S is not member of S . Then f is assumed to be discontinuous at that point.

Example: Let $f: \mathbb{R} - \{1\} \rightarrow \mathbb{R}$ be defined as $f(x) = \sin x$. Then at $x = 1$, f is discontinuous function.

- (c) At isolated points, $f(x)$ is said to be continuous if it is defined there at.

Example: Let $f: [0, 1] \cup \{3\} \rightarrow \mathbb{R}$ defined by $f(x) = \sin x$. Then f is continuous at $x = 3$.

Example: The function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \begin{cases} x \sin \frac{1}{x}, & x \neq 0 \\ 0 & x = 0 \end{cases}$, is continuous at $x = 0$.

Solution: We know, $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \left(x \sin \frac{1}{x} \right) = 0$. Also, $f(0) = 0$.

So $\lim_{x \rightarrow 0} f(x) = f(0)$.

Hence, f is continuous at $x = 0$.

Example: The function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \begin{cases} \frac{xe^{1/x}}{1+e^{1/x}}, & x \neq 0 \\ 0 & x = 0 \end{cases}$, is continuous at $x = 0$.

Solution: We have $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{xe^{1/x}}{1+e^{1/x}} = 0$

$\therefore \lim_{x \rightarrow 0} f(x) = f(0)$ ($\because f(0) = 0$)

Hence, f is continuous at $x = 0$.

CHAPTER 6

CONTINUITY

6.1. Definitions:

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Example: Let $f: [0, 1] \cup \{3\} \rightarrow \mathbb{R}$ defined by $f(x) = \sin x$. Then f is continuous at $x = 3$.

Example: The function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \begin{cases} x \sin \frac{1}{x}, & x \neq 0 \\ 0 & x = 0 \end{cases}$, is continuous at $x = 0$.

Solution: We know, $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \left(x \sin \frac{1}{x} \right) = 0$. Also, $f(0) = 0$.

So $\lim_{x \rightarrow 0} f(x) = f(0)$.

Hence, f is continuous at $x = 0$.

Example: The function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \begin{cases} \frac{xe^{1/x}}{1+e^{1/x}}, & x \neq 0 \\ 0 & x = 0 \end{cases}$, is continuous at $x = 0$.

Solution: We have $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{xe^{1/x}}{1+e^{1/x}} = 0$

$\therefore \lim_{x \rightarrow 0} f(x) = f(0)$ ($\because f(0) = 0$)

Hence, f is continuous at $x = 0$.

Example: The function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \begin{cases} \frac{x}{1+e^{1/x}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$, is

continuous at $x = 0$.

Solution: We have, $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{x}{1+e^{1/x}} = \frac{0}{1+0} = 0$, $\left(\because \lim_{x \rightarrow 0^-} e^{1/x} = 0 \right)$

$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{x}{1+e^{1/x}} = \lim_{x \rightarrow 0^+} \frac{xe^{-1/x}}{e^{-1/x} + 1} = \frac{0 \cdot 0}{0 + 1} = 0$. $\left(\because \lim_{x \rightarrow 0^+} e^{-1/x} = 0 \right)$

Also, $f(0) = 0$. Hence f is continuous at $x = 0$.

Example: Let f be the function defined on $\mathbb{R}$ by $f(x) = [x]$, for all $x \in \mathbb{R}$, where $[x]$ denote the greatest integer not exceeding x . Then f is discontinuous at the points $x = 0, \pm 1, \pm 2, \pm 3, \dots$ and is continuous at every other point.

Example: Let f be the function defined on $\mathbb{R}$ as $f(x) = |x| + [x]$, for all $x \in \mathbb{R}$. Determine the points of discontinuity of f .

Solution: We have,

$f(x) = x + 0 = x$, for $0 \leq x < 1$; $f(x) = x + 1$, for $1 \leq x < 2$;

$f(x) = x + 2$, for $2 \leq x < 3$; $f(x) = x + 3$, for $3 \leq x < 4$;

...

...

$f(x) = -x - 1$, for $-1 \leq x < 0$; $f(x) = -x - 2$, for $-2 \leq x < -1$;

and so on.

We have

$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (-x - 1) = -1$, $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x = 0$;

$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} x = 1$, $\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (x + 1) = 2$;

$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (x + 1) = 3$, $\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (x + 2) = 4$;

$\lim_{x \rightarrow -1^-} f(x) = \lim_{x \rightarrow -1^-} (-x - 2) = -3$, $\lim_{x \rightarrow -1^+} f(x) = \lim_{x \rightarrow -1^+} (-x - 1) = -2$;

and so on.

Thus $\lim_{x \rightarrow n^-} f(x) \neq \lim_{x \rightarrow n^+} f(x) \quad \forall n \in \mathbb{Z}$.

Hence the points of discontinuity of f are $0, \pm 1, \pm 2, \pm 3, \dots$

6.1.1. Second Definition of Continuity

Let $S \subseteq \mathbb{R}$ and $f: S \rightarrow \mathbb{R}$ be a function then we say f is continuous at the point $\alpha \in \bar{S} \Leftrightarrow$ for every sequence $\langle x_n \rangle$ in S such that $\langle x_n \rangle \rightarrow \alpha \Rightarrow f(x_n) \rightarrow f(\alpha)$

Remarks:

- (i) If we can find two sequences $\langle a_n \rangle, \langle b_n \rangle$ in S such that $\langle a_n \rangle, \langle b_n \rangle$ converges to α but $f(a_n), f(b_n)$ converges to different limits. Then f is not continuous at $x = \alpha$.
- (ii) If we can find a sequence $\langle a_n \rangle$ which converges to α but $f(a_n)$ does not converges at all. Then f is not continuous at $x = \alpha$.

Examples related to above definitions:

- (a) Let $f : [0, 1] \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} \frac{1}{q} & x = \frac{p}{q} \in [0, 1], p, q \in \mathbb{Z}^+ \\ 0 & \text{otherwise} \end{cases}$$

Then f is continuous at every irrational number in $[0, 1]$ and 0 but discontinuous at all rationals except 0.

- (b) Let $f : [-1, 1] \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} x & x \in [-1, 1] \cap \mathbb{Q}^c \text{ and } x \neq 0 \\ \sqrt{\frac{1+p^2}{1+q^2}} & x = \frac{p}{q} \in [-1, 1] \end{cases}$$

Then limit does not exist on $[-1, 0)$, so not continuous on $[-1, 0)$

Hence, f is continuous on $(0, 1) \cap \mathbb{Q}^c$ and at $x = 0, 1$

- (c) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ -1 & x \in \mathbb{Q}^c \end{cases}$$

Then, f is not continuous at any real number.

- (d) Let $f : [0, 1] \rightarrow [0, 1]$ defined by

$$f(x) = \begin{cases} p \sin\left(\frac{1}{q}\right) & x = \frac{p}{q} \in [0, 1], p, q \in \mathbb{Z}^+ \\ x & x = 0 \text{ or } x \in [0, 1] \cap \mathbb{Q}^c \end{cases}$$

Then f is continuous at all irrational in $[0, 1]$ & 0 and not continuous at all rationals except 0.

Note: Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by

$$f(x) = \begin{cases} \phi_1(x), & x \in S \\ \phi_2(x), & x \notin S \end{cases}$$

Where $\phi_1(x), \phi_2(x)$ are continuous function on their domain and S is any dense subset of $\mathbb{R}$.

Then roots of the equation $\phi_1(x) - \phi_2(x) = 0$ are only point of continuity of f .

Also if the roots are repeated then f is differentiable at that point also.

Example:

(a) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} x^6 - 1 & x \in \mathbb{Q} \\ 1 - x^6 & x \in \mathbb{Q}^c \end{cases}$$

Then f is continuous at $x = 1, -1$ only.

Reason: The equation $(x^6 - 1) - (1 - x^6) = 0$ gives $x^6 = 1$

Which have only two distinct real roots viz. 1 and -1.

(b) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} x & x \in \mathbb{Q} \\ -x & x \in \mathbb{Q}^c \end{cases}$$

Then f is continuous only at $x = 0$.

6.1.2. Third Definition of Continuity

A function $f: S \rightarrow \mathbb{R}$ is continuous at a point $c \in S$ iff for given any ε -neighbourhood $V_\varepsilon(f(c))$ of $f(c)$ there exist a δ -neighbourhood $V_\delta(c)$ of c such that $f(S \cap V_\delta(c)) \subseteq V_\varepsilon(f(c))$ i.e. if x is any point of $S \cap V_\delta(c)$, then $f(x)$ belongs to $V_\varepsilon(f(c))$.

6.2. Types of Discontinuity

1. Removable discontinuity (simple discontinuity): Let $S \subseteq \mathbb{R}$ and $f: S \rightarrow \mathbb{R}$ be a function. If $\lim_{x \rightarrow \alpha} f(x)$ exist but either $f(x)$ is not defined at $x = \alpha$ or it is defined and $\lim_{x \rightarrow \alpha} f(x) \neq f(\alpha)$. Then we say that the function has removable discontinuity at $x = \alpha$.

Example:

(a) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} \frac{\sin x}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

Then f has removable type discontinuity at $x = 0$.

Reason: $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) = 1$ and $f(0) = 0$.

(b) Let $f: (0, 1) \rightarrow \mathbb{R}$ defined by

$$f(x) = x \quad \forall x \in (0, 1).$$

Then f is discontinuous at $x = 0$ and 1.

Note: This type of discontinuity can be removed by defining a new function g as

$$g(x) = \begin{cases} f(x) & \text{when } x \neq \alpha \\ \lim_{x \rightarrow \alpha} f(x) & \text{when } x = \alpha \end{cases}$$

Then g is continuous at $x = \alpha$.

Put Your Own Notes

2. **Discontinuity of first kind or (Jump discontinuity):** A function $f: S \rightarrow \mathbb{R}$ is said to have discontinuity of first kind (or Jump discontinuity) at $x = \alpha$ if at this point left hand limit (L.H.L.) and right hand limit (R.H.L.) both exist but not equal furthermore.

- (i) A function f is said to have a discontinuity of the first kind from the left to α if L.H.L exist but it is not equal to $f(\alpha)$.
- (ii) A function f is said to have a discontinuity of first kind from the right to α if R.H.L. exist but it is not equal to $f(\alpha)$.

Note: Jump at $(x = \alpha) = \left| \lim_{x \rightarrow \alpha^-} f(x) - \lim_{x \rightarrow \alpha^+} f(x) \right|$

Example:

- (a) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = [x] \quad \forall x \in \mathbb{R}$

Then f has jump kind of discontinuity at every $\alpha \in \mathbb{Z}$.

- (b) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \{x\} \quad \forall x \in \mathbb{R}$, where $\{x\}$ is fractional part of x .

Then at $x = 1$ f has jump kind of discontinuity.

- (c) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$$

Then at $x = 0$, f has jump kind of discontinuity.

3. **Discontinuity of second kind (Mixed discontinuity):** A function $f(x)$ is said to have a discontinuity of second kind (or mixed discontinuity) at $x = \alpha$, if any of Right hand limit or Left hand limit or both does not exist at $x = \alpha$.

Examples:

- (a) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ -1 & x \in \mathbb{Q}^c \end{cases}$$

Then for any $\alpha \in \mathbb{R}$.

$$\exists \delta > 0 \text{ and } x_1 \in \mathbb{Q} \cap (\alpha, \alpha + \delta), x_2 \in \mathbb{Q}^c \cap (\alpha, \alpha + \delta) \text{ such that } |f(x_1) - f(x_2)| = 2.$$

$\Rightarrow$ R.H.L. does not exist

$\Rightarrow f$ has discontinuity of second kind at every $\alpha \in \mathbb{R}$.

- (b) Let $f: [0, 1] \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

Then at $x = 0$ L.H.L. and R.H.L. both does not exist

$\Rightarrow f$ has a discontinuity of second kind at $x = 0$.

(c) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} \frac{e^{1/x} \sin(1/x)}{1+e^{1/x}} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

Then f has discontinuity of second kind at $x = 0$.

Reason: We know that

$$x \rightarrow 0^- \Rightarrow e^{1/x} \rightarrow 0 \text{ and } x \rightarrow 0^+ \Rightarrow e^{-1/x} \rightarrow 0.$$

$$\therefore \lim_{x \rightarrow 0^+} \frac{e^{1/x} \sin(1/x)}{1+e^{1/x}} = \lim_{x \rightarrow 0^+} \frac{\sin(1/x)}{e^{-1/x} + 1} = \lim_{x \rightarrow 0^+} \sin \frac{1}{x},$$

which does not exist.

Hence, f has a discontinuity of the second kind from the right at $x = 0$.

6.3. Algebra of Continuity

Let $S \subseteq \mathbb{R}$ and f & g real valued be functions on S . Let $a, b \in \mathbb{R}$ suppose that $\alpha \in S$ and f & g are continuous at α . Then

- (i) Linear combination of f and g is also continuous at $x = \alpha$ i.e. $af + bg$ is continuous at $x = \alpha$.
- (ii) Product of f and g is continuous at α i.e. $f \cdot g$ is continuous at $x = \alpha$.
- (iii) $\frac{f}{g}$ (provided $g(x) \neq 0 \forall x \in S$) is continuous at $x = \alpha$.
- (iv) $|f|$ is also continuous at $x = \alpha$.
- (v) $\max\{f, g\} = \frac{(f+g) + |f-g|}{2}$ is also continuous at $x = \alpha$.
- (vi) $\min\{f, g\} = \frac{(f+g) - |f-g|}{2}$ is also continuous at $x = \alpha$.

Remark:

(a) Converse of the above results may not be true.

Example: Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be functions defined as $f(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ -1 & x \notin \mathbb{Q} \end{cases}$

$$\text{and } g(x) = \begin{cases} -1 & x \in \mathbb{Q} \\ 1 & x \notin \mathbb{Q} \end{cases}$$

Clearly both f and g are discontinuous at each real number.

But $(f+g)(x) = 0 \forall x \in \mathbb{R}$ is continuous function

$(f \cdot g)(x) = -1 \forall x \in \mathbb{R}$ is continuous function

$$\left(\frac{f}{g}\right)(x) = -1 \forall x \in \mathbb{R} \text{ is continuous function.}$$

(b) If f is continuous function at $x = \alpha$ and g is discontinuous function at $x = \alpha$ then $f \pm g$ cannot be continuous function at $x = \alpha$.

(c) If f and g are two continuous functions at $x = \alpha$ such that $g \circ f$ and $f \circ g$ is defined. Then $f \circ g$ and $g \circ f$ both are continuous at $x = \alpha$.
Converse need not be true.

Example: Consider two functions f and g defined as $f(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ -1 & x \in \mathbb{Q}^c \end{cases}$

$$\text{and } g(x) = \begin{cases} -1 & x \in \mathbb{Q} \\ 1 & x \in \mathbb{Q}^c \end{cases}$$

Then f and g are nowhere continuous but $f \circ g$ and $g \circ f$ both are continuous on $\mathbb{R}$.

6.4. Some Important Theorems on Continuous Functions

- 1. Boundedness Theorem:** Let $I = [a, b]$ and $f: I \rightarrow \mathbb{R}$ be a continuous function, then f is bounded on I .

Remark: Each hypothesis of the Boundedness Theorem is needed, we can construct examples which show that the conclusion may fail if any one of the hypothesis is relaxed.

- The interval must be bounded: The function $f(x) = x$ for x in the unbounded interval $A = [0, \infty)$ is continuous but not bounded on A .
- The interval must be closed: The function $g(x) = \frac{1}{x}$ for x in the half-open interval $B = (0, 1]$ is continuous but not bounded on B .
- The function must be continuous: The function $h(x)$ defined on the closed interval $C = [0, 1]$ by $h(x) = \frac{1}{x}$ for $x \in (0, 1]$ and $h(0) = 1$ is discontinuous and unbounded on C .
- Converse of the theorem need not be true.

Example: Let $f: [0, 1] \rightarrow \mathbb{R}$ defined by $f(x) = \begin{cases} 1 & x \in \mathbb{Q} \cap [0, 1] \\ -1 & x \in \mathbb{Q}^c \cap [0, 1] \end{cases}$

Then f is bounded but not continuous.

Definitions:

- (i) Absolute Maximum:** Let $A \subseteq \mathbb{R}$ and $f: A \rightarrow \mathbb{R}$ be a function, then we say that f has an **Absolute Maximum** on A if there is a point $x^* \in A$ such that $f(x^*) \geq f(x)$ for all $x \in A$.
- (ii) Absolute Minimum:** Let $A \subseteq \mathbb{R}$ and $f: A \rightarrow \mathbb{R}$ be a function, then we say that f has an **Absolute Minimum** on A if there is a point $x_* \in A$ such that $f(x_*) \leq f(x)$ for all $x \in A$.

We say that x^* is an **absolute maximum point** for f on A and that x_* is an **absolute minimum point** for f on A if they exist.

- 2. The Maximum-Minimum Theorem:** Let $I = [a, b]$ and $f: I \rightarrow \mathbb{R}$ be continuous on I , then f has an absolute maximum and an absolute minimum on I .

Remark:

- (a) If I is not closed interval then result may not hold.

Example: Let $f : (0,1) \rightarrow \mathbb{R}$ defined by $f(x) = \frac{1}{x}$ is continuous on $(0,1)$, but does not have absolute maximum on I .

- (b) If I is not bounded interval then result may not hold.

Example: Let $f : [0, \infty) \rightarrow \mathbb{R}$ defined by $f(x) = x \quad \forall x \in [0, \infty)$

Then f is continuous on $I = [0, \infty)$ but does not have absolute maximum on I .

- (c) Condition of continuity of f on closed and bounded interval cannot be relaxed.

Example: Let $f : [0,1] \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1 & x = 0 \\ \frac{1}{x} & x \neq 0 \end{cases}$$

Then f is discontinuous and unbounded on $I = [0,1]$

3. **The Location of Roots Theorem:** Let $I = [a,b]$ and $f : I \rightarrow \mathbb{R}$ be continuous on I and if $f(a) < 0 < f(b)$ or $f(a) > 0 > f(b)$ i.e. $f(a).f(b) < 0$ then there exist a number $c \in (a,b)$ such that $f(c) = 0$.

Remark:

- (a) The condition $f(a).f(b) < 0$ cannot be relaxed.

Example: Let $f : [1,2] \rightarrow \mathbb{R}$ defined by $f(x) = x$

Then f is continuous on $I = [1,2]$ but there does not exist any $c \in (1,2)$ such that $f(c) = 0$.

- (b) The condition of continuity of f cannot be relaxed.

Example: Let $f : [-1,1] \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} -1 & x \leq 0 \\ 1 & x > 0 \end{cases}$$

Then $f(-1).f(1) < 0$, but f does not vanish anywhere in $[-1,1]$

- (c) If $I = [a,b]$ and $f : I \rightarrow \mathbb{R}$ be such that $f(a).f(b) < 0$ and there exist $c \in (a,b)$ such that $f(c) = 0$. Then f may not be continuous.

Example: Let $f : [-1,1] \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} -1 & -1 \leq x < 0 \\ 0 & x = 0 \\ 1 & 0 < x \leq 1 \end{cases}$$

Put Your Own Notes

- (d) The converse of the above theorem need not be true i.e. Let $I = [a, b]$ and $f: I \rightarrow \mathbb{R}$ is continuous on I and if $\exists c \in (a, b)$ such that $f(c) = 0$ then $f(a), f(b)$ need not be negative.

Example: Let $f: [-1, 1] \rightarrow \mathbb{R}$ defined by $f(x) = x^2 \quad \forall x \in [-1, 1]$.

- (e) If $f: [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$ and there exist $\alpha, \beta \in [a, b]$ such that $f(\alpha)f(\beta) < 0$ then there exist $c \in (\alpha, \beta)$ such that $f(c) = 0$

4. Bolzano's Intermediate Value Theorem (I.V.P.):

Let I be an interval and $f: I \rightarrow \mathbb{R}$ be continuous on I and if there exist $a, b \in I$ and $k \in \mathbb{R}$ satisfying $f(a) < k < f(b)$, then there exists a point $c \in I$ between a and b such that $f(c) = k$.

Remark:

- (a) The condition of continuity of f cannot be relaxed.

Example: Let $f: [0, 1] \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} x & x \in (0, 1) \\ 1 & x = 0 \\ 0 & x = 1 \end{cases}$$

Then f is continuous on $(0, 1)$ but not continuous at 0 and 1.

$$\text{Also, } f\left(\frac{1}{2}\right) = \frac{1}{2} < \frac{3}{4} < f(0) = 1.$$

But there does not exist any $c \in \left(0, \frac{1}{2}\right)$ such that $f(c) = \frac{3}{4}$

- (b) Continuity of f on $[a, b]$ is sufficient condition for I.V.P. i.e. If a function f satisfy the I.V.P. on $[a, b]$ then it may not be continuous on $[a, b]$.

Example: Let $f: \left[-\frac{2}{\pi}, \frac{2}{\pi}\right] \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases} \quad \text{Then satisfy I.V.P but it is not continuous on } \left[-\frac{2}{\pi}, \frac{2}{\pi}\right]$$

- (c) I.V.P. is a necessary condition for continuity of a function f on $[a, b]$ i.e. if a function f defined on $[a, b]$ does not satisfy I.V.P. on $[a, b]$, then it cannot be continuous on $[a, b]$

5. Preservation of Interval Theorem :

1. Let I be a closed and bounded interval and let $f: I \rightarrow \mathbb{R}$ be continuous on I . Then the set $S = \{f(x) : x \in I\}$ is closed and bounded interval.

i.e., image of a closed and bounded interval under continuous map is closed and bounded interval.

Warning: If $I := [a, b]$ is an interval and $f: I \rightarrow \mathbb{R}$ is continuous on I , we have proved that $f(I)$ is the interval $[m, M]$. We have not proved (and it is not always true) that $f(I)$ is the interval $[f(a), f(b)]$.

2. Let I be an interval and let $f: I \rightarrow \mathbb{R}$ be continuous on I . Then the set $f(I)$ is an interval.

Remark:

- (a) Condition of closeness of I cannot be relaxed.

Example: Let $f: (0, 1) \rightarrow \mathbb{R}$ be defined as

$$f(x) = \frac{1}{x} \quad \forall x \in (0, 1)$$

Then f is continuous on $(0, 1)$

But $S = \{f(x) : x \in (0, 1)\}$ is $(1, \infty)$ which is neither bounded nor closed.

- (b) Condition of boundedness cannot be relaxed.

Example: Let $f: [0, \infty) \rightarrow \mathbb{R}$ be defined as

$$f(x) = x \quad \forall x \in [0, \infty)$$

Then f is continuous on $I = [0, \infty)$ but $S = \{f(x) : x \in I\}$ is $[0, \infty)$, which is unbounded.

- (c) Image of closed and bounded interval under continuous map f is closed and bounded interval and the interval is $[m, M]$, where $m = \inf \{f(x) : x \in I\}$, $M = \sup \{f(x) : x \in I\}$

6. Sign Retainment Property of a Continuous function

If a function f is continuous at an interior point c of an interval $[a, b]$ and $f(c) \neq 0$ then $\exists \delta > 0$ such that $f(x)$ has the same sign as $f(c)$, for every $x \in (c - \delta, c + \delta)$

Remark:

- (a) If f is continuous at the end point b of $[a, b]$ and $f(b) \neq 0$ then there exist an interval $(b - \delta, b)$ such that $f(x)$ has the sign of $f(b)$ for all x in $(b - \delta, b]$
- (b) A similar result hold for continuity and point a .
- (c) This result can be extended for any subset S of $\mathbb{R}$.

i.e., If $S \subseteq \mathbb{R}$, and $f: S \rightarrow \mathbb{R}$ is continuous function at $x = \alpha$ and $f(\alpha) \neq 0$.

Then there exist $\delta > 0$ such that $f(x)$ has the sign of $f(\alpha)$ for all x in $(\alpha - \delta, \alpha + \delta) \cap S$

7. Identity Theorem

(First Form)

Let f be a continuous function on an interval I and $f(x) = 0 \forall x \in I \cap \mathbb{Q}$. Then $f(x) = 0 \forall x \in I$

(Second Form)

If $f(x)$ and $g(x)$ are continuous on an interval I and $f(x) = g(x) \forall x \in I \cap \mathbb{Q}$. Then $f(x) = g(x) \forall x \in I$

Generalized form: Let f and g are two continuous functions on $S \subseteq \mathbb{R}$ and $f(x) = g(x) \forall x \in T$ (where T is a dense subset of S). Then $f(x) = g(x) \forall x \in S$.

Remarks:

- If f is monotonic on I . Then $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$ both exist may not be equal.
i.e., A monotonic function can have only jump kind of discontinuity
- If f is monotonic and satisfy I.V.P. on I then it is continuous function.
- The set of point of discontinuity of the first-kind of real valued function is countable.
- There is no real valued function which is continuous on every rational point and discontinuous at every irrational point.

Some Important Examples:

- Let f be a function defined on $\mathbb{R}$ which satisfy the condition $|f(x)| \leq |x| \forall x \in \mathbb{R}$

Then f is continuous at $x = 0$.

Reason: Since $0 \leq |f(0)| \leq |0|$

$$\Rightarrow f(0) = 0$$

Let $\varepsilon > 0$ be given.

Consider,

$$|f(x) - f(0)| = |f(x)| \leq |x| = |x - 0| < \varepsilon$$

Take $\delta = \varepsilon$. Then, $|f(x) - f(0)| < \varepsilon$, whenever $|x - 0| < \delta$

Thus, f is continuous at $x = 0$.

- Let f be a function defined on $\mathbb{R}$ which satisfy the condition $f(x+y) = f(x) + f(y) \forall x, y \in \mathbb{R}$

Then, if f is continuous at a point c then f is continuous at all points of $\mathbb{R}$

- Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function which satisfy the property

$$(i) |f(x)| \leq |x| \forall x \in \mathbb{R}$$

$$(ii) f(x+y) = f(x) + f(y) \forall x, y \in \mathbb{R}$$

Then $f(x) = cx \forall x \in \mathbb{R}$ where $c = f(1)$

4. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function satisfying the condition

(i) $|f(x)| \leq |x| \forall x \in \mathbb{R}$

(ii) $f(x+y) = f(x)f(y), \forall x, y \in \mathbb{R}$

Then, $f(x) = c^x \forall x \in \mathbb{R}$, where $c = f(1)$

5. If f is a continuous on $\mathbb{R}$ and any of the following conditions are satisfied then f must be constant function.

(i) $f(x) = f(mx) \forall x \in \mathbb{R}, |m| \neq 1, m \in \mathbb{R}$

(ii) $f(x) = f(2x+1) \forall x \in \mathbb{R}$

(iii) $f(x) = f(x^2) \forall x \in \mathbb{R}$.

Fixed Point: Let $S \subseteq \mathbb{R}$ and $f: S \rightarrow \mathbb{R}$ be a any function. Then $\alpha \in S$ is said to be a fixed point of f if $f(\alpha) = \alpha$.

Remark:

(a) Fixed point of $y = f(x)$ are the point of intersection of the curve $y = f(x)$ & $y = x$.

(b) If $h(x) = f(x) - x$, then the zeroes of h are the fixed point of f i.e., number of zeros of h is equal to numbers of fixed point of f .

Example:

(a) The function $f: \mathbb{R} - \{0\} \rightarrow \mathbb{R}$ defined by $f(x) = 2x + \frac{1}{x} - \sin x$ has no fixed point

Reason: Define a function h as

$$h(x) = f(x) - x = x + \frac{1}{x} - \sin x$$

Then h has no zero, since Arithmetic Mean $\geq$ Geometric Mean

$$\Rightarrow \frac{x + \frac{1}{x}}{2} \geq \sqrt{x \cdot \frac{1}{x}}$$

$$\Rightarrow x + \frac{1}{x} \geq 2$$

$$\Rightarrow x + \frac{1}{x} - \sin x \geq 2 - \sin x \geq 1$$

$$\Rightarrow x + \frac{1}{x} - \sin x \neq 0 \forall x$$

$$\Rightarrow x + \frac{1}{x} \neq \sin x \forall x$$

Thus f has no fixed point.

(b) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x + \sin x$

Let $A = \{x \in \mathbb{R} / f(x) = x\}$. Then card (A) is countably infinite

Reason: Let us define a function h as

$$h(x) = f(x) - x$$

$$= \sin x + x - x$$

$$= \sin x$$

Then $h(x) = 0 \quad \forall x = n\pi$ where $n \in \mathbb{Z}$

Thus, card A is countably infinite.

(c) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined as $f(x) = e^x \quad \forall x \in \mathbb{R}$. Then f has no fixed point.

Reason: Since the line $y = x$ and $y = f(x)$ does not intersect anywhere.

(d) Let $f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ be a function defined as $f(x) = |\log_2 |x|| \quad \forall x \in \mathbb{R} \setminus \{0\}$.

Then f has 1 fixed point.

Reason: Since the line $y = x$ and $y = f(x)$ intersect at exactly 1 points.

8. Fixed Point Existence Theorem

If $f: [a, b] \rightarrow [a, b]$ is a continuous function. Then f has a fixed point.
i.e., $\exists x_0 \in [a, b]$ such that $f(x_0) = x_0$

Remark:

(a) Condition of closeness cannot be relaxed.

Example: Let $f: (0, 1) \rightarrow (0, 1)$ be a function defined by $f(x) = x^2$.
Then f is continuous on $(0, 1)$, but it does not have any fixed point.

(b) Condition of boundedness of the interval cannot be relaxed.

Example: Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = e^x$. Then f is continuous on $\mathbb{R}$ but f does not have any fixed point.

(c) If domain and range are not equal then function may not have fixed point

Example: Let $f: (0, 1) \rightarrow (0, \infty)$ be a function defined by $f(x) = \frac{1}{x}$.

Then f is continuous on $(0, 1)$ but does not have any fixed point.

(d) If $f: [a, b] \rightarrow [c, d]$ is a continuous function where $[c, d] \subseteq [a, b]$.
Then f must have a fixed point.

Reason: Take the restriction of f on $[c, d]$.

(e) This theorem gives only the existence of a fixed point. i.e., if $f: [a, b] \rightarrow [a, b]$ is a continuous function then f may have countably infinite, finite uncountable numbers of fixed point.

(i) Let $f: [-1, 1] \rightarrow [-1, 1]$ be a function defined as

$$f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases} \text{ has countably infinite fixed points.}$$

(ii) Let $f: [-1, 1] \rightarrow [-1, 1]$ be a function defined as $f(x) = x^2$ has two fixed point.

(iii) Let $f: [-1, 1] \rightarrow [-1, 1]$ be a function defined as $f(x) = x$ has uncountable numbers of fixed point.

Result: If f is differentiable on $\mathbb{R}$ & $|f'(x)| \leq \alpha < 1$ (where $f'(x)$ denote the derivative of f). Then either f has no fixed point or unique fixed point.

6.5. Uniform Continuity

A function f defined on an interval I is said to be uniformly continuous on I if for each $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$|f(x_2) - f(x_1)| < \varepsilon, \text{ for arbitrary points } x_1, x_2 \text{ of } I \text{ for which } |x_1 - x_2| < \delta$$

Note: The uniform continuity of f on arbitrary set S can be defined by replacing the interval I by S .

Example: The function $f(x) = x/(x+1)$ is uniformly continuous for $x \in [0, 2]$

Reason: Let x, y be two arbitrary points in $[0, 2]$. Then $x \geq 0, y \geq 0$.

$$\Rightarrow x+1 \geq 1 \text{ and } y+1 \geq 1$$

$$\Rightarrow (x+1)(y+1) \geq 1 \quad \dots(1)$$

$$\text{Now, } |f(x) - f(y)| = \left| \frac{x}{x+1} - \frac{y}{y+1} \right| = \frac{|x-y|}{(x+1)(y+1)} \leq |x-y| \text{ by (1)}$$

Let $\varepsilon > 0$ be given. Taking $\delta = \varepsilon$, we get

$$|f(x) - f(y)| < \varepsilon, \text{ whenever } |x-y| < \delta \quad \forall x, y \in [0, 2]$$

Hence, f is uniformly continuous in $[0, 2]$.

Example: Let $f(x) = x^2, x \in \mathbb{R}$. Then f is uniformly continuous on every closed and finite interval but is not uniformly continuous on $\mathbb{R}$.

Example: The function $f(x) = 1/x^2$ is uniformly continuous on $[a, \infty[$, where $a > 0$, but not uniformly continuous on $]0, \infty[$.

Example: The function $f(x) = \sin x^2$ is not uniformly continuous on $[0, \infty[$.

Example: The function $f(x) = \sqrt{x}$ is uniformly continuous in $[0, 1]$.

Example: The function $f(x) = \sin \frac{1}{x}$ is not uniformly continuous on $]0, \infty[$.

Example: The function $f(x) = 1/x$ is not uniformly continuous on $]0, 1[$, but it is uniformly continuous on $[a, \infty[$, where $a > 0$.

6.5.1. Non-Uniform Continuity Criterion

Let $I \subseteq \mathbb{R}$ and let $f: I \rightarrow \mathbb{R}$. Then the following statements are equivalent:

- (i) f is not uniformly continuous on I .
- (ii) There exists an $\varepsilon_0 > 0$ such that for every $\delta > 0$ there are points x_δ, u_δ in I such that $|x_\delta - u_\delta| < \delta$ and $|f(x_\delta) - f(u_\delta)| \geq \varepsilon_0$.
- (iii) There exists an $\varepsilon_0 > 0$ and two sequences (x_n) and (u_n) in I such that $\lim(x_n - u_n) = 0$ and $|f(x_n) - f(u_n)| \geq \varepsilon_0$ for all $n \in \mathbb{N}$.

Application: We can apply this result to claim that $f(x) = 1/x$ is not uniformly continuous on $I = (0, \infty)$. For, if $x_n = 1/n$ and $u_n = 1/(n+1)$, then we have $\lim(x_n - u_n) = 0$, but $|f(x_n) - f(u_n)| = 1$ for all $n \in \mathbb{N}$.

Result: A function which is uniformly continuous on an interval is continuous on that interval.

Note: Converse of the result need not be true.

Example: Let $f: (0,1) \rightarrow \mathbb{R}$ be a function defined as $f(x) = \frac{1}{x}$. Then f is continuous but not uniformly continuous on $(0,1)$.

6.5.2. Uniform Continuity Theorem

Let I be a closed and bounded interval and let $f: I \rightarrow \mathbb{R}$ be continuous on I . Then f is uniformly continuous on I .

Alternative solution: The function $f(x) = x/(x+1)$ is uniformly continuous for $x \in [0,2]$

Here $f(x)$ is a rational function and so it is continuous for every real number other than zeros of the denominator of polynomial function $x+1$. But $x+1=0 \Rightarrow x=-1$, which does not belong to $[0,2]$. Hence $f(x)$ is continuous on the closed interval $[0,2]$.

Since every function which is continuous in a closed interval is also uniformly continuous in that interval, it follows that $f(x) = x/(x+1)$ is uniformly continuous in $[0,2]$.

6.5.3. Lipschitz Function

Let $A \subseteq \mathbb{R}$ and let $f: A \rightarrow \mathbb{R}$ be a function if there exists a constant $K > 0$ such that $|f(x) - f(u)| \leq K|x - u|$ for all $x, u \in A$. Then f is said to be a **Lipschitz function** (or to satisfy a **Lipschitz condition**) on A .

Result: If $f: I \rightarrow \mathbb{R}$ is a Lipschitz function, then f is uniformly continuous on I .

Remark:

Converse of the above result need not be true. i.e., A uniformly continuous function on the interval I may not be Lipschitz function on I .

Example: Let $g(x) = \sqrt{x}$ for x in the closed and bounded interval $I = [0,2]$. Since g is continuous on I . Then g is uniformly continuous on I . However, g is not a Lipschitz function on I .

6.5.4. Continuous Extension Theorem

Let $I = (a, b)$ be an interval. Then, $f : I \rightarrow \mathbb{R}$ is uniformly continuous on I if and only if it can be defined at the point a and b such that the extended function is continuous on $[a, b]$

Application: The function defined by $f(x) = x \sin\left(\frac{1}{x}\right)$ is uniformly continuous on $(0, 1)$

Reason: Since the extension function $g : [0, 1] \rightarrow \mathbb{R}$ defined by

$$g(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases} \text{ is continuous on } [0, 1]$$

6.5.5. Some Important Sufficient Conditions For Uniform Continuity

1. If f is continuous on $\mathbb{R}$ then it is uniformly continuous on every interval of finite length. But not conversely.

Example: A function $f(x) = x^2$ is uniformly continuous on every interval of finite length but not uniformly continuous on $\mathbb{R}$.

2. If f is differentiable on (a, b) and derivative of f is bounded. Then f is uniformly continuous on (a, b) but not conversely (we will prove this by LMVT).

Example: A function $f : (0, 1) \rightarrow \mathbb{R}$ defined as $f(x) = \sqrt{x}$ is uniformly continuous on $(0, 1)$ but derivative is not bounded.

3. If f is periodic on $\mathbb{R}$ and continuous on $\mathbb{R}$. Then it is uniformly continuous on $\mathbb{R}$.
4. Graph of a function is parallel to x -axis then f must be uniformly continuous on its given domain.
5. Graph of f is parallel to x -axis (or fluctuate) then f cannot be uniformly continuous on its given domain.
6. If f is continuous on $[a, \infty)$, ($a \in \mathbb{R}$) and limit at ∞ exist finitely. Then f is uniformly continuous on $[a, \infty)$.
7. If f is uniformly continuous on (a, b) & (b, c) and continuous at $x = b$. Then f is uniformly continuous on (a, c)

Result: Let $f : I \rightarrow \mathbb{R}$ be a uniformly continuous function on I and $\langle x_n \rangle$ be a Cauchy sequence in I . Then $\langle f(x_n) \rangle$ is also Cauchy Sequence.

Remark:

- (a) This result tells us that image of a Cauchy Sequence under uniformly continuous map is Cauchy sequence.
- (b) Uniform continuity $\nRightarrow$ boundedness i.e., A uniformly continuous function may not be bounded.

Example: A function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x$.

- (c) Continuity + boundedness $\nRightarrow$ Uniform continuity i.e., A continuous bounded function may not be uniform continuous.

Example: A function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \sin(x^2)$.

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6.5.6. Algebra of Uniformly Continuous Functions

Let f and g are uniformly continuous functions on the interval I . Then

- (i) $\alpha f + \beta g$ is uniformly continuous on I , $\forall \alpha, \beta \in \mathbb{R}$
- (ii) If f and g both are bounded functions. Then $f \cdot g$ is uniformly continuous function.
- (iii) If $f(x) \geq k > 0 \quad \forall x \in I$ where $k \in \mathbb{R}$. Then $\frac{1}{f}$ is uniformly continuous on I .

Remark:

- (a) Condition of boundedness of f and g cannot be relaxed in (ii)

Example: Let us consider functions $f, g: \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = g(x) = x \quad \forall x \in \mathbb{R}$.

Then both f and g are uniformly continuous on $\mathbb{R}$ but $(f \cdot g)(x) = x^2$ is not uniformly continuous on $\mathbb{R}$.

- (b) The condition $f(x) \geq k > 0$ cannot be relaxed in (iii)

Example: Let us consider a functions $f: (0,1) \rightarrow \mathbb{R}$ defined as $f(x) = x \quad \forall x \in (0,1)$. Then f is uniformly continuous. But $\frac{1}{f}: (0,1) \rightarrow \mathbb{R}$ which is defined as $\left(\frac{1}{f}\right)(x) = \frac{1}{x}$ is not uniformly continuous

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CHAPTER 7

DIFFERENTIABILITY

7.1. Definition

Let f be a real valued function defined on an interval $[a, b]$ i.e., $f: [a, b] \rightarrow \mathbb{R}$. Let $a < c < b$.

(i) **Left Hand Derivative:** The left hand derivative of a function f at $x = c$, denoted by $Lf'(c)$ or $f'(c-0)$ is defined as

$$Lf'(c) = \lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{x - c}, \text{ provided the limit exists.}$$

(ii) **Right Hand Derivative:** The right hand derivative of a function f at $x = c$, denoted by $Rf'(c)$ or $f'(c+0)$, is defined as

$$Rf'(c) = \lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{x - c}, \text{ provided the limit exists.}$$

(iii) If $Lf'(c) = Rf'(c)$, we say that the function f is derivable at $x = c$. The common limit is called the derivative of f at $x = c$ and is denoted by

$$f'(c). \text{ Thus } f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}.$$

(iv) If $Lf'(c) \neq Rf'(c)$, we say that the function f is not derivable at $x = c$.

Example:

(a) Let us define $f: \mathbb{R} \rightarrow \mathbb{R}$ as $f(x) = \begin{cases} x^2 & x \in \mathbb{Q} \\ 2x-1 & x \in \mathbb{Q}^c \end{cases}$. Then f is differentiable at $x = 1$.

(b) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined as $f(x) = \begin{cases} x^\alpha \sin\left(\frac{1}{x^\beta}\right), & \alpha, \beta > 0, x \neq 0 \\ 0, & x = 0 \end{cases}$. Then f is differentiable at $x = 0 \Leftrightarrow \alpha > 1$.

(c) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined as $f(x) = |x| + |x-1| \forall x \in \mathbb{R}$. Then f is not differentiable at $x = 0, 1$.

(d) The function $f(x) = x|x|$ is derivable at the origin.

Reason: We have

$$Lf'(0) = \lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^-} \frac{x(-x)}{x} = 0,$$

$$Rf'(0) = \lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{x(x)}{x} = 0.$$

$$\therefore Lf'(0) = Rf'(0) = 0.$$

$$\text{Hence } f'(0) = 0$$

(e) The function f defined by $f(x) = \begin{cases} x^2 + 3x + a, & \text{if } x \leq 1 \\ bx + 2, & \text{if } x > 1 \end{cases}$ is given to

be derivable for every x . Find a and b .

Solution: Since f is derivable for every x , f must be derivable at $x=1$ and hence, f must be continuous at $x=1$. Let $h > 0$. Then f is continuous at $x=1$.

$$\Rightarrow f(1) = f(1+0) = f(1-0)$$

$$\Rightarrow 4 + a = \lim_{h \rightarrow 0} f(1+h) = \lim_{h \rightarrow 0} f(1-h)$$

$$\Rightarrow 4 + a = \lim_{h \rightarrow 0} \{b(1+h) + 2\} = \lim_{h \rightarrow 0} \{(1-h)^2 + 3(1-h) + a\}$$

$$\Rightarrow 4 + a = b + 2 = 4 + a \Rightarrow a - b = -2 \quad \dots(1)$$

Again, f is differentiable at $x=1 \Rightarrow Rf'(1) = Lf'(1)$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h}$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{b(1+h) + 2 - (4+a)}{h} = \lim_{h \rightarrow 0} \frac{(1-h)^2 + 3(1-h) + a - (4+a)}{-h}$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{b - a + bh - 2}{h} = \lim_{h \rightarrow 0} \frac{h^2 - 5h}{-h}$$

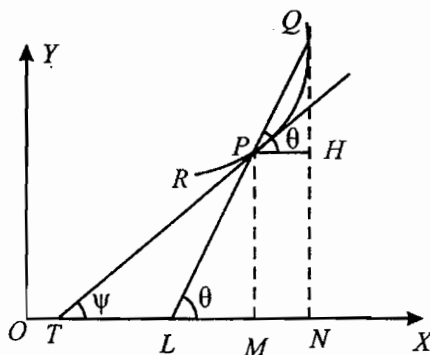
$$\Rightarrow \lim_{h \rightarrow 0} \frac{b - (b-2) + bh - 2}{h} = \lim_{h \rightarrow 0} (5 - h), \text{ using (1)}$$

$$\Rightarrow b = 5$$

Since $b = 5$, so (1) gives $a = b - 2 = 5 - 2 = 3$. Thus, $a = 3, b = 5$.

7.2. Geometrical Meaning of the Derivative

Let $f: [a, b] \rightarrow \mathbb{R}$ be derivable at $c \in [a, b]$. Let RQ be the graphical representation of the function $y = f(x)$. We take two neighboring points $P[c, f(c)]$ and $Q[c+h, f(c+h)]$. On the curve such that $a < c+h < b$. Join PQ and produce it to meet the x -axis at L . Let QL make an angle θ with x -axis. Draw the tangent PT at P which makes an angle ψ with x -axis. Draw PM and QN perpendiculars from P and Q on the x -axis. Also draw PH perpendicular from P on QN . Then, we have $QH = QN - HN = QN - PM = f(c+h) - f(c)$ and $PH = MN = ON - OM = (c+h) - c = h$.



$$\therefore \text{Slope of } PQ = \tan \theta = \frac{QH}{PH} = \frac{f(c+h) - f(c)}{h}$$

From the figure, we see that as $h \rightarrow 0$, the point Q approaches the point P along the graph of f and, hence, the chord PQ approaches the tangent line PT' as its limiting position. Thus, as $h \rightarrow 0, \theta \rightarrow \psi$.

$$\therefore \tan \psi = \lim_{Q \rightarrow P} \tan \theta = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h} = f'(c),$$

Showing that $f'(c)$ is the slope of the tangent to the curve $y = f(x)$ at the point $(c, f(c))$.

Note: If there is a sharp edge in the curve then at that f is not diff.

Remark:

1. Continuity is a necessary but not a sufficient condition for the existence a finite derivative.

i.e., every differentiable function is continuous but converse need not be true.

Proof: Let f be derivable at the point c so that $\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$ exists.

$$\text{We have } f(x) - f(c) = \left[\frac{f(x) - f(c)}{x - c} \right] (x - c); x \neq c$$

$$\Rightarrow \lim_{x \rightarrow c} [f(x) - f(c)] = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \lim_{x \rightarrow c} (x - c)$$

$$\Rightarrow \lim_{x \rightarrow c} f(x) - f(c) = f'(c) \cdot 0 = 0$$

$$\Rightarrow \lim_{x \rightarrow c} f(x) = f(c)$$

$$\Rightarrow f \text{ is continuous at } c.$$

Thus, continuity is a necessary condition for differentiability but it is not a sufficient condition for the existence of a finite derivative as shown in the following example. Thus, the converse may not be true.

Example: Let f be defined by $f(x) = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$

Then f is continuous at 0, but it is not derivable at 0.

2. If at $x = c$, f has Left Hand Derivative and Right Hand Derivative both (may or may not be equal)

Then f is continuous at $x = c$.

3. If $f: [a, b] \rightarrow \mathbb{R}$ such that f is differentiable on (a, b) and Right Hand

Limit of $\frac{f(x) - f(b)}{x - b}$ exist finitely at $x = b$ and Left Hand Limit of

$\frac{f(x) - f(a)}{x - a}$ exist finitely at $x = a$. Then f is differentiable on $[a, b]$

7.3. Algebra of Differentiability

Let f and g are two differentiable functions at $c \in S \subseteq \mathbb{R}$. Then

(i) $\alpha f + \beta g$ is differentiable at $c \quad \forall \alpha, \beta \in \mathbb{R}$ and

$$(\alpha f + \beta g)'(c) = \alpha f'(c) + \beta g'(c)$$

(ii) $(f \cdot g)$ is differentiable at c and $(fg)'(c) = f(c)g'(c) + f'(c)g(c)$

(iii) If $g(c) \neq 0$. Then $\frac{f}{g}$ is differentiable at c and

$$\left(\frac{f}{g}\right)'(c) = \frac{g(c)f'(c) - f(c)g'(c)}{(g(c))^2}$$

Remark:

(a) Converse of the above result need not be True.

Example: Let $f: \mathbb{R} \rightarrow \mathbb{R}$, and $g: \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ -1 & x \notin \mathbb{Q} \end{cases}, g(x) = \begin{cases} -1 & x \in \mathbb{Q} \\ 1 & x \notin \mathbb{Q} \end{cases}$$

Clearly, f and g are not differentiable at $x=0$. But $f+g, f \cdot g, \frac{f}{g}$ all are differentiable at $x=0$.

(b) Above results can be extended for finite numbers of differentiable functions (By induction).

(c) If f is differentiable on $S \subseteq \mathbb{R}$ and g is not differentiable on S . Then $f \pm g$ cannot be differentiable on S . Also if $f(x) \neq 0 \quad \forall x \in S$, then $f \cdot g$ is not differentiable on S .

Result: Let $f: [a, b] \rightarrow \mathbb{R}$ be a differentiable function. If c is an interior point of $[a, b]$ and $f'(c) > 0$ then $\exists \delta > 0$ such that $f(x) > f(c) \quad \forall x \in [c, c+\delta]$, $f(x) < f(c) \quad \forall x \in [c-\delta, c]$.

Similarly if $f'(c) < 0$ then $\exists \delta > 0$ such that $f(x) < f(c) \quad \forall x \in [c, c+\delta]$

Also, it may be shown that

$$(a) \quad f'(a) > 0 \Rightarrow f(x) > f(a) \quad \forall x \in [a, a+\delta]$$

$$(b) \quad f'(a) < 0 \Rightarrow f(x) < f(a) \quad \forall x \in [a, a+\delta]$$

$$(c) \quad f'(b) > 0 \Rightarrow f(x) < f(b) \quad \forall x \in [b-\delta, b]$$

$$(d) \quad f'(b) < 0 \Rightarrow f(x) > f(b) \quad \forall x \in [b-\delta, b]$$

7.3.1 Darboux's Theorem

If a function is derivable in a closed interval $[a, b]$ and $f'(a), f'(b)$ are of opposite signs, then there exist at least one point c in the open interval (a, b) such that $f'(c) = 0$.

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Remark:

1. If f is derivable in $[a, b]$ and $f'(a) \neq f'(b)$, then for each number k lying between $f'(a)$ and $f'(b)$, $\exists$ some point $c \in (a, b)$ such that $f'(c) = k$.

Reason: We are given that $f'(a) < k < f'(b) \dots (1)$

$$\text{Let } g(x) = f(x) - kx \quad \forall x \in [a, b] \dots (2)$$

Since, f is derivable in $[a, b]$ and kx is also derivable in $[a, b]$, therefore, by (2), g is derivable in $[a, b]$.

$$\text{Now } g'(x) = f'(x) - k \dots (3)$$

$$\Rightarrow f'(c) - k = 0, \text{ by (3)}$$

$$\text{Hence, } f'(c) = k$$

2. If f be defined and derivable on $[a, b]$, $f(a) = f(b) = 0$ and $f'(a)$ and $f'(b)$ are of the same sign, then f must vanish at least once in (a, b) .

Reason: We are given that $f(a) = f(b) = 0$.

Since $f'(a)$ and $f'(b)$ are of the same sign, we may take

$$f'(a) > 0 \text{ and } f'(b) > 0.$$

Now $f'(a) > 0 \Rightarrow \exists$ some $\delta_1 > 0$ such that

$$f(x) > f(a) = 0 \quad \forall x \in [a, a + \delta_1].$$

Again $f'(b) > 0 \Rightarrow \exists$ some $\delta_2 > 0$ such that

$$f(x) > f(b) = 0 \quad \forall x \in [b - \delta_2, b].$$

$$\text{Thus } f(a + \delta_1) > 0 \text{ and } f(b - \delta_2) < 0. \dots (1)$$

$$\text{Obviously, } [a + \delta_1, b - \delta_2] \subset [a, b] \dots (2)$$

Since f is derivable in $[a, b]$, f is continuous in $[a, b]$ and consequently by (2), f is continuous in $[a + \delta_1, b - \delta_2]$, where $f(a + \delta_1)$ and $f(b - \delta_2)$ are of opposite signs, by (1). Thus, there exists some $c \in [a + \delta_1, b - \delta_2] \subset [a, b]$ such that $f(c) = 0$. Hence f must vanish at least once in $]a, b[$.

3. If f is derivable on $[a, b]$, $f(a) = f(b) = 0$ and $f(x) \neq 0$ for any x in (a, b) then $f'(a)$ and $f'(b)$ must have opposite signs.

Reason: Let, if possible, $f'(a)$ and $f'(b)$ be of the same sign. Also $f(a) = f(b) = 0$. Then, there exists some $c \in]a, b[$ such that $f(c) = 0$.

This contradicts the given hypothesis that $f(x) \neq 0$ for all x in $]a, b[$.

Hence $f'(a)$ and $f'(b)$ must be of opposite signs.

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4. If f is derivable at a point c , then $|f|$ is also derivable at c , provided $f(c) \neq 0$. However result may not hold when $f(c) = 0$.

5. Let $f: [-1, 1] \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} 0, & -1 \leq x < 0 \\ 1, & 0 < x \leq 1 \end{cases}$

Then there does not exist a function g such that $g'(x) = f(x) \forall x \in [-1, 1]$.

Reason: Let, if possible, there exist a function $g: [-1, 1] \rightarrow \mathbb{R}$ such that

$$g'(x) = f(x) = \begin{cases} 0, & -1 \leq x < 0 \\ 1, & 0 < x \leq 1. \end{cases}$$

It follows that g is derivable on $[-1, 1]$ and $g'(-1) \neq g'(1)$
[$\because g'(-1) = 0$ and $g'(1) = 1$].

$\Rightarrow g'$ must assume every value between 0 and 1.
[$\because g'(-1) = 0$ and $g'(1) = 1$]

But this is impossible, since $g'(x) = 1 \forall x \in [0, 1]$.

Hence there does not exist any function g such that $g'(x) = f(x) \forall x \in [-1, 1]$.

7.4. Local Maxima & Local Minima

Let $f: I \rightarrow \mathbb{R}$ and c be an interior point of the interval I . Then

1. $f(c)$ is said to be a local maximum value of the function f , if there exist some neighbourhood $(c-\delta, c+\delta)$ of c , such that $f(c) \geq f(x) \forall x \in (c-\delta, c+\delta)$ other than c and c is called point of local maxima.
2. $f(c)$ is said to be a minimum value of the function f , if there exists some neighbourhood $]c-\delta, c+\delta[$ of c , such that $f(c) \leq f(x) \forall x \in]c-\delta, c+\delta[$ other than c and c is called point of local minima.
3. $f(c)$ is said to be an extreme value of f , if it is either a maximum or a minimum value.
4. **Point of inflexion:** An inflection point, point of inflection is a point on a curve at which the curve changes from being concave (concave downward) to convex (concave upward), or vice versa.

Example: Let $f(x) = x^3$. Then $(0, 0)$ is point of inflexion of f .

Note: For $f(c)$ to be an extreme value, $f(c) - f(x)$ must keep the same sign for every point x other than c in some neighbourhood of c .

Interior Extremum Theorem: Let c be an interior point of the interval I at which $f: I \rightarrow \mathbb{R}$ has an extremum. If the derivative of f at c exist then $f'(c) = 0$

Remark:

- (a) This theorem is applicable only on interior points of the domain.

Reason: Let $f : [0,1] \rightarrow \mathbb{R}$ be defined as $f(x) = x$. Then at $x=0$, f has minimum value 1 and at $x=1$, f has maximum value but neither point is a zero of the derivative of f .

- (b) Converse of the theorem need not be true i.e., if $f'(c) = 0$, then f may have neither a maximum value nor a minimum value.

Example: Let $f : [0,1] \rightarrow \mathbb{R}$ defined as $f(x) = x^3$. Such that $f'(x) = 3x^2$

Then $f'(0) = 0$

But f has neither a maximum value nor a minimum value at $x=0$.

- (c) A function may have maximum or a minimum value at a point without being derivable at that point.

Example: Let us define a function $f(x) = |x|$.

Then f is not derivable at $x=0$ but f has the minimum at $x=0$.

- (d) Let $f : I \rightarrow \mathbb{R}$ be continuous on an interval I and suppose that f has a relative extremum at an interior point c of I . Then either the derivative of f at c does not exist, or it is equal to zero.

Results:

1. If a function f is continuous in $[a, b]$ such that

- (i) $f'(x) > 0 \forall x \in]a, b[$, then f is increasing in $[a, b]$,
- (ii) $f'(x) < 0 \forall x \in]a, b[$, then f is decreasing in $[a, b]$. Further if f is defined in $[a, b]$ and $a < c < b$, then
- (iii) $f'(c) > 0 \Rightarrow f$ is increasing in $]c - \delta_1, c + \delta_1[$ for some $\delta_1 > 0$.
- (iv) $f'(c) < 0 \Rightarrow f$ is decreasing in $]c - \delta_2, c + \delta_2[$ for some $\delta_2 > 0$.

2. **First derivative test for extreme values**

Let a function f be derivable in a neighbourhood of c , where f has an extreme value at c . Then $f(c)$ is a maximum value if the sign of f' changes from plus to minus and $f(c)$ is a minimum value if the sign of f' changes from minus to plus as x passes through c .

Applications:

Example: Examine the function $x^3 - 6x^2 + 9x - 4$ for maximum and minimum values.

Solution: Let $f(x) = x^3 - 6x^2 + 9x - 4$. Then

$$f'(x) = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3)$$

$$\text{or } f'(x) = 3(x-1)(x-3)$$

$$\therefore f'(x) = 0 \text{ for } x = 1, 3.$$

Put Your Own Notes

Now for $x < 1$, $f'(x) = (-) \cdot (-) > 0$,

and for $x > 1$, $f'(x) = (+) \cdot (-) < 0$.

Thus $f'(x)$ changes sign from plus to minus in going through $x = 1$.

Hence the function has a maximum at $x = 1$ and $f(1) = 0$ is a maximum value of f .

Now for $x < 3$, $f'(x) = (+) \cdot (-) < 0$,

and for $x > 3$, $f'(x) = (+) \cdot (+) > 0$.

Thus $f'(x)$ changes sign from minus to plus in going through $x = 3$.

Hence the function has minimum at $x = 3$ and $f(3) = -4$ is a minimum value of f .

3. Second derivative test for extreme values

Let f be derivable on an interval $[c - \varepsilon, c + \varepsilon]$ and let $f'(c) = 0$.

(i) If $f'(c) = 0$ and $f''(c) < 0$, then f has a maximum value at $x = c$.

(ii) If $f'(c) = 0$ and $f''(c) > 0$, then f has a minimum value at $x = c$.

Example: Find the maximum and minimum values of the function $f(x) = 12x^5 - 45x^4 + 40x^3 + 6$, $x \in \mathbb{R}$.

Solution: We have

$$\begin{aligned} f'(x) &= 60x^4 - 180x^3 + 120x^2 \\ &= 60x^2(x^2 - 3x + 2) = 60x^2(x-1)(x-2). \end{aligned}$$

The extreme values of the function f are given by $f'(x) = 0$

i.e., $x = 0, 1, 2$.

$$\text{Now } f''(x) = 240x^3 - 540x^2 + 240x$$

$$\text{or } f''(x) = 60(4x^3 - 9x^2 + 4x).$$

$$\therefore f''(2) = 240 > 0 \Rightarrow f \text{ has a minimum at } x = 2.$$

$$\text{and } f''(1) = -60 < 0 \Rightarrow f \text{ has a maximum at } x = 1.$$

Since $f''(0) = 0$, we find out the third derivative. We have

$$f'''(x) = 720x^2 - 1080x + 240 \Rightarrow f'''(0) = 240 \neq 0.$$

So f has neither a maximum nor a minimum at $x = 0$.

$$\text{Minimum value} = f(2) = -10.$$

$$\text{Maximum value} = f(1) = 12 - 45 + 40 + 6 = 13.$$

Remark:

- (a) Let $f(x) = x^3 \forall x \in \mathbb{R}$. Then $f'(0) = 0$ and $f''(0) = 0$, but the function f has neither a maximum nor a minimum at $x = 0$.
- (b) Let $f(x) = x^4 \forall x \in \mathbb{R}$. Then $f'(0) = 0$ and $f''(0) = 0$. Clearly, the curve $y = x^4$ is symmetrical about the y -axis, passes through the origin and lies above the x -axis only. Hence the function f has a minimum at $x = 0$.
- (c) Let $f(x) = -x^4 \forall x \in \mathbb{R}$. Then $f'(0) = 0, f''(0) = 0$, and f has a maximum at $x = 0$.
4. Let f be a function defined on an interval $[a, b]$ and $a < c < b$. Let $f'(c) = f''(c) = \dots = f^{n-1}(c) = 0$ and $f^n(c) \neq 0$.
- (i) If n is odd, then f has neither a maximum nor a minimum at $x = c$.
- (ii) If n is even, then f has a maximum at $x = c$ if $f^n(c) < 0$, and f has a minimum at $x = c$ if $f^n(c) > 0$.

7.5. Mean Value Theorems

1. Rolle's Theorem:

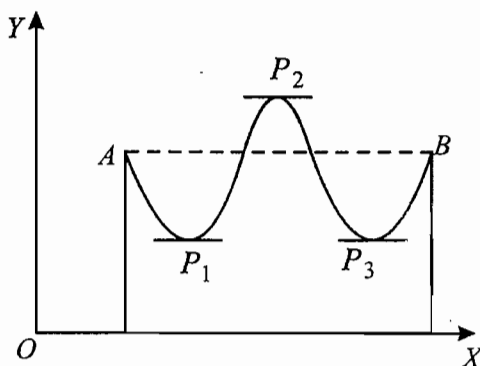
Let f be a function defined on $[a, b]$ such that

- (i) f is continuous on $[a, b]$
- (ii) f is differentiable on (a, b)
- (iii) $f(a) = f(b)$

Then $\exists c \in (a, b)$ such that $f'(c) = 0$.

Geometrical interpretation of Rolle's Theorem:

Let A and B be the points on the graph $AP_1P_2P_3B$ of the function $y = f(x)$ corresponding to $x = a$ and $x = b$ respectively. Then, geometrically, Rolle's theorem asserts that there is at least one point between $x = a$ and $x = b$, at which the tangent to the curve of the function, is parallel to x -axis. In the figure, we have shown the possibility of three points P_1, P_2 and P_3 where the tangent is parallel to x -axis.



Put Your Own Notes

Remark:

- (a) This theorem is useful to produce the zero of derivative of the function.
- (b) A non constant function satisfy the conditions of Rolle's theorem cannot be monotonic.
- (c) This theorem tells us that between two distinct zeros of a differentiable function f there lies a zero of f'
- (d) Third condition of Rolle's theorem cannot be relaxed.

Example: Let $f : [2, 3] \rightarrow \mathbb{R}$ be defined as $f(x) = x$.

Then f is continuous on $[2, 3]$ and differentiable on $(2, 3)$ but $\nexists c \in (2, 3)$ such that $f'(c) = 0$.

Infact Rolle's theorem fails to hold good for a function which does not satisfy any condition of the theorem.

- (e) This theorem only gives the existence of a root of a derivative of the function. i.e., it may be possible that derivative of function has finitely many or infinite zeroes.
- (f) The condition of Rolle's theorem are only sufficient but not necessary for $f'(x)$ to vanish at some point in $[a, b]$

Example: Consider the function $f(x) = \begin{cases} 0 & \text{when } 0 \leq x \leq 1 \\ x+1 & \text{when } 1 < x \leq 2 \end{cases}$. We

can easily show that $f(x)$ is discontinuous and not derivable at $x=1$. Thus $f(x)$ is not continuous on $[0, 2]$ and $f(x)$ is not derivable on $(0, 2)$. Also $f(0) \neq f(2)$ but $f'(x) = 0$ $\forall x \in [0, 1] \subset [0, 2]$

2. Lagrange's Mean Value Theorem (LMVT)

Let f be a function defined on $[a, b]$ such that

- (i) f is continuous on $[a, b]$
- (ii) f is differentiable on (a, b)

Then $\exists c \in (a, b)$ such that $\frac{f(b) - f(a)}{b - a} = f'(c)$

Remark:

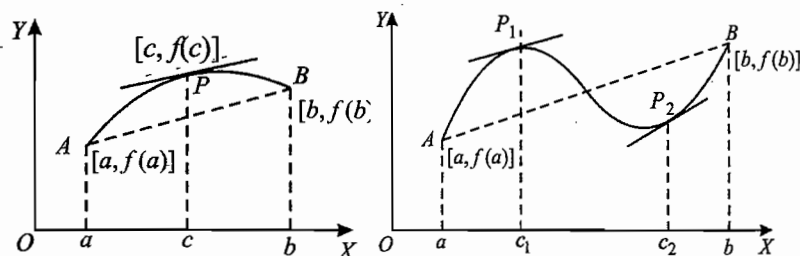
- (a) The c obtained in LMVT need not to be unique i.e., it may be finite or countably infinite or uncountable
- (b) The collection of such c is non empty.
- (c) LMVT is obtained by Rolle's theorem.

Geometrical Interpretation of Lagrange's Mean Value Theorem:

Let A and B be the points on the graph of the function $y = f(x)$ corresponding to $x = a$ and $x = b$ respectively. Then the coordinates of A and B are $(a, f(a))$ and $(b, f(b))$ respectively.

Now, the slope of chord $AB = \frac{f(b) - f(a)}{b - a}$.

Since $f(x)$ is continuous on $[a, b]$ and differentiable on $]a, b[$, figure shows that there is at least one point between A and B the tangent at which is parallel to the chord AB .



If ' c ' be the abscissa of P , then slope of tangent at P is $f'(c)$. Since chord AB is parallel to the tangent at P , we have $[f(b) - f(a)] / (b - a) = f'(c)$.

Thus, interpreted geometrically, Lagrange's mean value theorem says that the tangent to the graph $y = f(x)$ at some suitable point between a and b is parallel to the chord joining the points on the graph with abscissa a and b .

3. Generalized form of Lagrange's Mean Value Theorem:

If a function f defined in $[a, a+h]$ is such that

- (i) f is continuous in $[a, a+h]$
- (ii) f is derivable in $]a, a+h[$

Then there exists some $\theta \in]0, 1[$ such that $f(a+h) - f(a) = hf'(a+\theta h)$.

Some useful deduction from the Mean Value Theorem:

- (i) If f is defined and continuous on $[a, b]$ and is derivable on $]a, b[$, and if $f'(x) = 0$ for all x in $]a, b[$, then $f(x)$ has a constant value throughout $[a, b]$.
- (ii) If $f(x)$ and $g(x)$ are both defined and continuous on $[a, b]$, and are derivable on $]a, b[$, and if $f'(x) = g'(x) \forall x \in]a, b[$, then $f(x)$ and $g(x)$ differ by a constant on $[a, b]$.
- (iii) If f is continuous on $[a, b]$, and $f'(x) \geq 0$ in $]a, b[$, then f is increasing in $[a, b]$.
- (iv) If f is continuous on $[a, b]$ and $f'(x) > 0$ in $]a, b[$, then f is strictly increasing in $[a, b]$.
- (v) If f is continuous on $[a, b]$ and $f'(x) \leq 0$ in $]a, b[$, then f is decreasing in $[a, b]$.
- (vi) If f is continuous on $[a, b]$ and $f'(x) < 0$ in $]a, b[$, then f is strictly decreasing in $[a, b]$.

Put Your Own Notes

Example: Let a function f be defined on $\mathbb{R}$ such that $|f(x) - f(y)| < (x - y)^2$ for all x and y . Then f must be constant function

Solution: Given $|f(x) - f(y)| < (x - y)^2 \quad \forall x, y \in \mathbb{R}$... (1)

Let c be any real number. Then, for $x \neq c$, (1) gives

$$\left| \frac{f(x) - f(c)}{x - c} \right| \leq |x - c| \quad \dots (2)$$

Let $\varepsilon > 0$ be given. Then, we choose $\delta = \varepsilon$ and so (2) gives

$$\left| \frac{f(x) - f(c)}{x - c} - 0 \right| < \varepsilon, \text{ whenever } |x - c| < \delta$$

$$\Rightarrow \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = 0 \Rightarrow f'(c) = 0 \quad \dots (3)$$

Since c is any point on $\mathbb{R}$, (3) shows that $f'(x) = 0 \quad \forall x \in \mathbb{R}$.

Hence, f must be constant function.

4. Cauchy Mean Value Theorem:

Let f and g be two functions defined on $[a, b]$ such that

- (i) f and g are continuous in $[a, b]$
- (ii) f and g are derivable in (a, b)
- (iii) $g'(x) \neq 0$ for each $x \in (a, b)$ and $g(a) \neq g(b)$

Then there exist at least one point $c \in (a, b)$ such that

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$$

7.6. Higher Order Derivatives

We know that the existence of the derivative f' of a function f at a point c implies the existence and continuity of the function in a neighbourhood of c . The derivative of the function f' at c (in case it exists), is called the second derivative of f at c and denoted by $f''(c)$. Evidently, the existence of $f''(c)$ implies the existence and continuity of f' in a neighbourhood of c .

Higher order derivatives can be similarly defined. The derivative of f^{n-1} at c (in case it exists) is called the n th derivative of f at c and is denoted by $f^n(c)$.

1. Taylor's Theorem with Lagrange's Remainder

If a function f defined on $[a, a+h]$ is such that

(i) The $(n-1)$ th derivative $f^{(n-1)}$ is continuous in $[a, a+h]$,

(ii) The $(n-1)$ th derivative $f^{(n-1)}$ is derivable in $]a, a+h[$,

then there exists some $\theta, 0 < \theta < 1$, such that

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \dots + \frac{h^{n-1}}{(n-1)!} f^{(n-1)}(a) + R_n \quad \text{where}$$

$$R_n = \frac{h^n}{n!} f^n(a+\theta h).$$

[Lagrange's Remainder]

2. Taylor's Theorem with Cauchy's Remainder

If a function f defined on $[a, a+h]$ is such that

(i) The $(n-1)$ th derivative $f^{(n-1)}$ is continuous in $[a, a+h]$,

(ii) The $(n-1)$ th derivative $f^{(n-1)}$ is derivable in $]a, a+h[$,

then there exists some $\theta, 0 < \theta < 1$ such that

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \dots + \frac{h^{n-1}}{(n-1)!} f^{(n-1)}(a) + R_n, \quad \text{Where}$$

$$R_n = \frac{h^n (1-\theta)^{n-1}}{(n-1)!} f^n(a+\theta h).$$

[Cauchy's Remainder]

3. Maclaurin's Theorem (With Lagrange's Remainder)

On taking $a=0$ in Taylor Theorem, we get **Maclaurin's Theorem**, which may be stated as follows:

If a function f defined on $[0, h]$ is such that

(i) $f^{(n-1)}$ is continuous in $[0, h]$,

(ii) $f^{(n-1)}$ is derivable in $]0, h[$,

then for each x in $[0, h]$, there exists a real number θ between 0 and 1 such that

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \dots + \frac{x^{n-1}}{(n-1)!} f^{(n-1)}(0) + \frac{x^n}{n!} f^n(\theta x)$$

(Lagrange's remainder)

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \dots + \frac{x^{n-1}}{(n-1)!} f^{(n-1)}(0)$$

$$+ \frac{x^n}{(n-1)!} (1-\theta)^{n-1} f^n(\theta x)$$

(Cauchy's remainder)

7.7. Indeterminate Forms

We shall now discuss the evaluation of limits of functions generally known as Indeterminate forms. They are not indeterminate but have acquired this name by usage of the word.

In general, the limit of $\frac{\phi(x)}{\psi(x)}$ when $x \rightarrow a$, in case the limits of both the functions exist, is equal to the limit of the numerator divided by the limit of the denominator. But what happens when both these limits are zero? The division $\frac{0}{0}$ then becomes meaningless. A case like is known as

Indeterminate form. Other such forms are $\frac{\infty}{\infty}$, $0 \times \infty$, $\infty - \infty$, 0^0 , 1^∞ , and ∞^0 .

Ordinary methods of evaluating the limits are of little help. Particular methods are required to evaluate these limits. We shall now discuss these particular methods, generally called L'Hospital rule, due to the French mathematician L'Hospital.

It should, however, be clearly understood, that in what follows, we do not find the value of $\frac{0}{0}$ or of any of the other indeterminate forms. We only find the limits of combinations of functions which assume these forms when the limits of functions are taken separately.

L'Hospital's Rule for $\frac{0}{0}$ Form

If f and g be two functions such that

- (i) $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$,
- (ii) $f'(x), g'(x)$ exist and $g(x) \neq 0, g'(x) \neq 0 \forall x \in]a - \delta, a + \delta[$, $\delta > 0$, except possibly at a ,
- (iii) $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ exists,

$$\text{Then } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}.$$

Remark: It may happen that $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ is $\frac{0}{0}$ form. In this case, we can extend the L'Hospital's rule as follows:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow a} \frac{f''(x)}{g''(x)},$$

where f' and g' satisfy the conditions of L'Hospital's rule.

This rule can be generalized as follows:

If $f(x)$ and $g(x)$ are functions such that $\lim_{x \rightarrow a} \frac{f^{(n-1)}(x)}{g^{(n-1)}(x)}$ represents the indeterminate form $\frac{0}{0}$ and the functions $f^{(n)}(x)$ and $g^{(n)}(x)$ satisfy the conditions of L'Hospital's rule, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = \dots = \lim_{x \rightarrow a} \frac{f^{(n)}(x)}{g^{(n)}(x)}.$$

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Examples:

1. $\lim_{x \rightarrow 0} \frac{1 - \cos x^2}{x^2 \sin x^2} = \frac{1}{2}$

Solution: We have $\frac{1 - \cos x^2}{x^2 \sin x^2} = \frac{1 - \cos x^2}{x^4} \cdot \frac{x^2}{\sin x^2}$

Since $\lim_{x \rightarrow 0} \frac{x^2}{\sin x^2} = 1$, $\therefore \lim_{x \rightarrow 0} \frac{1 - \cos x^2}{x^2 \sin x^2} = \lim_{x \rightarrow 0} \frac{1 - \cos x^2}{x^4} \left(\frac{0}{0} \right)$

$= \lim_{x \rightarrow 0} \frac{2x \sin x^2}{4x^3}$

(by L'Hospital's rule)

$= \lim_{x \rightarrow 0} \frac{1}{2} \cdot \frac{\sin x^2}{x^2} = \frac{1}{2} \cdot 1 = \frac{1}{2}$

2. $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^2 \tan x} = \frac{1}{3}$

Solution: We have $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^2 \tan x} = \lim_{x \rightarrow 0} \left(\frac{\tan x - x}{x^3} \cdot \frac{x}{\tan x} \right)$

$= \lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$, since $\lim_{x \rightarrow 0} \frac{x}{\tan x} = 1$

$= \lim_{x \rightarrow 0} \frac{\sec^2 x - 1}{3x^2} \left(\frac{0}{0} \right)$

$= \lim_{x \rightarrow 0} \frac{2 \sec^2 x \tan x}{6x} = \frac{1}{3} \lim_{x \rightarrow 0} \sec^2 x \left(\frac{\tan x}{x} \right)$

$= \frac{1}{3} \cdot 1 \cdot 1 = \frac{1}{3}$

3. $\lim_{x \rightarrow 0} \frac{a^x - 1}{b^x - 1} = \frac{\log a}{\log b}$, where $b \neq 1$

Solution: The given limit is $0/0$ form and so

$\lim_{x \rightarrow 0} \frac{a^x - 1}{b^x - 1} = \lim_{x \rightarrow 0} \frac{a^x \log a}{b^x \log b} = \frac{\log a}{\log b}$

4. $\lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e + \frac{1}{2}ex}{x^2} = \frac{11e}{24}$

Solution: Since $\lim_{x \rightarrow 0} (1+x)^{1/x} = e$, the given limit is $0/0$ form.

Before we evaluate the limit, we find the expansion of $(1+x)^{1/x}$.

Let $y = (1+x)^{1/x}$, so that, $\log y = \frac{1}{x} \log(1+x) = \frac{1}{x} \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \right)$

$= \left(1 - \frac{x}{2} + \frac{x^2}{3} - \dots \right) = 1 + \alpha$, where $\alpha = -\frac{x}{2} + \frac{x^2}{3} - \dots$

Put Your Own Notes

Now $\log y = 1 + \alpha \Rightarrow y = e^{1+\alpha} = e \cdot e^\alpha \Rightarrow y = e \left(1 + \alpha + \frac{\alpha^2}{2!} + \dots \right)$

or $y = e \left[1 + \left(-\frac{x}{2} + \frac{x^2}{3} - \dots \right) + \frac{1}{2!} \left(-\frac{x}{2} + \frac{x^2}{3} - \dots \right)^2 + \dots \right]$

$\therefore (1+x)^{1/x} = e \left[1 - \frac{x}{2} + \frac{11}{24}x^2 - \dots \right]$ Now $\lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e + \frac{1}{2}ex}{x^2}$

$$= \lim_{x \rightarrow 0} \frac{e \left(1 - \frac{x}{2} + \frac{11}{24}x^2 - \dots \right) - e + \frac{1}{2}ex}{x^2} = \lim_{x \rightarrow 0} \left(\frac{11e}{24} + \text{terms containing } x \right) = \frac{11e}{24}$$

Examples of the indeterminate forms of the type $\infty - \infty, \frac{\infty}{\infty}, 0 \cdot \infty$

5. $\lim_{x \rightarrow 0^+} \frac{\cot x}{\log \sin x}$ does not exist.

Solution: The given limit is ∞/∞ form. By L'Hospital's rule for ∞/∞ form, we have

$$= \lim_{x \rightarrow 0^+} \frac{\cot x}{\log \sin x} = \lim_{x \rightarrow 0^+} \frac{-\operatorname{cosec}^2 x}{\cot x} \left(\frac{\infty}{\infty} \right)$$

$$= \lim_{x \rightarrow 0^+} \frac{2\operatorname{cosec}^2 x \cot x}{-\operatorname{cosec}^2 x} = \lim_{x \rightarrow 0^+} (-2\cot x) = -\infty$$

6. $= \lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{\sin x} \right) = 0..$

Solution: $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{\sin x} \right)$ is $\infty - \infty$ form. Such a form can be reduced to $0/0$ or ∞/∞ form. Now

$$\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{\sin x} \right) = \lim_{x \rightarrow 0^+} \frac{\sin x - x}{x \sin x} = \lim_{x \rightarrow 0^+} \frac{\sin x - x}{x^2} \left(\frac{x}{\sin x} \right)$$

$$= \lim_{x \rightarrow 0^+} \frac{\sin x - x}{x^2} \left(\frac{0}{0} \right) \quad \left(\because \lim_{x \rightarrow 0^+} \frac{x}{\sin x} = 1 \right)$$

$$= \lim_{x \rightarrow 0^+} \frac{\cos x - 1}{2x} \left(\frac{0}{0} \right) = \lim_{x \rightarrow 0^+} \frac{-\sin x}{2} = 0.$$

7. $\lim_{x \rightarrow \frac{1}{2}\pi} (\sec x - \tan x) = 0.$

Solution: The given limit is $\infty - \infty$ form. We have

$$\lim_{x \rightarrow \frac{1}{2}\pi} (\sec x - \tan x) = \lim_{x \rightarrow \frac{1}{2}\pi} \left(\frac{1}{\cos x} - \frac{\sin x}{\cos x} \right)$$

$$= \lim_{x \rightarrow \frac{1}{2}\pi} \frac{1 - \sin x}{\cos x} \left(\frac{0}{0} \right) = \lim_{x \rightarrow \frac{1}{2}\pi} \frac{-\cos x}{-\sin x} = \frac{0}{1} = 0.$$

8. $\lim_{x \rightarrow 1} \sec\left(\frac{\pi}{2}x\right) \log\left(\frac{1}{x}\right) = \frac{2}{\pi}$.

Solution: The given limit is $0 \cdot \infty$ form. We can write it as

$$\lim_{x \rightarrow 1} \frac{\log(1/x)}{\cos\left(\frac{\pi}{2}x\right)} \left(\frac{0}{0}\right) = \lim_{x \rightarrow 1} \frac{x\left(-\frac{1}{x^2}\right)}{-\frac{\pi}{2} \sin\left(\frac{\pi}{2}x\right)} = \frac{2}{\pi}.$$

The Indeterminate Forms of the type 0^0 , 1^∞ and ∞^0

To evaluate $\lim_{x \rightarrow a} [f(x)]^{g(x)}$,

where $\lim_{x \rightarrow a} f(x) = 0, 1$ or ∞ and $\lim_{x \rightarrow a} g(x) = 0, \infty$ or 0 .

We write $y = [f(x)]^{g(x)}$

$$\Rightarrow \log y = g(x) \log f(x)$$

$$\Rightarrow \lim_{x \rightarrow a} \log y = \lim_{x \rightarrow a} [g(x) \cdot \log f(x)].$$

The R.H.S. can be reduced to the form $0/0$ or ∞/∞ , which can be evaluated. Suppose $\lim_{x \rightarrow a} [g(x) \cdot \log f(x)] = l$.

$$\text{Then } \lim_{x \rightarrow a} \log y = l \Rightarrow \log \lim_{x \rightarrow a} y = l \Rightarrow \lim_{x \rightarrow a} y = e^l.$$

$$\text{Hence } \lim_{x \rightarrow a} [f(x)]^{g(x)} = e^l.$$

9. $\lim_{x \rightarrow 0^+} (\cot x)^{\sin x} = 1.$

Solution: The given limit is ∞^0 form.

Let $y = (\cot x)^{\sin x}$, so that $\log y = \sin x \log \cot x$.

$$\therefore \lim_{x \rightarrow 0^+} \log y = \lim_{x \rightarrow 0^+} \sin x \log \cot x = \lim_{x \rightarrow 0^+} \frac{\log \cot x}{\operatorname{cosec} x} \left(\frac{\infty}{\infty}\right)$$

$$= \lim_{x \rightarrow 0^+} \frac{-\operatorname{cosec}^2 x (1/\cot x)}{-\cot x \operatorname{cosec} x},$$

$$= \lim_{x \rightarrow 0^+} \frac{\operatorname{cosec} x}{\cot^2 x} = \lim_{x \rightarrow 0^+} \frac{\sin x}{\cos^2 x} = 0$$

$$\therefore \lim_{x \rightarrow 0^+} \log y = 0 \Rightarrow \log \lim_{x \rightarrow 0^+} y = 0 \Rightarrow \lim_{x \rightarrow 0^+} y = e^0 = 1.$$

$$\text{Hence } \lim_{x \rightarrow 0^+} (\cot x)^{\sin x} = 1.$$

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10. $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{1/x^2} = e^{1/3}$.

Solution: The given limit is 1^∞ form.

Let $y = \left(\frac{\tan x}{x} \right)^{1/x^2}$, so that $\log y = \frac{1}{x^2} \log \left(\frac{\tan x}{x} \right)$.

$$\begin{aligned} \therefore \lim_{x \rightarrow 0} \log y &= \lim_{x \rightarrow 0} \frac{\log(\tan x / x)}{x^2} \left(\frac{0}{0} \right) = \lim_{x \rightarrow 0} \frac{\frac{x}{\tan x} \left(\frac{x \sec^2 x - \tan x}{x^2} \right)}{2x} \\ &= \lim_{x \rightarrow 0} \frac{x \sec^2 x - \tan x}{2x^2 \tan x} \left(\frac{0}{0} \right) = \lim_{x \rightarrow 0} \frac{\sec^2 x + 2x \sec^2 x \tan x - \sec^2 x}{4x \tan x + 2x^2 \sec^2 x} \\ &= \lim_{x \rightarrow 0} \frac{\sec^2 x \tan x}{2 \tan x + x \sec^2 x} \left(\frac{0}{0} \right) = \lim_{x \rightarrow 0} \frac{\sec^2 x + 2 \sec x (\sec x \tan x) \tan x}{2 \sec^2 x + \sec^2 x + 2x \sec^2 x \tan x} = \frac{1}{3} \\ \therefore \lim_{x \rightarrow 0} \log y &= \frac{1}{3} \Rightarrow \lim_{x \rightarrow 0} y = e^{1/3}. \end{aligned}$$

Hence $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{1/x^2} = e^{1/3}$

11. $\lim_{x \rightarrow 0} (1+x)^{1/x} = e$.

Solution: Let $y = (1+x)^{1/x}$, so that $\log y = \frac{1}{x} \log(1+x)$.

$$\therefore \lim_{x \rightarrow 0} \log y = \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} \left(\frac{0}{0} \right) = \lim_{x \rightarrow 0} \frac{1/(1+x)}{1} = 1.$$

$$\therefore \lim_{x \rightarrow 0} \log y = \log \lim_{x \rightarrow 0} y = 1 \text{ Hence, } \lim_{x \rightarrow 0} (1+x)^{1/x} = e^1 = e.$$

12. $\lim_{x \rightarrow a} \left(2 - \frac{x}{a} \right)^{\tan\left(\frac{\pi x}{2a}\right)} = e^{2/\pi}$, where $a \in \mathbb{R}$.

Solution: The given limit is 1^∞ form. Let $y = \left(2 - \frac{x}{a} \right)^{\tan\left(\frac{\pi x}{2a}\right)}$ Then

$$\log y = \tan\left(\frac{\pi x}{2a}\right) \log\left(2 - \frac{x}{a}\right)$$

$$\therefore \lim_{x \rightarrow a} \log y = \lim_{x \rightarrow a} \frac{\log\left(2 - \frac{x}{a}\right)}{\cot(\pi x / 2a)} \left(\frac{0}{0} \right) = \lim_{x \rightarrow a} \frac{-\frac{1}{a} \cdot \frac{1}{(2-x/a)}}{-(\pi/2a) \operatorname{cosec}^2(\pi x / 2a)} = \frac{2}{\pi}.$$

$$\therefore \log \lim_{x \rightarrow a} y = \frac{2}{\pi}.$$

Hence $\lim_{x \rightarrow a} y = e^{2/\pi}$

13. $\lim_{x \rightarrow 0} (1 + \sin x)^{\cot x} = e.$

Solution: The given limit is 1^∞ form. Let $y = (1 + \sin x)^{\cot x}$, so that
 $\log y = \cot x \log(1 + \sin x).$

$$\therefore \lim_{x \rightarrow 0} \log y = \lim_{x \rightarrow 0} \frac{\log(1 + \sin x)}{\tan x} \left(\frac{0}{0} \right) = \lim_{x \rightarrow 0} \frac{\cos x}{(1 + \sin x) \sec^2 x} = 1.$$

Hence $\lim_{x \rightarrow 0} y = e^1 = e$ or $\lim_{x \rightarrow 0} (1 + \sin x)^{\cot x} = e.$

7.8. Convex Set

A set S of real numbers is said to be convex if $x, y \in S, \lambda \in (0, 1)$ implies
 $x + (1 - \lambda)y \in S.$

7.8.1. Convex Function

Let $S \subseteq \mathbb{R}$ be a subset of $\mathbb{R}$. Then a function $f: S \rightarrow \mathbb{R}$ is said to be convex function. If for any $x_1, x_2 \in S$, $f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2)$, where $\lambda \in [0, 1]$ i.e., Image of convex combination of points of $S \leq$ Convex combination of images of corresponding points.

Example: Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined as $f(x) = x^2$. Then f is a convex function.

Reason: As $(x_1 - x_2)^2 \geq 0 \Rightarrow x_1^2 + x_2^2 \geq 2x_1x_2 \Rightarrow 2x_1^2 + 2x_2^2 \geq (x_1 + x_2)^2$

$$\Rightarrow \frac{x_1^2 + x_2^2}{2} \geq \left(\frac{x_1 + x_2}{2} \right)^2 \Rightarrow \frac{1}{2}(f(x_1) + f(x_2)) \geq f\left(\frac{1}{2}(x_1 + x_2)\right)$$

7.8.2. Concave Function

Let $S \subseteq \mathbb{R}$ be a subset of $\mathbb{R}$. Then a function $f: S \rightarrow \mathbb{R}$ is said to be a concave function if for any $x_1, x_2 \in S$,
 $f(\lambda x_1 + (1 - \lambda)x_2) \geq \lambda f(x_1) + (1 - \lambda)f(x_2)$, where $\lambda \in [0, 1]$ i.e., Image of convex combination of points of $S \geq$ Convex combination of images of corresponding points.

Note:

- The condition of convexity is equivalent to the geometric condition that the curve is below (may be identical in special case) the segment joining any two points of the graph of the convex function.
- For a concave function the chord should lie below the graph.
- Convexity and concavity are geometric characters of a function.
- It is not always easy to determine whether a function is convex or concave. But for differentiable function we shall derive a simple criterion to determine this. But clearly if the tangents at every point of a curve lies below curve, the curve is convex and if the tangents lie above, the curve is concave.

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- (e) A function may be convex on one part of its domain while it is concave or linear on other parts.
- (f) If f is a convex function then $-f$ is concave function and conversely.
- (g) Constant function is concave as well as convex infact (any straight line is convex as well as concave function)

Result: A differentiable function $f:S \rightarrow \mathbb{R}$ is convex function on S
 $\Leftrightarrow f''(x) \geq 0 \quad \forall x \in S$

Result: A differentiable function $f:S \rightarrow \mathbb{R}$ is concave function on S
 $\Leftrightarrow f''(x) \leq 0 \quad \forall x \in S$

Note: Differentiability is not a necessary condition. i.e., a non differentiable function may convex

Example: Let $f:\mathbb{R} \rightarrow \mathbb{R}$ be a function defined as $f(x) = |x| \quad \forall x \in \mathbb{R}$

Then $f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda)f(x_2)$

$(\because |\lambda x_1 + (1-\lambda)x_2| \leq \lambda |x_1| + (1-\lambda)|x_2|)$

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CHAPTER 8

RIEMANN INTEGRAL

8.1. Definition

Integration is concerned with the problem of finding the area of a region under a curve.

Let us start with a simple problem: Find the area A of the region enclosed by a circle of radius r .

For an arbitrary n , consider the n equal inscribed and super-scribed triangles as shown in Figure 1.

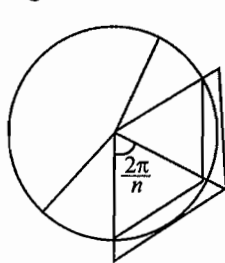


Figure 1

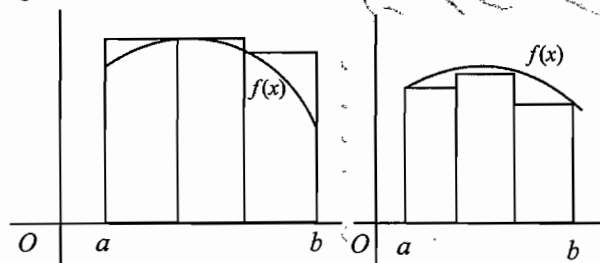


Figure 2

Since A is between the total areas of the inscribed and super-scribed triangles, we have

$$nr^2 \sin\left(\frac{\pi}{n}\right) \cos\left(\frac{\pi}{n}\right) \leq A \leq nr^2 \tan\left(\frac{\pi}{n}\right)$$

By sandwich theorem, $A = \pi r^2$. We will use this idea to define and evaluate the area of the region under a graph of a function.

Suppose f is a non-negative function defined on the interval $[a, b]$. We first subdivide the interval into a finite number of subintervals. Then we squeeze the area of the region under the graph of f between the areas of the inscribed and super-scribed rectangles constructed over the subintervals as shown in Figure 2. If the total areas of the inscribed and super-scribed rectangles converge to the same limit as we make the partition of $[a, b]$ finer and finer then the area of the region under the graph of f can be defined as this limit and f is said to be integrable.

Let us define whatever has been explained above formally.

Definition: Let $[a, b]$ be a given interval. A partition P of $[a, b]$ is a finite set of points $x_0, x_1, x_2, \dots, x_n$ such that $a = x_0 \leq x_1 \leq \dots \leq x_{n-1} \leq x_n = b$. We write $P = \{x_0, x_1, x_2, \dots, x_n\}$.

If P is a partition of $[a, b]$ we write $\Delta x_i = x_i - x_{i-1}$ for $1 \leq i \leq n$. Let f be a bounded real valued function on $[a, b]$. For a partition P of $[a, b]$, we define

$$M_i = \sup\{f(x) : x_{i-1} \leq x \leq x_i\} \text{ and } m_i = \inf\{f(x) : x_{i-1} \leq x \leq x_i\}.$$

$$U(P, f) = \sum_{i=1}^n M_i \Delta x_i \text{ and } L(P, f) = \sum_{i=1}^n m_i \Delta x_i.$$

The numbers $U(P, f)$ & $L(P, f)$ are called upper and lower Riemann sums for the partition P (see Figure 2).

Since f is bounded, there exist real numbers m and M such that $m \leq f(x) \leq M$, for all $x \in [a, b]$. Thus for every partition P ,

$$m(b-a) \leq L(P, f) \leq U(P, f) \leq M(b-a).$$

We define

$$\int_a^b f(x) dx = \inf U(P, f) \quad \dots(1)$$

$$\text{And } \int_a^b f(x) dx = \sup L(P, f). \quad \dots(2)$$

(1) and (2) are called upper and lower Riemann integrals of f over $[a, b]$ respectively.

If the upper and lower integrals are equal, we say that f is Riemann integrable or integrable.

In this case the common value of (1) and (2) is called the Riemann integral

of f and is denoted by $\int_a^b f dx$ or $\int_a^b f(x) dx$.

Example 1: Consider the function $f: [0, 1] \rightarrow \mathbb{R}$ defined by

$$f\left(\frac{1}{2}\right) = 1 \text{ and } f(x) = 0 \text{ for all } x \in [0, 1] \setminus \left\{\frac{1}{2}\right\}.$$

Then f is integrable. We show this using the definition as follows. For any partition P of $[0, 1]$, $L(P, f)$ is always 0 and hence the lower integral is 0.

Let us evaluate the upper integral. Let $P = \{x_1, x_2, \dots, x_n\}$ be any partition of

$[0, 1]$ and $\frac{1}{2} \in [x_i, x_{i+1}]$ for some i . Then $U(P, f) \leq 2 \max \Delta x_j$. Since we can

always choose a partition P such that $\max \Delta x_j$ is as small as possible, the

upper integral, which is the infimum of $U(P, f)$'s, is 0. Hence, f is

integrable and $\int_0^1 f(x) dx = 0$.

Example 2: Not every bounded function is integrable. For example the function

$$f(x) = 1 \text{ if } x \text{ is rational and } 0 \text{ otherwise}$$

Is not integrable over any interval $[a, b]$ (check this).

In general, determining whether a bounded function on $[a, b]$ is integrable, using the definition, is difficult. For the purpose of checking the integrability, we give a criterion for integrability, called Riemann criterion, which is analogous to the Cauchy criterion for the convergence of a sequence.

Let us define some concepts and results before presenting the criterion. Throughout, we will assume that f is a bounded real function on $[a, b]$.

Definition: A partition P_2 of $[a, b]$ is said to be finer than a partition P_1 if $P_2 \supset P_1$. In this case we say that P_2 is a refinement of P_1 . Given two partition P_1 and P_2 , the partition $P_1 \cup P_2 = P$ is called their common refinement.

The following theorem illustrates that refining partition increases lower terms and decreases upper terms.

Theorem 1: Let P_2 be a refinement of P_1 then $L(P_1, f) \leq L(P_2, f)$ and $U(P_2, f) \leq U(P_1, f)$.

Proof: First we assume that P_2 contains just one more point than P_1 . Let this extra point be x^* . Suppose $x_{i-1} < x^* < x_i$, where x_{i-1} and x_i are consecutive points of P_1 . Let

$$w_1 = \inf \{f(x) : x_{i-1} \leq x \leq x^*\} \text{ and}$$

$$w_2 = \inf \{f(x) : x^* \leq x \leq x_i\}$$

Then $w_1 \geq m_i$ and $w_2 \geq m_i$ where $m_i = \inf \{f(x) : x_{i-1} \leq x \leq x_i\}$. Then

$$L(P_2, f) - L(P_1, f) = w_1(x^* - x_{i-1}) + w_2(x_i - x^*) - m_i(x_i - x_{i-1}) \geq 0.$$

If P_2 contains k more points then we repeat this process k -times. The other inequality is analogously proved. (Prove it).

The geometric interpretation suggests that the lower integral is less than or equal to the upper integral. So the next result is also anticipated.

Corollary 2: $\int_a^b f dx \geq \int_a^b f dx$.

Proof: Let P_1, P_2 be two partitions and let P be their common refinement. Then

$$L(P_1, f) \leq L(P, f) \leq U(P, f) \leq U(P_2, f).$$

Thus for any two partitions P_1 and P_2 , we have $L(P_1, f) \leq U(P_2, f)$.

Fix P_2 and take sup over all P_1 . Then $\int_a^b f dx \leq U(P_2, f)$. Now take inf over all P_2 to get the desired result.

In the following result we present the Riemann criterion (a necessary and sufficient condition for the existence of the integral of a bounded function).

Theorem 3: (Riemann's criterion for integrability): f is integrable on $[a, b] \Leftrightarrow$ for every $\varepsilon > 0$ there exists a partition P such that

$$U(P, f) - L(P, f) < \varepsilon. \quad \dots(3)$$

Proof: For any P , we have

$$L(P, f) \leq \int_a^b f dx \leq \int_a^b f dx \leq U(P, f).$$

Therefore (3) implies that

$$\int_a^b f dx - \int_a^b f dx < \varepsilon, \quad \forall \varepsilon > 0.$$

Hence $\int_a^b f dx = \int_a^b f dx$ i.e. f is integrable. Conversely, suppose f is integrable and $\varepsilon > 0$. Then there exist partitions P_1 and P_2 . Then $U(P, f) - L(P, f) < \varepsilon$.

The proof of the following corollary is immediate from the previous theorem.

Corollary 3: Let $f: [a, b] \rightarrow \mathbb{R}$ be a bounded function. Suppose (P_n) is a sequence of partitions of $[a, b]$ such that the sequence $\{\|P_n\|\}$ converges to 0, and $U(P_n, f) - L(P_n, f) \rightarrow 0$, then f is integrable.

Problem: Let $f: [0, 1] \rightarrow \mathbb{R}$ such that $f(x) = \begin{cases} \frac{1}{n} & \text{if } x = \frac{1}{n} \\ 0 & \text{otherwise.} \end{cases}$ Show that f is

integrable and find $\int_0^1 f(x) dx$.

Solution: We will use the Riemann criterion to show that f is integrable on $[0, 1]$. Let $\varepsilon > 0$ be given. We will choose a partition P such that $U(P, f) - L(P, f) < \varepsilon$. Since $\frac{1}{n} \rightarrow 0$, there exists N such that $1/n \in [0, \varepsilon]$ for all $n > N$. So only finite number of $\frac{1}{n}$'s lies in the interval $[\varepsilon, 1]$. Cover these finite number of $\frac{1}{n}$'s by the intervals $[x_1, x_2], [x_3, x_4], \dots, [x_{m-1}, x_m]$ such that $x_i \in [\varepsilon, 1]$ for all $i = 1, 2, \dots, m$ and the sum of the length of these m intervals is less than ε . Consider the partition $P = \{x_1, x_2, \dots, x_m\}$. It is clear that $U(P, f) - L(P, f) < 2\varepsilon$. Hence by the Riemann criterion the function is integrable. Since the lower integral is 0 and the function is integrable, $\int_0^1 f(x) dx = 0$.

We will apply the Riemann criterion for integrability to prove the following two existence theorems.

Theorem 4: If f is continuous on $[a, b]$ then f is integrable.

Proof: Let $\varepsilon > 0$. Since f is uniformly continuous, choose $\delta > 0$ such that $|s - t| < \delta \Rightarrow |f(s) - f(t)| < \varepsilon$ for $s, t \in [a, b]$.

Let P be a partition of $[a, b]$ such that $\Delta x_i < \delta \quad \forall i = 1, 2, \dots, n$. Then

$$M_i - m_i \leq \varepsilon \quad \forall i = 1, 2, \dots, n.$$

Hence

$$U(P, f) - L(P, f) = \sum_{i=1}^n (M_i - m_i) \Delta x_i \leq \varepsilon(b - a).$$

This implies that f is integrable.

Theorem 5: If f is a monotone function on $[a, b]$ then f integrable.

Proof: Suppose f is monotonically increasing (the proof is similar in the other case)

Choose a partition P such that $\Delta x_i = \frac{b-a}{n}$. Then $M_i = f(x_i)$ and $m_i = f(x_{i-1})$. Therefore

$$\begin{aligned} U(P, f) - L(P, f) &= \frac{b-a}{n} \sum_{i=1}^n [f(x_i) - f(x_{i-1})] \\ &= \frac{b-a}{n} [f(b) - f(a)] \end{aligned}$$

$\epsilon < \epsilon$ for large n .

Hence f is integrable.

In the following problem we will see that limit and integral cannot be interchanged.

Problem: Let $g_n(y) = \begin{cases} \frac{ny^{n-1}}{1+y} & \text{if } 0 \leq y < 1 \\ 0 & \text{if } y = 1 \end{cases}$ Then prove that

$$\lim_{n \rightarrow \infty} \int_0^1 g_n(y) dy = \frac{1}{2} \text{ whereas } \int_0^1 \lim_{n \rightarrow \infty} g_n(y) dy = 0.$$

Solution: From the ratio test for sequences we can show that $\lim_{n \rightarrow \infty} \frac{ny^{n-1}}{1+y} = 0$,

for each $0 < y < 1$. Therefore $\int_0^1 \lim_{n \rightarrow \infty} g_n(y) dy = 0$.

For the other part, use integration by parts to see that $\int_0^1 \frac{ny^{n-1}}{1+y} dy = \frac{1}{2} + \int_0^1 \frac{y^n}{(1+y)^2} dy$. Note that $\int_0^1 \frac{y^n}{(1+y)^2} dy \leq \int_0^1 y^n = \frac{1}{n+1} \rightarrow 0$ as

$n \rightarrow \infty$. Therefore, $\lim_{n \rightarrow \infty} \int_0^1 g_n(y) dy = \frac{1}{2}$.

8.2. Quick Review for Riemann Integrability

1. If a bounded function f is Riemann integrable on $[a, b]$ then

$$\int_a^b f dx = \int_a^{\bar{b}} f dx = \int_{\underline{a}}^b f dx$$

2. Every bounded function need not be Riemann integrable.

3. If M, m are bounds of a bounded function f over $[a, b]$, then

$$m(b-a) \leq L(P, f) \leq U(P, f) \leq M(b-a).$$

4. If $f(x) = \begin{cases} 0, & \text{when } x \text{ is rational} \\ 1, & \text{when } x \text{ is irrational} \end{cases}$, f is not integrable on any bounded interval but $f(x)$ bounded.

5. Every constant function is integrable.

6. Let f be a bounded function on $[a, b]$ and P a partition of $[a, b]$. If P^* is any refinement of P , then
 - (i) $U(P^*, f) \leq U(P, f)$
 - (ii) $L(P^*, f) \geq L(P, f)$
7. Let P^* be a refinement of P containing r points more than P , where P is a partition of $[a, b]$ with norm $\mu(P) < \delta$ and $|f(x)| < k \forall x \in [a, b]$. Then
 - (i) $U(P^*, f) \geq U(P, f) - 2rk\delta$
 - (ii) $L(P^*, f) \leq L(P, f) + 2rk\delta$
8. If P_1 and P_2 are any two partitions of $[a, b]$, then $L(P_1, f) \leq U(P_2, f)$ i.e. an upper sum can never be less than any lower sum.
9. If M, m are the bounds of an integrable function f over $[a, b]$, then

$$m(b-a) \leq \int_a^b f dx \leq M(b-a).$$
10. If f a bounded function on $[a, b]$ then for each $\varepsilon > 0$ there exists $\delta \geq 0$ such that
 - (a) $U(P, f) < \int_a^b f dx + \varepsilon$
 - (b) $U(P, f) > \int_a^b f dx - \varepsilon$

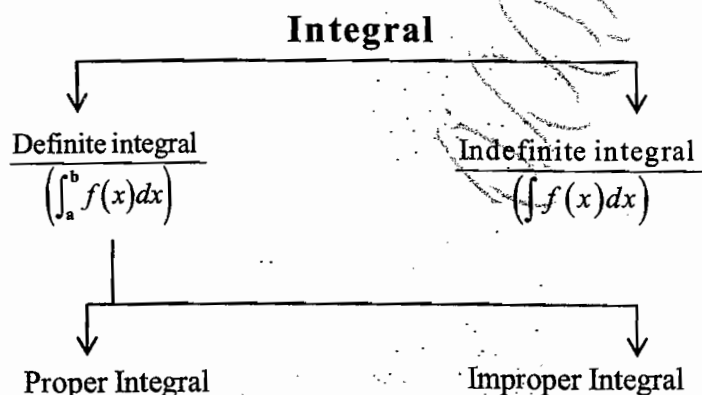
For all partitions P of $[a, b]$ with norm $\mu(P) < \delta$
11. If f is continuous and integrable on $[a, b]$, then there exists a number c lying between a and b such that $\int_a^b f dx = (b-a)f(c)$.
12. If f is bounded and integrable on $[a, b]$ such that $|f(x)| < k \forall x \in [a, b]$, then

$$\left| \int_a^b f dx \right| < k|b-a| \quad k > 0.$$
13. Necessary and sufficient condition for the Integrability of a bounded function f is that to every $\varepsilon > 0$, there corresponds $\delta > 0$ such that for every partition P of $[a, b]$ with norm $\mu(P) < \delta$, $U(P, f) - L(P, f) < \varepsilon$.
14. A necessary and sufficient condition for a bounded function f to be integrable over $[a, b]$ is that $\lim_{\mu(P) \rightarrow 0} w(P, f) = 0$, where $w(p, f)$ is oscillatory sum of $[a, b]$.
15. Every continuous function on $[a, b]$ is integrable on $[a, b]$.
16. An integrable function on $[a, b]$ need not be continuous function.
17. Every monotonic function f on $[a, b]$ is integrable on $[a, b]$.
18. A bounded function f having only finite number of points of discontinuity in $[a, b]$ is integrable on $[a, b]$.

19. If a function f is bounded in $[a, b]$ and the set of its points of discontinuity has a finite number of limit points, then f is integrable in $[a, b]$.
20. If a bounded function f is integrable at $[a, b]$ implies f is also integrable on $[a, c]$ and $[c, b]$, $a < c < b$ and $\int_a^b f dx = \int_a^c f dx + \int_c^b f dx$.
21. If f is bounded and integrable on $[a, c]$, $[c, b]$ then f is integrable over $[a, b]$ and $\int_c^b f dx = \int_a^c f dx + \int_c^b f dx$.
22. If f_1 and f_2 are two bounded and integrable functions on $[a, b]$, then
 - (i) $f_1 + f_2$ is bounded and integrable on $[a, b]$ and $\int_a^b (f_1 + f_2) dx = \int_a^b f_1 dx + \int_a^b f_2 dx$.
 - (ii) $f_1 - f_2$ is bounded and integrable on $[a, b]$ and $\int_a^b (f_1 - f_2) dx = \int_a^b f_1 dx - \int_a^b f_2 dx$.
23. If f is integrable on $[a, b]$, then f^2 is also integrable on $[a, b]$.
24. If f_1 and f_2 are both integrable on $[a, b]$, then $f_1 f_2$ is also integrable on $[a, b]$.
25. If f_1 and f_2 are two bounded and integrable functions on $[a, b]$ and let $\lambda > 0$ such that $|f_2(x)| > \lambda$ for all x in $[a, b]$, then $\frac{f_1}{f_2}$ is bounded and integrable on $[a, b]$.
26. If f is bounded and integrable on $[a, b]$, then $|f|$ is also bounded and integrable on $[a, b]$ and $\left| \int_a^b f dx \right| \leq \int_a^b |f| dx$.
27. If $|f|$ is integrable, then f need not be integrable.
28. A function f is bounded and integrable on $[a, b]$ and there exists a function F such that $F' = f$ on $[a, b]$, and $\int_a^b f dx = F(b) - F(a)$.
29. If f and g are integrable on $[a, b]$ and g keeps the same sign over $[a, b]$ then there exists a number μ lying between the bounds of f such that $\int_a^b f g dx = \mu \int_a^b g dx$.
30. If $\int_a^b \phi dx$ and $\int_a^b g dx$ both exists and ϕ is positive and monotonically decreasing on $[a, b]$, then there exists a point $\xi \in [a, b]$ such that $\int_a^b \phi dx = \phi(\xi) \int_a^b g dx$.
31. If $\int_a^b f dx$ and $\int_a^b g dx$ both exist and f is monotonic on $[a, b]$, then there exists some $\xi \in [a, b]$ such that $\int_a^b f g dx = f(a) \int_a^\xi g dx + f(b) \int_\xi^b g dx$.

32. If f is a non-negative continuous function on $[a, b]$ such that $\int_a^b f dx = 0$, then $f(x) = 0, \forall x \in [a, b]$.
33. Let f be bounded integrable on $[a, b]$ then $\int_a^b [f(x)]^2 dx = 0 \Leftrightarrow f(c) = 0$ at every point c of continuity of f .
34. If the function f and ϕ are bounded and integrable in $[a, b]$ then $\left[\int_a^b f(x)\phi(x) dx \right]^2 \leq \int_a^b [f(x)]^2 dx \cdot \int_a^b [\phi(x)]^2 dx$ and equality hold only if f & g are constant.
35. Let f be integrable on $[a, b]$ and for each $x \in [a, b]$, let $F(x) = \int_a^x f(t) dt$ then F is uniformly continuous on $[a, b]$.
36. If f is bounded and integrable on $[a, b]$, then $F(x) = \int_a^x f(t) dt$, ($a \leq x \leq b$) is continuous on $[a, b]$ and If f is continuous on $[a, b]$ then $F' = f$.

8.3. Improper Integral



$\int_a^b f(x) dx$ is said to be proper integrals if $[a, b]$ If either of a or b is ∞ or $-\infty$ but $f(x)$ is bounded

Finite & $f(x)$ is bounded on $[a, b]$ such type of integral or a & b are finite but $f(x)$ is unbounded is always convergent.

Introduction

In the chapter of Riemann Integral the theory of integration was developed under two assumptions .

- (i) The interval of integration was required to be a closed & bounded interval.
- (ii) The integrand was required to be bounded on the interval.
 - The scope of the theory of integration may be widened by relaxing these restrictions. If these restrictions are relaxed we have the following two type of integrals, called improper integrals or infinite integrals.

(a.) Improper integrals on a finite interval where the integrand is unbounded.

(b.) Improper integrals on an unbounded interval.

Now we define convergence of improper integrals & discuss the properties of each type separately.

A. Improper integrals on a closed & bounded interval, the integrand having infinite discontinuities:

Point of infinite discontinuity

A point $x=a$ is called point of infinite discontinuity (P.I.D.) if $\lim_{x \rightarrow a} f(x) = +\infty$ or $-\infty$.

1. Convergence at the left end point

(i) Let the left end point 'a' of the closed & bounded interval $[a, b]$ be the only point of infinite discontinuity of a function $f(x)$ which is bounded and integrable on $[a+\epsilon, b]$ for every ϵ satisfying $0 < \epsilon < b-a$.

The we say that improper integral $\int_a^b f(x) dx$ exists, or converges at a ; if $\lim_{\epsilon \rightarrow 0^+} \int_{a+\epsilon}^b f(x) dx$ exists.

Thus, $\int_a^b f(x) dx = \lim_{\epsilon \rightarrow 0^+} \int_{a+\epsilon}^b f(x) dx$; provided that limit on R.H.S. exists.

If the limit does not exist, we say that improper integral $\int_a^b f(x) dx$ does not exist or is divergent.

Example (i) : $\int_0^1 f(x) dx = \int_0^1 \frac{1}{x} dx$; here '0' is the only point of infinite discontinuity of the integrand.

The integrand is bounded & integrable on $[0+\epsilon, 1]$: for all ϵ satisfying $0 < \epsilon < 1$.

$$\lim_{\epsilon \rightarrow 0^+} \int_{0+\epsilon}^1 \frac{1}{x} dx = \lim_{\epsilon \rightarrow 0^+} [-\log \epsilon] = \infty$$

Therefore, the improper integral $\int_0^1 \frac{1}{x} dx$ is divergent.

(ii) $\int_0^1 \frac{1}{\sqrt{x}} dx$; 0 is a point of infinite discontinuity of the integrand.

The integrand is bounded and integrable on $[0+\epsilon, 1]$; for all ϵ satisfying $0 < \epsilon < 1$.

$$\lim_{\epsilon \rightarrow 0^+} \int_{0+\epsilon}^1 \frac{1}{\sqrt{x}} dx = \lim_{\epsilon \rightarrow 0^+} [2-2\sqrt{\epsilon}] = 2$$

Therefore, the improper integral $\int_{0+\epsilon}^1 \frac{1}{\sqrt{x}} dx$ is convergent &

$$\int_0^1 \frac{1}{\sqrt{x}} dx = 2.$$

2. Convergence at the right end point

Let the right end point 'b' of the closed & bounded interval $[a, b]$ be the only point of infinite discontinuity of a function $f(x)$ which is bounded & integrable on $[a, b-\epsilon]$ for every ϵ satisfying $0 < \epsilon < b-a$.

Then we say that the improper integral $\int_a^b f(x) dx$ exists, or converges at b , if $\lim_{\epsilon \rightarrow 0^+} \int_a^{b-\epsilon} f(x) dx$ exists.

Thus, $\int_a^b f(x) dx = \lim_{\epsilon \rightarrow 0^+} \int_a^{b-\epsilon} f(x) dx$; provided that the limit on R.H.S. exists.

If the limit does not exist, we say the improper integral $\int_a^b f(x) dx$ does not exist or is divergent.

Example:

- (i) The integral $\int_0^1 \frac{1}{1-x} dx$ is improper, since 1 is a point of infinite discontinuity of the integrand.

Clearly, the integrand is bounded and integrable on $[0, 1-\epsilon]$ for all ϵ satisfying $0 < \epsilon < 1$.

$$\lim_{\epsilon \rightarrow 0^+} \int_0^{1-\epsilon} \frac{1}{1-x} dx = \lim_{\epsilon \rightarrow 0^+} [-\log \epsilon] = \infty.$$

Therefore, the improper integral $\int_0^1 \frac{1}{1-x} dx$ is divergent.

- (ii) $\int_0^1 \frac{1}{\sqrt{1-x^2}} dx$; 1 is a point of infinite discontinuity of the integrand.

Clearly, the integrand is bounded & integrable on $[0, 1-\epsilon]$ for all ϵ satisfying $0 < \epsilon < 1$.

$$\lim_{\epsilon \rightarrow 0^+} \int_0^{1-\epsilon} \frac{1}{\sqrt{1-x^2}} dx = \lim_{\epsilon \rightarrow 0^+} [\sin^{-1}(1-\epsilon)] = \frac{\pi}{2}.$$

Therefore, the improper integral $\int_0^1 \frac{1}{\sqrt{1-x^2}} dx$ is convergent &

$$\int_0^1 \frac{1}{\sqrt{1-x^2}} dx = \frac{\pi}{2}.$$

3. Convergence at the both end point

Let the end points a & b be the only points of infinite discontinuity of a function $f(x)$ which is bounded & integrable on $[a+\epsilon, b-\epsilon]$ for every ϵ, ϵ' satisfying $0 < \epsilon, \epsilon' < b-a$.

Let c be any point such that $a < c < b$. If the improper integrals and $\int_c^b f(x) dx$ both converge at 'a' & 'b' respectively. Then we say the

improper integral $\int_a^b f(x) dx$ converges & write

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx.$$

Example: $\int_0^2 \frac{1}{\sqrt{x(2-x)}} dx$ is improper integral, since 0 & 2 are points of infinite discontinuity of the integrand.

Clearly, the integrand is bounded & integrable on $[0+\epsilon, 2-\epsilon]$ for all $\epsilon \in \mathbb{R}^+$ satisfying $0 < \epsilon < 2$.

$$\lim_{\epsilon \rightarrow 0^+} \int_{0+\epsilon}^1 \frac{1}{\sqrt{x(2-x)}} dx = \lim_{\epsilon \rightarrow 0^+} [\sin^{-1}(x-1)]_{0+\epsilon}^1 = \frac{\pi}{2} \quad \&$$

$$\lim_{\epsilon \rightarrow 0^+} \int_1^{2-\epsilon} \frac{1}{\sqrt{x(2-x)}} dx = \lim_{\epsilon \rightarrow 0^+} [\sin^{-1}(x-1)]_1^{2-\epsilon} = \frac{\pi}{2}$$

Therefore, $\int_0^2 \frac{1}{\sqrt{x(2-x)}} dx$ is convergent & $\int_0^2 \frac{1}{\sqrt{x(2-x)}} dx = \pi$.

4. Convergence at Any of the mid point:

Let a point $c (a \leq c \leq b)$ of the closed & bounded interval $[a, b]$ be the only point of infinite discontinuity of a function $f(x)$

If the improper integrals $\int_a^c f(x) dx$ & $\int_c^b f(x) dx$ be both convergent according to the definitions given above, then the improper integral $\int_a^b f(x) dx$ is said to be convergent & we write

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

Therefore, if both the limits $\lim_{\epsilon \rightarrow 0^+} \int_a^{c-\epsilon} f(x) dx$ & $\lim_{\epsilon \rightarrow 0^+} \int_{c+\epsilon}^b f(x) dx$ exists then

$$\int_a^b f(x) dx = \lim_{\epsilon \rightarrow 0^+} \int_a^{c-\epsilon} f(x) dx + \lim_{\epsilon \rightarrow 0^+} \int_{c+\epsilon}^b f(x) dx.$$

If the improper integral $\int_a^b f(x) dx$ is convergent, its value is also equal to the symmetric limit $\left[\lim_{\epsilon \rightarrow 0^+} \int_a^{c-\epsilon} f(x) dx + \lim_{\epsilon \rightarrow 0^+} \int_{c+\epsilon}^b f(x) dx \right]$.

Note: It may happen that the improper integral $\int_a^b f(x) dx$ is divergent but the limit $\lim_{\epsilon \rightarrow 0^+} \left[\int_a^{c-\epsilon} f(x) dx + \int_{c+\epsilon}^b f(x) dx \right]$ exists, then this symmetric limit is called the Cauchy principal value of the improper integral $\int_a^b f(x) dx$ and it is denoted by $P \int_a^b f(x) dx$.

Example: $\int_{-1}^1 f(x) dx$; where $f(x) = \begin{cases} \frac{1}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$

Here, $\lim_{\epsilon \rightarrow 0^+} \int_{-1}^{0-\epsilon} f(x) dx + \lim_{\epsilon \rightarrow 0^+} \int_{0+\epsilon}^1 f(x) dx$
 $= \lim_{\epsilon \rightarrow 0^+} \log \epsilon + \lim_{\epsilon \rightarrow 0^+} (-\log \epsilon)$ & this limit does not exist if $\epsilon \rightarrow 0^+, \epsilon' \rightarrow 0^+$ independently.

$$\text{But } \lim_{\epsilon \rightarrow 0^+} \left[\int_{-1}^{0-\epsilon} f(x) dx + \int_{0+\epsilon}^1 f(x) dx \right] = \lim_{\epsilon \rightarrow 0^+} (\log \epsilon - \log \epsilon) = 0$$

Therefore, the integral $\int_{-1}^1 f(x) dx$ is divergent but $P \int_{-1}^1 f(x) dx = 0$.

B. Improper Integrals On An Unbounded Interval:

1. The improper integral $\int_a^\infty f(x)dx$; where f is integrable on $[a, t]$; $t > a$ and

$$\int_a^\infty f(x)dx = \lim_{t \rightarrow \infty} \int_a^t f(x)dx; \text{ provided that limit exists.}$$

If limit exist finitely then the improper integral $\int_a^\infty f(x)dx$ is said to be convergent. If limit does not exist finitely then the improper integral $\int_a^\infty f(x)dx$ is said to be divergent.

Example: $\int_0^\infty f(x)dx = \lim_{t \rightarrow \infty} \int_0^t e^{-x} dx = \lim_{t \rightarrow \infty} (1 - e^{-t}) = 1$

Therefore, $\int_0^\infty e^{-x} dx$ is convergent & $\int_0^\infty e^{-x} dx = 1$.

Similarly, $\int_0^\infty \frac{1}{1+x} dx$ is divergent.

2. The improper integral $\int_{-\infty}^b f(x)dx$; where f is integrable on $[t, b]$; $t < b$ &

$$\int_{-\infty}^b f(x)dx = \lim_{t \rightarrow -\infty} \int_t^b f(x)dx; \text{ provided that limit exists. If limit exist finitely}$$

then $\int_{-\infty}^b f(x)dx$ is convergent otherwise $\int_{-\infty}^b f(x)dx$ is said to be divergent.

3. Convergence of the improper integral $\int_{-\infty}^\infty f(x)dx$; where f is integrable on $[t_1, t_2]$; $t_1, t_2 \in \mathbb{R}$ such that $t_1 < t_2$.

let $c \in \mathbb{R}$. If both the integrals $\int_{-\infty}^c f(x)dx$ & $\int_c^\infty f(x)dx$ be convergent according to the definition I & II above, then the improper integral $\int_{-\infty}^\infty f(x)dx$ is said to be convergent & we write

$$\int_{-\infty}^\infty f(x)dx = \int_{-\infty}^c f(x)dx + \int_c^\infty f(x)dx.$$

Example: $\int_{-\infty}^\infty \frac{1}{1+x^2} dx$; the integrand is integrable on any closed interval $[t_1, t_2]$; $t_1 < t_2$.

$$\int_{-\infty}^a \frac{1}{1+x^2} dx + \int_a^\infty \frac{1}{1+x^2} dx; a \in \mathbb{R}.$$

$$\int_{-\infty}^a \frac{dx}{1+x^2} = \tan^{-1} a - \tan^{-1} t \Rightarrow \lim_{t \rightarrow -\infty} (\tan^{-1} a - \tan^{-1} t) = \tan^{-1} a + \frac{\pi}{2}.$$

$$\& \int_a^t \frac{dx}{1+x^2} = \tan^{-1} t - \tan^{-1} a \Rightarrow \lim_{t \rightarrow \infty} (\tan^{-1} a - \tan^{-1} t) = \frac{\pi}{2} - \tan^{-1} a$$

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \left(\tan^{-1} a + \frac{\pi}{2} \right) + \left(\frac{\pi}{2} - \tan^{-1} a \right) = \pi$$

Thus, $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$ is convergent.

Comparison Test

1. Let $I_1 = \int_a^b f(x)dx$; $I_2 = \int_a^b g(x)dx$ be two improper integrals where.

- (i) f and g have only P.I.D. at $x=a$
- (ii) f and g are non-ve
- (iii) $f(x) \leq g(x); \forall x \in [a, b]$

Then

(a) If $\int_a^b g(x)dx$ is convergent then $\int_a^b f(x)dx$ is convergent.

(b) If $\int_a^b f(x)dx$ is divergent then $\int_a^b g(x)dx$ is divergent.

2. Now if $b \rightarrow \infty$ then all the result is same for $\int_a^{\infty} f(x)dx$.

Limit Form Test of Comparison Test:

1. If $I_1 = \int_a^b f(x)dx$; $I_2 = \int_a^b g(x)dx$ be two improper integrals. Where f, g are define on $[a, b]$ & $x=a$ is only point of infinite discontinuity such that

- (i) f & g are non-ve
- (ii) $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \ell$ (finite & non-zero)

Then both integrals converge or diverge together.

2. Now, if $b \rightarrow \infty$ then all the result is same for $\int_a^{\infty} f(x)dx$ using $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \ell$; in $[a, t]$.

Note: If $\ell=0$ & $\int_a^b g(x)dx$ convergent then $\int_a^b f(x)dx$ convergent & if

$\ell=\infty$ & $\int_a^b g(x)dx$ divergent then $\int_a^b f(x)dx$ divergent.

Results:

(i) $\int_0^1 \frac{1}{x^P} dx$ is convergent if $P < 1$ and is divergent if $P \geq 1$

(ii) $\int_a^\infty \frac{1}{x^P} dx$ is convergent if $P > 1$ and is divergent if $P \leq 1$
 ($a > 0$)

(iii) $\int_a^b \frac{1}{(x-a)^P} dx$ is convergent if $P < 1$ and is divergent if $P \geq 1$

(iv) $\int_a^b \frac{1}{(b-x)^P} dx$ is convergent if $P < 1$ and is divergent if $P \geq 1$

Example: $\int_0^{\frac{\pi}{2}} \frac{\sin x}{x^P} dx$

$$\therefore \frac{\sin x}{x^P} = \frac{\sin x}{x} \cdot \frac{1}{x^{P-1}} \leq \frac{1}{x^{P-1}}$$

$$\Rightarrow \frac{\sin x}{x^P} \leq \frac{1}{x^{P-1}} \quad \text{and } 0 \text{ is only P.I.D.}$$

(f) (g)

$$\int_0^{\frac{\pi}{2}} \frac{1}{x^{P-1}} dx \text{ is convergent if } P-1 < 1 \Rightarrow P < 2$$

$$\Rightarrow \int_0^{\frac{\pi}{2}} \frac{\sin x}{x^P} dx \text{ convergent if } P < 2.$$

By limit form test: $\lim_{x \rightarrow 0} \frac{\frac{\sin x}{x^P}}{\frac{1}{x^{P-1}}} = 1$ (finite and non-zero)

$$\Rightarrow \int_0^{\frac{\pi}{2}} \frac{1}{x^{P-1}} dx \& \int_0^{\frac{\pi}{2}} \frac{\sin x}{x^P} dx \text{ behaves a like}$$

$$\Rightarrow \frac{1}{x^{P-1}} \text{ is convergent for } p-1 < 1 \Rightarrow p < 2 \Rightarrow \int_0^{\frac{\pi}{2}} \frac{\sin x}{x^p} dx \text{ Convergent if } p < 2.$$

$$\Rightarrow \frac{1}{x^{P-1}} \text{ is divergent for } p-1 \geq 1 \Rightarrow p \geq 2 \Rightarrow \int_0^{\frac{\pi}{2}} \frac{\sin x}{x^p} dx \text{ Divergent if } p \geq 2$$

Absolutely convergent: The improper integral $\int_a^\infty f(x) dx$ is said to be

absolutely convergent if the integral $\int_a^\infty |f(x)| dx$ is convergent.

Definition is same if $a \rightarrow -\infty$

- The improper integral which is convergent but not absolutely convergent in called conditionally convergent.

Example: $(x) = \begin{cases} \frac{1}{x^2}; x \in Q \\ -\frac{1}{x^2}; x \in Q^c \end{cases}$ & $f(x)$ is defined on $[1, \infty]$

$$\Rightarrow |f(x)| = \frac{1}{x^2}; x \in [1, \infty] \Rightarrow \int_1^{\infty} |f(x)| dx \text{ convergent}$$

$$\Rightarrow \int_1^{\infty} f(x) dx \text{ is A.C.} \Rightarrow \int_1^{\infty} f(x) dx \text{ convergent}$$

Test for Convergence of the Integrals of a Product:

- Let a function f be integrable on $[a, X]; \forall x > a$ & the integral $\int_a^{\infty} f(x) dx$ is absolutely convergent & a function ϕ be bounded on $[a, \infty]$ & integrable on $[a, X]; \forall x > a$.

Then the integral $\int_a^{\infty} f(x) \cdot \phi(x) dx$ is absolutely convergent.

2. Abel's Test:

Let

(i) a function ϕ be monotonic & bd on $[a, \infty]$ &

(ii) The integral $\int_a^{\infty} f(x) dx$ be convergent

Then the integral $\int_a^{\infty} f(x) \cdot \phi(x) dx$ is convergent.

3. Dirichlet's test:

Let

(i) a function ϕ be monotonic & bd on $[a, \infty]$ & $\lim_{x \rightarrow \infty} \phi(x) = 0$

(ii) The integral $\int_a^X f(x) dx$ be bd on $[a, X]; \forall x > a$.

Then the integral $\int_a^{\infty} f(x) \phi(x) dx$ is convergent.

Example: $\int_0^{\infty} \frac{\sin^3 x}{x^5} dx = \int_a^{\infty} \sin^3 x \cdot \frac{1}{x^5} dx$

Let $f = \frac{1}{x^5}; g = \sin^3 x$

Here, $\left| \int_0^t \sin^3 x dx \right| \leq m; \forall t \in [a, \infty]$

$\lim_{x \rightarrow \infty} \frac{1}{x^5} = 0$; then $\int_0^{\infty} \frac{\sin^3 x}{x^5} dx$ is convergent

Cauchy - Maclaurin Integral Test: If f in a monotonic decreasing function on $[1, \infty]$ & $f(x) > 0 \quad \forall x \in [1, \infty]$, then the improper integral $\int_1^{\infty} f(x) dx$ & the infinite series $\sum_{n=1}^{\infty} f(n)$ converge or diverge together.

Beta Function: The integral $\int_0^1 x^{m-1} (1-x)^{n-1} dx; m > 0, n > 0$ is called the Beta function & it is denoted by $B(m, n)$.

$$\text{Thus, } B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx; m > 0; n > 0$$

The improper integral $\int_0^1 x^{m-1} (1-x)^{n-1} dx$ is convergent if $m > 0, n > 0$.

Gamma Function: The integral $\int_0^{\infty} e^{-x} x^{n-1} dx; n > 0$ is called the gamma function & it is denoted by $\Gamma(x)$.

$$\text{Thus, } \Gamma(x) = \int_0^{\infty} e^{-x} x^{n-1} dx; n > 0$$

The improper integral $\int_0^{\infty} e^{-x} x^{n-1} dx$ is convergent if $n > 0$.

Result: $\int_0^{\infty} \frac{\sin mx}{x^n} dx (m > 0, n > 0)$ is convergent if $0 < n < 2$ & absolutely convergent if $1 < n < 2$.

- The improper integral $\int_1^{\infty} \sin x^p dx$ is convergent if $p > 1$
- The improper integral $\int_1^{\infty} \frac{\sin x}{x^p} dx$ is convergent absolutely if $p > 1$
- The improper integral $\int_0^{\infty} \frac{\sin x^m}{x^n} dx$ convergent if $n + m > 1$

μ -Test: Let a be the only point of infinite discontinuity of a function f which is integrable on $[a + \epsilon, b]$ for $0 < \epsilon < b - a$ & $f(x) > 0; \forall x \in (a, b]$

if $\lim_{x \rightarrow a^+} f(x)(x-a)^{\mu} = \ell; (\ell \neq 0)$ then $\int_a^b f(x) dx$ is convergent iff $\mu < 1$.

Note: If $\lim_{x \rightarrow b^-} f(x)(b-x)^{\mu} = \ell$; where ℓ is a finite non-zero no. then

$\int_a^b f(x) dx$ is convergent iff $\mu < 1$.

CHAPTER 9

UNIFORM CONVERGENCE

9.1. Definition

Previously, we have studied sequences of real numbers. Now we discuss the topic of sequence of real valued functions. A sequence of functions $\{f_n\}$ is a list of functions $(f_1, f_2, \dots)$ such that each f_n maps a given subset D of $\mathbb{R}$ into $\mathbb{R}$.

1. Pointwise convergence

Definition. Let D be a subset of $\mathbb{R}$ and let $\{f_n\}$ be a sequence of functions defined on D . We say that $\{f_n\}$ converges pointwise on D if

$$\lim_{n \rightarrow \infty} f_n(x) \text{ exists for each point } x \text{ in } D.$$

If $\{f_n\}$ is pointwise convergent then the function defined by $f(x) = \lim_{n \rightarrow \infty} f_n(x)$, for every x in D , is called the pointwise limit of the sequence $\{f_n\}$.

Example 1: Let $\{f_n\}$ be the sequence of functions on $\mathbb{R}$ defined by $f_n(x) = nx$. This sequence does not converge pointwise on $\mathbb{R}$ because $\lim_{n \rightarrow \infty} f_n(x) = \infty$ for any $x > 0$.

Example 2: Let $\{f_n\}$ be the sequence of functions on $\mathbb{R}$ defined by $f_n(x) = x/n$. This sequence converges pointwise to the zero function on $\mathbb{R}$.

Example 3: Consider the sequence $\{f_n\}$ of functions defined by

$$f_n(x) = \frac{nx + x^2}{n^2} \text{ for all } x \text{ in } \mathbb{R}.$$

Show that $\{f_n\}$ converges pointwise.

Solution: For every real number x , we have :

$$\lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \frac{x}{n} + \frac{x^2}{n^2} = x \left(\lim_{n \rightarrow \infty} \frac{1}{n} \right) + x^2 \left(\lim_{n \rightarrow \infty} \frac{1}{n^2} \right) = 0 + 0 = 0$$

Thus, $\{f_n\}$ converges pointwise to the zero function on $\mathbb{R}$.

Example 4: Consider the sequence $\{f_n\}$ of functions defined by

$$f_n(x) = \frac{\sin(nx+3)}{\sqrt{n+1}} \text{ for all } x \text{ in } \mathbb{R}.$$

Show that $\{f_n\}$ converges pointwise.

Solution: For every x in $\mathbb{R}$, we have

$$\frac{-1}{\sqrt{n+1}} \leq \frac{\sin(nx+3)}{\sqrt{n+1}} \leq \frac{1}{\sqrt{n+1}}$$

Moreover,

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n+1}} = 0.$$

Applying the squeeze theorem for sequences, we obtain that

$$\lim_{n \rightarrow \infty} f_n(x) = 0 \text{ for all } x \text{ in } \mathbb{R}.$$

Therefore, $\{f_n\}$ converges pointwise to the function $f \equiv 0$ on $\mathbb{R}$.

Example 5: Consider the sequence $\{f_n\}$ of functions defined by $f_n(x) = n^2 x^n$ for $0 \leq x \leq 1$. Determine whether $\{f_n\}$ is pointwise convergent.

Solution: First of all, observe that $f_n(0) = 0$ for every n in $\mathbb{N}$. So the sequence $\{f_n(0)\}$ is constant and converges to zero. Now suppose $0 < x < 1$ then $n^2 x^n = n^2 e^{n \ln(x)}$. But $\ln(x) < 0$ when $0 < x < 1$, it follows that

$$\lim_{n \rightarrow \infty} f_n(x) = 0 \text{ for } 0 < x < 1$$

Finally, $f_n(1) = n^2$ for all n . So, $\lim_{n \rightarrow \infty} f_n(1) = \infty$. Therefore, $\{f_n\}$ is not pointwise convergent on $[0, 1]$.

Example 6: Let $\{f_n\}$ be the sequence of functions defined by $f_n(x) = \cos^n(x)$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$. Discuss the pointwise convergence of the sequence.

Solution: For, $-\frac{\pi}{2} \leq x < 0$ and for $0 < x \leq \frac{\pi}{2}$, we have

$$0 < \cos(x) < 1.$$

It follows that

$$\lim_{n \rightarrow \infty} (\cos(x))^n = 0 \text{ for } x \neq 0.$$

Moreover, since $f_n(0) = 1$ for all n in $\mathbb{N}$, one gets $\lim_{n \rightarrow \infty} f_n(0) = 1$.

Therefore, $\{f_n\}$ converges pointwise to the function f defined by

$$f(x) = \begin{cases} 0 & \text{if } -\frac{\pi}{2} \leq x < 0 \text{ or } 0 < x \leq \frac{\pi}{2} \\ 1 & \text{if } x = 0 \end{cases}$$

Example 7: Consider the sequence of function defined by

$$f_n(x) = nx(1-x)^n \text{ on } [0, 1].$$

Show that $\{f_n\}$ converges pointwise to the zero function.

Solution: Note that $f_n(0) = f_n(1) = 0$, for all $n \in \mathbb{N}$. Now suppose $0 < x < 1$, then

$$\lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} nxe^{n \ln(1-x)} = x \lim_{n \rightarrow \infty} ne^{n \ln(1-x)} = 0$$

Because $\ln(1-x) < 0$ when $0 < x < 1$. Therefore, the given sequence converges pointwise to zero.

Example 8: Let $\{f_n\}$ be the sequence of functions on $\mathbb{R}$ defined by

$$f_n(x) = \begin{cases} n^3 & \text{if } 0 < x \leq \frac{1}{n} \\ 1 & \text{otherwise} \end{cases}$$

Show that $\{f_n\}$ converges pointwise to the constant function $f=1$ on $\mathbb{R}$.

Solution: For any x in $\mathbb{R}$ there is a natural number N such that x does not belong to the interval $(0, 1/N)$. The intervals $(0, 1/n)$ get smaller as $n \rightarrow \infty$. Therefore, $f_n(x) = 1$ for all $n > N$. Hence,

$$\lim_{n \rightarrow \infty} f_n(x) = 1 \quad \text{for all } x.$$

For Formal Definition of Pointwise Convergence

Let D be a subset of $\mathbb{R}$ and let $\{f_n\}$ converges pointwise to f if given any x in D and given any $\varepsilon > 0$, there exists a natural number $N = N(x, \varepsilon)$ such that

$$|f_n(x) - f(x)| < \varepsilon \quad \text{for every } n > N.$$

Note: The notation $N = N(x, \varepsilon)$ means that the natural number N depends on the choice of x and ε .

2. Uniform convergence

Definition: Let D be a subset of $\mathbb{R}$ and let $\{f_n\}$ be a sequence of real valued functions defined on D . Then $\{f_n\}$ converges uniformly to f if given any $\varepsilon > 0$, there exists a natural number $N = N(\varepsilon)$ such that

$$|f_n(x) - f(x)| < \varepsilon \quad \text{for every } n > N \text{ and for every } x \text{ in } D.$$

Note: In the above definition the natural number N depends only ε .

Therefore, uniform convergence implies pointwise convergence. But the converse is false as we can see from the following counter-example.

Example 9: Let $\{f_n\}$ be the sequence of functions on $(0, \infty)$ defined by

$$f_n(x) = \frac{nx}{1+n^2x^2}.$$

This function converges pointwise to zero. Indeed, $(1+n^2x^2) \sim n^2x^2$ as n gets larger and larger. So,

$$\lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \frac{nx}{1+n^2x^2} = \frac{1}{x} \lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

But for any $\varepsilon < \frac{1}{2}$, we have

$$\left| f_n\left(\frac{1}{n}\right) - f\left(\frac{1}{n}\right) \right| = \frac{1}{2} - 0 > \varepsilon.$$

Hence $\{f_n\}$ is not uniformly convergent.

Theorem: Let D be a subset of $\mathbb{R}$ and let $\{f_n\}$ be a sequence of continuous function on D which converges uniformly to f on D . Then f is continuous on D .

Problem 1: Let $\{f_n\}$ be the sequence of functions on $[0, 1]$ defined by $f_n(x) = nx(1-x^4)^n$. Show that $\{f_n\}$ converges pointwise. Find its pointwise limit.

Problem 2: Is the sequence of functions on $(0, 1]$ defined by $f_n(x) = (1-x)^{\frac{1}{n}}$ pointwise convergent? Justify your answer.

Problem 3: Consider the sequence $\{f_n\}$ of functions defined by

$$f_n(x) = \frac{n + \cos(nx)}{2n+1} \text{ for all } x \text{ in } \mathbb{R}.$$

Show that $\{f_n\}$ is pointwise convergent. Find its pointwise limit.

Problem 4: Consider the sequence $\{f_n\}$ of functions defined on $[0, \pi]$ by $f_n(x) = \sin^n(x)$. Show that $\{f_n\}$ converges pointwise. Find its pointwise limit. Using the above theorem, show that $\{f_n\}$ is not uniformly convergent.

9.2. Quick Review for Uniform Convergence

1. The sequence of function x^n converges uniformly to 0 on $[0, a]$ where $0 < a < 1$, but not on $[0, 1]$.
2. If $f_n(x)$ converges point-wise to $f(x)$ on S and $M_n = \sup_{x \in S} |f_n(x) - f(x)|$, then $f_n(x)$ converges uniformly to $f(x)$ on S iff $M_n \rightarrow 0$, as $n \rightarrow \infty$.
3. A sequence of functions $f_n(x)$ defined as a set S converges uniformly on S iff to each $\varepsilon > 0 \exists m \in \mathbb{N}$ such that $|f_{n+p}(x) - f_n(x)| < \varepsilon \forall n \geq m, p \geq 0$ and $x \in S$.
4. The sequence x^n converges on $[0, 1]$ to 0 if $0 < x < 1$ and to 1 iff $x = 1$. However, the convergence is not uniform on $[0, 1]$.
5. A sequence $f_n(x)$ defined on a set S is said to be **uniformly bounded** on S if there exists $k > 0$ such that $|f_n(x)| < k \forall x \in S$ and $n \in \mathbb{N}$.
6. If the sequence of continuous functions $f_n(x)$ converges to a continuous function $f(x)$ on a compact set E and $f_n(x) \geq f_{n+1}(x) \forall n \in \mathbb{N}$ and $x \in E$, then $f_n(x) \rightarrow f(x)$ uniformly on E .
7. If $f_n(x)$ converges uniformly to a function $f(x)$ on S , and x be a limit point of S , and $\lim_{t \rightarrow x} f_n(t) = a_n$, then a_n converges and $\lim_{t \rightarrow x} f(t) = \lim_{n \rightarrow \infty} a_n$.

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8. If $f_n(x)$ converges uniformly to a function $f(x)$ on a set S and if each $f_n(x)$ is continuous at $c \in S$ then $f(x)$ is continuous at c on S .
9. $\langle \tan^{-1} nx \rangle$ is uniformly convergent on $[a, b], a > 0$, but it is only point-wise convergent on $(0, b]$.
10. If $f_n(x)$ is monotonically non-decreasing sequence of continuous functions on $[a, b]$ and point-wise converges, then the convergence is uniform on $[a, b]$.
11. If $f_n(x)$ is monotonically non-decreasing sequence of continuous functions on $[a, b]$ and the sum function $\sum f_n(x)$ is continuous on $[a, b]$, then $\sum f_n(x)$ is uniformly convergent on $[a, b]$.
12. If $f(x)$ is continuous on $[a, b]$, then there exists a sequence of polynomials $P_n(x)$ such that $P_n(x)$ converges uniformly to $f(x)$ on $[a, b]$.
13. If $f_n(x)$ is a sequence of integrable functions on $[a, b]$ such that it converges uniformly to a function $f(x)$ on $[a, b]$, then $f(x)$ is also integrable on $[a, b]$ and $\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = \int_a^b f(x) dx$. Moreover, $\int_a^x f_n(t) dt$ converges to $\int_a^x f(t) dt$ uniformly on $[a, b]$.
14. If $f_n(x)$ converges for a point in $[a, b]$ and $f'_n(x)$ converges uniformly on $[a, b]$, then $f_n(x)$ converges uniformly to a function $f(x)$ on $[a, b]$, and $f'(x) = \lim_{n \rightarrow \infty} f'_n(x) \quad \forall x \in [a, b]$.
15. If $f_n(x), g_n(x)$ converge uniformly to $f(x), g(x)$ respectively on a set S , then $f_n(x) \pm g_n(x)$ converge uniformly to $f(x) \pm g(x)$ respectively on S . Also $k f_n(x)$ converges uniformly to $k f(x)$ on S , where $k \in \mathbb{R}$.
16. If $f_n(x), g_n(x)$ converges uniformly on a set S then the product sequence $f_n(x) g_n(x)$ may, or may not converge uniformly on S .
17. If the sequence of monotonically non-decreasing continuous functions $f(x)$ converges point-wise to a continuous function $f(x)$ on $[a, b]$, then the convergence is uniform on $[a, b]$.
18. If $f_n(x) \rightarrow f(x)$ uniformly on $[a, b]$, where each $f_n(x)$ and $g(x)$ are continuous on $[a, b]$, then

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) g(x) dx = \int_a^b f(x) g(x) dx.$$
19. If each $f_n(x)$ is integrable on $[a, b]$ and $f_n(x) \rightarrow f(x)$ uniformly on $[a, b]$ then $f(x)$ is integrable on $[a, b]$ and $\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx$ exists, and is equal to $\int_a^b f(x) dx$.

20. If $f_n \rightarrow f$ uniformly on a set S and $x \in S'$ so as each $\lim_{t \rightarrow x} f_n$ exists on S , then $\lim_{t \rightarrow x} f$ exists on S and $\lim_{t \rightarrow x} f(t) = \lim_{n \rightarrow \infty} \lim_{t \rightarrow x} f_n(t)$ on S .
21. If $f_n(x)$ converges for a point in $[a, b]$ and $f'_n(x)$ converges uniformly on $[a, b]$, then $f_n(x)$ converges uniformly to a function $f(x)$ on $[a, b]$ and $f'(x) = \lim_{n \rightarrow \infty} f'_n(x) \forall x \in [a, b]$.
22. $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly on S iff for every $\varepsilon > 0 \exists m \in \mathbb{N}$ such that $|f_{n+1}(x) + f_{n+2}(x) + \dots + f_{n+p}(x)| < \varepsilon \forall n \geq m, p \geq 0$ and $x \in S$.
23. If $\sum f_n(x) \rightarrow f(x)$ uniformly on a set S and x be a limit point of S and $\lim_{t \rightarrow x} f_n(t) = a_n$, then $\sum a_n$ converges and $\lim_{t \rightarrow x} f(t) = \sum_{n=1}^{\infty} a_n$.
24. If $f_n(x)$ are continuous at $c \in S$ and $\sum f_n(x) \rightarrow f(x)$ uniformly on S , then $f(x)$ is continuous at $c \in S$.
25. If each $f_n(x) \in R[a, b]$ and $\sum f_n(x)$ converges uniformly on $[a, b]$, then $\int_a^b \sum_{n=1}^{\infty} f_n(x) dx = \sum_{n=1}^{\infty} \int_a^b f_n(x) dx$.
26. If $f_n(x) \in R[a, b] \forall n$ and $\sum f_n(x)$ converges uniformly to $f(x)$ on $[a, b]$, then $\sum_{n=1}^{\infty} \int_a^x f_n(x) dx$ converges uniformly to $\int_a^x f(x) dx$ on $[a, b]$.
27. If $f_n(x) \in R[a, b] \forall n$ and $\sum f_n(x)$ converges uniformly to $f(x)$ on $[a, b]$ and $g(x) \in R[a, b]$, then $\sum_{n=1}^{\infty} \int_a^x g(x) f_n(x) dx$ converges uniformly to $\int_a^x g(x) f(x) dx$ on $[a, b]$.
28. If each $f_n(x)$ is differentiable on $[a, b]$ and $\sum f'_n(x)$ converges uniformly on $[a, b]$ and $\sum f_n(x)$ converges for some point $c \in [a, b]$, then $\sum f_n(x)$ converges uniformly on $[a, b]$ and $s'(x) = \sum f'_n(x)$ on $[a, b]$, where $s(x) = \sum f_n(x)$.
29. If $f_n(x)$ has continuous derivatives on $[a, b]$ and $\sum f_n(x), \sum f'_n(x)$ converge uniformly to $f(x)$ and $g(x)$ respectively on $[a, b]$, then $f(x)$ is differentiable and $f'(x) = g(x)$ on $[a, b]$.
30. If $|f_n(x)| \leq M_n \forall x \in S$ and $n \in \mathbb{N}$, then $\sum f_n(x)$ converges uniformly (and absolutely) on S if $\sum M_n$ converges.
31. The series $\sum 3^n \sin \frac{1}{4^n} x$ converges absolutely and uniformly on (a, ∞) , where $a > 0$.

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32. The function $f(x) = \sum_{n=1}^{\infty} b^n \cos(a^n \pi x)$ where $0 < b < 1$ and a is an odd positive integer such that $ab > 1 + \frac{3}{2}\pi$, is everywhere continuous but it has no derivative for any $x \in \mathbb{R}$.
33. If $\sum f_n(x)$ converges uniformly on a set S , then $f_n(x)$ converge uniformly to 0 on S .
34. If $\sum f_n(x)$, $\sum g_n(x)$ converge uniformly to $f(x)$, $g(x)$ respectively on a set S , then $\sum (f_n(x) \pm g_n(x))$ converge uniformly to $f(x) \pm g(x)$ respectively on S also $\sum k f_n(x)$ will converge uniformly to $k f(x)$ on S , where k is any constant real number.
35. $\sum x^n$ converges uniformly on $(-\delta, \delta)$, where $1 > \delta > 0$, but not on $(-1, 1)$.
36. If $\sum f_n(x)$ converges uniformly to $f(x)$ on $[a, b]$ and each $f_n(x) \in R[a, b]$, then $f(x) \in R[a, b]$ and $\int_a^b f(x) dx = \sum \int_a^b f_n(x) dx$.
37. If each $f'_n(x)$ is continuous on $[a, b]$ and $\sum f_n(x)$, $\sum f'_n(x)$ converge uniformly to $s(x)$ and $g(x)$ respectively on $[a, b]$ then, $s'(x)$ exists and $s'(x) = g(x)$ on $[a, b]$.
38. If $\sum f_n(x)$ converges uniformly on a set S and $g_n(x)$ is monotonic and uniformly bounded on S , then the series $\sum f_n(x) g_n(x)$ converges uniformly on S .
39. If $\sum f_n(x)$ is uniformly bounded on S and $g_n(x)$ is monotonic and converges uniformly to zero on S , then $\sum f_n(x) g_n(x)$ converges uniformly on S .
40. If $\sum f_n(x)$ converges uniformly to $f(x)$ on a set S and $\sum a_n$ converges to a , then $\sum (f_n(x) + a_n)$ converges uniformly to $f(x) + a$ on S .

CHAPTER 10

FUNCTION OF SEVERAL VARIABLES

10.1. Definition of n-tuples

The Euclidean n -space $\mathbb{R}^n$ is the set of all n -tuples $(x_1, \dots, x_n)$ of real numbers $x_i, i=1, \dots, n$ (called vectors or points) on which the two operations of vector addition and scalar multiplication are defined as follows:

(i) For any two points $(x_1, \dots, x_n)$ and $(y_1, \dots, y_n)$ in $\mathbb{R}^n$

$$(x_1, \dots, x_n) + (y_1, \dots, y_n) = (x_1 + y_1, \dots, x_n + y_n)$$

(ii) For any $a \in \mathbb{R}$ and $(x_1, \dots, x_n) \in \mathbb{R}^n$

$$a \cdot (x_1, \dots, x_n) = (ax_1, \dots, ax_n)$$

we can write $a \cdot (x_1, \dots, x_n) = a(x_1, \dots, x_n)$. If $x \in \mathbb{R}^n$ then $x = (x_1, \dots, x_n)$ for some real numbers $x_1, \dots, x_n$, and x_i is called the i^{th} component of x .

Neighbourhood: Let $S \subseteq \mathbb{R}^n$ and $\alpha = (a_1, a_2, \dots, a_n) \in \mathbb{R}^n$. Then we say S is neighbourhood of α , if $\exists \varepsilon > 0$ such that

$$B(\alpha, \varepsilon) = \left\{ (x_1, x_2, \dots, x_n) \in \mathbb{R}^n : \sqrt{(x_1 - a_1)^2 + (x_2 - a_2)^2 + \dots + (x_n - a_n)^2} < \varepsilon \right\} \subseteq S$$

Open Set: Let $S \subseteq \mathbb{R}^n$ be a subset of $\mathbb{R}^n$. Then S is said to be open if it is neighborhood of each of its point.

Closed Set: Let $S \subseteq \mathbb{R}^n$ be a subset of $\mathbb{R}^n$. Then S is said to be closed if complement of S in $\mathbb{R}^n$ is open set i.e., $\mathbb{R}^n - S$ is an open set.

Remarks:

- An open Ball in $\mathbb{R}^n$ is an open set.
- A closed ball in $\mathbb{R}^n$ is closed set.
- The two sets $\mathbb{R}^n$ and ϕ are both open and closed sets in $\mathbb{R}^n$ and moreover there is no other subsets of $\mathbb{R}^n$ which is both open and closed.

Important Results:

- The union of any collection of open sets in $\mathbb{R}^n$ is open.
- The intersection of a finite collection of open sets in $\mathbb{R}^n$ is open.
- The intersection of any collection of closed sets in $\mathbb{R}^n$ is closed.
- The union of finite collection of closed sets is closed.
- A subset A of $\mathbb{R}^n$ is open iff A is the union of a countable collection of open rectangles (balls).

6. A subset A of $\mathbb{R}^n$ is closed iff A is the intersection of a countable collection of closed sets.

7. Let A be a subset of $\mathbb{R}^n$ and w be a point in $\mathbb{R}^n$. The translation of A by w is denoted by $w+A$ and is defined by $w+A = \{w+v : v \in A\}$. Then

- (i) A is open if and only if $w+A$ is open.
- (ii) A is closed if and only if $w+A$ is closed.

Cluster Point: A point $x \in \mathbb{R}^n$ is called a cluster point of $A \subset \mathbb{R}^n$ if every open ball containing x has a nonempty intersection with $A - \{x\}$.

If $x \in A$ but it is not a cluster point of A then x is called an isolated point of A i.e. $x \in w \cap A - \{x\} \neq \emptyset$, $x \in A$, $w \cap A = \{x\}$.

8. A subset of $\mathbb{R}^n$ is closed iff the subset contains all its cluster points.

Product of Sets: Let $A \subset \mathbb{R}^n$ and $B \subset \mathbb{R}^m$. The product of A & B denoted by $A \times B$ is defined as $A \times B = \{(x_1, \dots, x_n, y_1, \dots, y_m) : (x_1, \dots, x_n) \in A \text{ and } (y_1, \dots, y_m) \in B\}$. If $x \in A$ & $y \in B$ then the point $(x_1, \dots, x_n, y_1, \dots, y_m)$ in $A \times B$ is denoted by (x, y) .

From definition it is evident that $A \times B \subset \mathbb{R}^{n+m}$, and in particular $\mathbb{R}^n \times \mathbb{R}^m = \mathbb{R}^{n+m}$.

9. Let $A \subset \mathbb{R}^n$ and $B \subset \mathbb{R}^m$. Then

- (i) A & B are open sets in $\mathbb{R}^n$ and $\mathbb{R}^m$ respectively iff $A \times B$ is open in $\mathbb{R}^{n+m}$.
- (ii) A and B are closed sets in $\mathbb{R}^n$ and $\mathbb{R}^m$ respectively iff $A \times B$ is closed in $\mathbb{R}^{n+m}$.

10. Let $A, B \subseteq \mathbb{R}^n$. Then

- (i) $\overline{A \cup B} = \overline{A} \cup \overline{B}$
- (ii) $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$

11. If $A \subset \mathbb{R}^n$, and $B \subset \mathbb{R}^n$, and x is a cluster point of $A \cap B$ then x is a cluster point of both A & B .

12. If $A \subset \mathbb{R}^n$, $B \subset \mathbb{R}^n$, and x is a cluster point of $A \cup B$ then x is a cluster point of A or of B .

10.2. Real Valued Function on n Variables

Let $S \subseteq \mathbb{R}^n$. Then a map $f: S \rightarrow \mathbb{R}$ is called real valued function on n variables $x_1, x_2, \dots, x_n$. Provided no two different images are mapped under same point in S i.e. for $x, y \in S$, $f(x) \neq f(y) \Rightarrow x \neq y$.

Limit of a real valued function on- n variables:

Let $S \subseteq \mathbb{R}^n$ be a subset of $\mathbb{R}^n$ and $f: S \rightarrow \mathbb{R}$ be a function. Let $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n)$ be limit point of S and $l \in \mathbb{R}$. Then we say limit of f as $x = (x_1, x_2, \dots, x_n) \rightarrow (\alpha_1, \alpha_2, \dots, \alpha_n)$ is l and we write $\lim_{x \rightarrow \alpha} f(x_1, x_2, \dots, x_n) = l$, if for each $\varepsilon > 0$, $\exists \delta > 0$ such that $|f(x_1, x_2, \dots, x_n) - l| < \varepsilon$, whenever $0 < \sqrt{(x_1 - \alpha_1)^2 + \dots + (x_n - \alpha_n)^2} < \delta$.

13. Let f & g be two real valued function with common domain $A \subset \mathbb{R}^n$, and c a cluster point of A . If $\lim_{x \rightarrow c} f(x)$ and $\lim_{x \rightarrow c} g(x)$ exists, then

$$(i) \lim_{x \rightarrow c} (f+g)(x) = \lim_{x \rightarrow c} f(x) + \lim_{x \rightarrow c} g(x)$$

$$(ii) \lim_{x \rightarrow c} (f \cdot g)(x) = \lim_{x \rightarrow c} f(x) \cdot \lim_{x \rightarrow c} g(x)$$

$$(iii) \lim_{x \rightarrow c} \left(\frac{f}{g} \right)(x) = \frac{\lim_{x \rightarrow c} f(x)}{\lim_{x \rightarrow c} g(x)} \text{ provided } g \text{ \& its limit at } c \text{ not vanish.}$$

Continuity of a Function at a Point:

Let $f: D \rightarrow \mathbb{R}$, where D is an open subset of $\mathbb{R}^n$. Then f is said to be continuous at a point $a = (a_1, a_2, \dots, a_n) \in D$ if $\lim_{x \rightarrow a} f(x) = f(a)$, where $x = (x_1, x_2, \dots, x_n)$.

In other words, f is continuous at a point $(a_1, a_2, \dots, a_n) \in D$ if given any $\varepsilon > 0$, there exists some $\delta > 0$ such that $|f(x_1, x_2, \dots, x_n) - f(a_1, a_2, \dots, a_n)| < \varepsilon$, when

$$\sqrt{(x_1 - a_1)^2 + \dots + (x_n - a_n)^2} < \delta.$$

Partial Derivatives: If f is a function of n variables, then its partial derivatives are the functions f_{x_i} 's defined by

$$f_{x_i}(x_1, x_2, \dots, x_n) = \lim_{h \rightarrow 0} \frac{f(x_1, \dots, x_i + h, \dots, x_n) - f(x_1, x_2, \dots, x_n)}{h} \text{ for all } 1 \leq i \leq n.$$

14. Let f and g are functions of n variables. Also if $f_{x_i}, f_{x_j}, g_{x_i}, g_{x_j}$ exist. Then

$$(i) (f \pm g)_{x_i} = f_{x_i} \pm g_{x_i}$$

$$(ii) (f \cdot g)_{x_i} = f_{x_i} g + g_{x_i} f$$

$$(iii) \left(\frac{f}{g} \right)_{x_i} = \frac{g \cdot f_{x_i} - f \cdot g_{x_i}}{g^2}$$

Differentiability at a Point: Let $f: D \rightarrow \mathbb{R}$ be a real-valued function defined on an open set $D \subset \mathbb{R}^2$. Then f is said to be differentiable at a point $(a, b) \in D$, if $f(a+h, b+k) - f(a, b) = Ah + Bk + \sqrt{h^2 + k^2} g(h, k)$, Where A and B are real numbers independent of h and k , and g is a real-valued function such that $\lim_{(h,k) \rightarrow (0,0)} g(h, k) = 0$.

Equivalently, f is differentiable at (a, b) , if $f(a+h, b+k) - f(a, b) = Ah + Bk + h\phi_1(h, k) + k\phi_2(h, k)$ where A and B are real numbers independent of h and k and $\phi(h, k)$, $\phi_1(h, k)$ and $\phi_2(h, k)$ converges to 0 as $(h, k) \rightarrow (0, 0)$.

Remark:

- (i) It may be noted that the constants A, B appearing in the definition of differentiability of f are given by : $A = f_x(a, b), B = f_y(a, b)$
 - (ii) If either of f_x, f_y does not exist at (a, b) , then f is not differentiable at (a, b) .
15. If a function $f: D \rightarrow \mathbb{R}$ is differentiable at $(a, b) \in D$, then f has partial derivatives f_x and f_y at (a, b) .
 16. If a function $f: D \rightarrow \mathbb{R}$ is differentiable at a point $(a, b) \in D$, then f is continuous at (a, b) .

Remark: The converse of the above theorem may not be true. A function may be continuous at a point without being differentiable at that point.

17. If (a, b) be a point of the domain of definition of a function f such that
 - (i) f_x is continuous at (a, b)
 - (ii) f_y exists at (a, b) ,

Then f is differentiable at (a, b) .

Note: In a similar way it can be shown that f is differentiable at (a, b) , if f_x exists and f_y is continuous at (a, b) . In fact, one of the partial derivatives is to be continuous and the other merely to exist at the point.

Remark:

- (i) If a function f is not differentiable at (a, b) , then f_x and f_y cannot be continuous at (a, b) .
- (ii) The conditions of the above theorem are sufficient, but are not necessary. If a function f is differentiable at a point (a, b) , then f_x, f_y may not be continuous at (a, b) .

Chain Rule 1: Suppose that $z = f(x, y)$ is a differentiable function of x & y , where $x = g(t)$ and $y = h(t)$ are both differentiable functions of t . Then z is a differentiable function of t and $\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$

Chain Rule 2: Suppose that $z = f(x, y)$ is a differentiable function of x and y , and where $x = g(s, t), y = h(s, t)$ and the partial derivatives g_x, g_t, h_s and h_t exist. Then

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s} \qquad \frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$

Directional Derivatives: Let f be a function of n variables defined on a disk D centered at $(x_1, x_2, \dots, x_n)$ and let $u = (a_1, a_2, \dots, a_n)$ be a unit vector. Then the directional derivative of f at $(x_1, x_2, \dots, x_n)$ in the direction of u , denoted $D_u f(x_1, x_2, \dots, x_n)$ is defined by

$$\lim_{h \rightarrow 0} \frac{f(x_1 + ha_1, x_2 + ha_2, \dots, x_n + ha_n) - f(x_1, x_2, \dots, x_n)}{h} \text{ If limit exists.}$$

18. A Sufficient Condition for Continuity:

A sufficient condition for a function f to be continuous at (a, b) is that one of the partial derivatives exists and is bounded in a neighbourhood of (a, b) and that the other exists at (a, b) .

19. If f is a differentiable function of x and y then f has a directional derivative in the direction of any unit vector $u = (a, b)$ and
- $$D_u f(x, y) = f_x(x, y)a + f_y(x, y)b = \nabla f(x, y) \cdot u.$$

Remark: If the unit vector u makes an angle θ with the positive x -axis then we can write $u = (\cos \theta, \sin \theta)$ and

$$D_u f(x_0, y_0) = f_x(x_0, y_0) \cos \theta + f_y(x_0, y_0) \sin \theta$$

Function from $\mathbb{R}^n$ to $\mathbb{R}^m$: A map $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined as $f(x_1, x_2, \dots, x_n) = (\phi_1, \phi_2, \dots, \phi_m)$ is called function in n -variables to m -variables, where for each $i = 1, 2, \dots, m$, $\phi_i: \mathbb{R}^n \rightarrow \mathbb{R}$ is a real valued function in n -variables.

10.3. Limit and Continuity of Vectors and Real Valued Functions

- (i) **Definition (Limit):** Let $f: A \rightarrow \mathbb{R}^m$, $A \subset \mathbb{R}^n$ and c is a cluster point of A . Then f is said to have a limit l in $\mathbb{R}^m$ at the point c in $\mathbb{R}^n$ if for every $\varepsilon > 0, \exists \delta > 0$ such that $\forall x \in A$
- $$0 < |x - c| < \delta \Rightarrow |f(x) - l| < \varepsilon, \quad \lim_{x \rightarrow c} f(x) = l$$

- (ii) **Definition (Continuity):** A function $f: A \rightarrow \mathbb{R}^m$, $A \subset \mathbb{R}^n$ is said to be continuous at a point $a \in A$ if a is not a cluster point of A or
- $$\lim_{x \rightarrow c} f(x) = f(a).$$

- (iii) **Definition (Uniform continuity):** Given a function $f: A \rightarrow \mathbb{R}^m$, f is said to be uniformly continuous on A iff $\forall \varepsilon > 0, \exists \delta > 0$ such that
- $$\forall x, y \in A: |x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon.$$

- (iv) **Definition (Differentiability):** Let A be an open set in $\mathbb{R}^n$ and $f: A \rightarrow \mathbb{R}^m$, then f is said to be differentiable at $a \in A$ if and only if there is a linear transformation $\lambda: \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that
- $$\lim_{h \rightarrow 0} \frac{|f(a+h) - f(a) - \lambda(h)|}{|h|} = 0$$
- If f is differentiable at every point of A , then f is said to be a differentiable function.

20. Let $f: A \rightarrow \mathbb{R}^m$, where $A \subset \mathbb{R}^n$ and c is cluster point of A then $\lim_{x \rightarrow c} f(x) = l$ iff $\lim_{x \rightarrow c} \phi_i(x) = l_i$ $i = 1, \dots, m$.
21. Suppose $f: A \rightarrow \mathbb{R}, A \subset \mathbb{R}^n$ and c is a cluster point of A . If $\lim_{x \rightarrow c} f(x)$ exists, then $\exists M > 0, \delta > 0$ such that $\forall x \in A$ $0 < |x - c| < \delta \Rightarrow |f(x)| < M$.
22. If $f: A \rightarrow \mathbb{R}^m, A \subset \mathbb{R}^n$ and c is a cluster point of A such that $\lim_{x \rightarrow c} f(x)$ exists then $\lim_{x \rightarrow c} |f(x)|$ exists and further more $\lim_{x \rightarrow c} |f(x)| = \left| \lim_{x \rightarrow c} f(x) \right|$.
23. Let $f: A \rightarrow \mathbb{R}^m, A \subset \mathbb{R}^n$ and $a \in A$, then f is continuous at a iff $\phi_i, i = 1, \dots, m$ is continuous at a .
24. Let $f, g: A \rightarrow \mathbb{R}^m, A \subset \mathbb{R}^n$ and $\phi: A \rightarrow \mathbb{R}$. If f, g and ϕ are continuous at $a \in A$, then $f + g, f \cdot g$ and $\phi \cdot f$ are all continuous at $a \in A$.
25. If $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation, then T is continuous function.
26. Let $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation, then f is differentiable.
27. Let $f: A \rightarrow \mathbb{R}^m$ and $g: B \rightarrow \mathbb{R}^k, A \subset \mathbb{R}^n$ and $B \subset \mathbb{R}^m$ such that $B \supseteq \text{range } f$. If $\lim_{x \rightarrow a} f(x) = b \in B$ and g is continuous at b then $\lim_{x \rightarrow a} (g \circ f)(x) = g(b)$.
28. Let $f: A \rightarrow \mathbb{R}^m, g: B \rightarrow \mathbb{R}^k, A \subset \mathbb{R}^n$ and such that $\text{range } f \subset B$. Suppose f is continuous at $a \in A$, and g is continuous at $f(a) \in B$. Then $g \circ f$ is continuous at a .
29. Let $f: A \rightarrow \mathbb{R}^m, A \subset \mathbb{R}^n$, then f is continuous on A iff for every open set $U \subset \mathbb{R}^m$ there is an open set $V \subset \mathbb{R}^n$ such that $f^{-1}(U) = V \cap A$.
30. Let $f: A \rightarrow \mathbb{R}^m, A \subset \mathbb{R}^n$, then f is continuous on A iff for every closed set $F \subset \mathbb{R}^m$ there is a closed set $H \subset \mathbb{R}^n$ such that $f^{-1}(F) = H \cap A$.
31. Let m and n be positive integers then the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{x^m y^n}{x^2 + y^2}$ Exists if and only if $m+n > 2$.
32. Let $A \subseteq \mathbb{R}^n$ be a subset and $f: A \rightarrow \mathbb{R}^m$ be a function differentiable at $a \in A$, then f is continuous at a .
33. Let $A \subseteq \mathbb{R}^n$ be open set and $f: A \rightarrow \mathbb{R}^m$ be a function defined as $f(x_1, x_2, \dots, x_n) = (\phi_1, \phi_2, \dots, \phi_m)$, where $\phi_i: \mathbb{R}^n \rightarrow \mathbb{R}$ be real valued function $\forall i = 1, 2, \dots, m$. Then f is differentiable at $a \in A \Leftrightarrow$ each ϕ_i is differentiable at a and its derivative is given by $f'(a) = (\phi'_1(a), \phi'_2(a), \dots, \phi'_m(a))$.

Jacobian Matrix: Let $f: D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined by $f(x_1, x_2, \dots, x_n) = (\phi_1, \phi_2, \dots, \phi_m)$ Where $\phi_i: \mathbb{R}^n \rightarrow \mathbb{R}$ is a real valued function, where $1 \leq i \leq m$. Then matrix of derivative at $x=c$ is denoted by $D(f(c))$ and it is defined as

$$D(f(c)) = \begin{pmatrix} \frac{\partial \phi_1}{\partial x_1} & \frac{\partial \phi_1}{\partial x_2} & \dots & \frac{\partial \phi_1}{\partial x_n} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial \phi_m}{\partial x_1} & \frac{\partial \phi_m}{\partial x_2} & \dots & \frac{\partial \phi_m}{\partial x_n} \end{pmatrix}_{m \times n} = (a_{ij}) \quad a_{ij} = \frac{\partial \phi_i}{\partial x_j} \begin{pmatrix} 1 \leq i \leq m \\ 1 \leq j \leq n \end{pmatrix}$$

10.4. Maxima and Minima of Functions of Two Variables

Definition:

(i) A function $f(x, y)$ is said to have a maximum at a point (a, b) if there exists a neighbourhood N of (a, b) such that $f(a+h, b+k) < f(a, b) \forall (a+h, b+k) \in N - \{(a, b)\}$

The set $N - \{(a, b)\}$ is called a deleted neighbourhood of (a, b) .

(ii) A function $f(x, y)$ is said to have a minimum at a point (a, b) if there exists a neighbourhood N of (a, b) such that $f(a+h, b+k) > f(a, b) \forall (a+h, b+k) \in N - \{(a, b)\}$

(iii) A function $f(x, y)$ is said to have an extremum at a point (a, b) if it has either a maximum or minimum at (a, b) .

(iv) A point (a, b) is said to be a critical point for a function $f(x, y)$ if f_x and f_y both exist at (a, b) and $f_x(a, b) = 0$ and $f_y(a, b) = 0$

(a) If a function $f(x, y)$ has an extremum at any point (a, b) of the domain of f if f admits partial derivatives f_x and f_y at (a, b) , then $f_x(a, b) = 0$ and $f_y(a, b) = 0$

Remark: The conditions of the above theorem are only necessary but are not sufficient.

(b) Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be a real valued function in two variables which possesses continuous second order partial derivatives and $f_{xx}(a, b) = A, f_{xy}(a, b) = B, f_{yy}(a, b) = C$ then

(i) f has maximum at (a, b) if $AC - B^2 > 0, A > 0$.

(ii) f has maximum at (a, b) if $AC - B^2 > 0, A < 0$.

(iii) f has neither maximum value nor minimum value at (a, b) if $AC - B^2 < 0$.

(iv) f may or may not have an extremum at (a, b) if $AC - B^2 = 0$

(This case is often called the doubtful case, which needs further investigation)

10.5. Compactness of Connectedness

1. A collection O of open sets is an open cover of A if every point x in A is in some open set in the collection O .
2. A set $A \subset \mathbb{R}^n$ is called compact if every open cover O of A has a finite sub cover of A .
3. **(Heine-Borel Theorem):** $A \subset \mathbb{R}^n$ is compact iff A is closed and bounded.
4. If $B \subset \mathbb{R}^m$ is compact and $x \in \mathbb{R}^n$ then $\{x\} \times B$ is compact in $\mathbb{R}^{n+m}$.
5. If $A \subset \mathbb{R}^n$ and $B \subset \mathbb{R}^m$ are compact then $A \times B \subset \mathbb{R}^{n+m}$ is compact.
6. If $A_i \subset \mathbb{R}^{n_i}, i = 1, \dots, k$ are compact then $A_1 \times A_2 \times \dots \times A_k \subset \mathbb{R}^{n_1 + \dots + n_k}$ is compact.

In particular any closed rectangle $W \subset \mathbb{R}^k, k \in \mathbb{N}$ is compact.

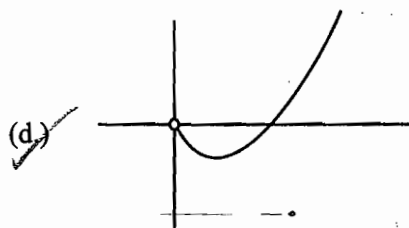
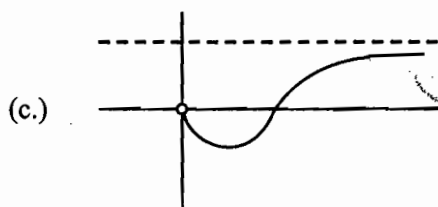
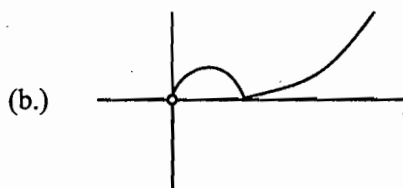
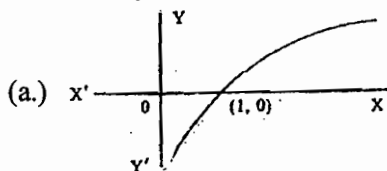
7. **(Bolzano-Weierstrass Theorem):** Every bounded infinite subset of $\mathbb{R}^n$ has a cluster point.
8. **(Cantor Intersection Theorem):** Suppose F_1 is a nonempty, closed and bounded subset of $\mathbb{R}^n$, and $\{F_i\}_{i \in \mathbb{N}}$ is a sequence of nonempty closed sets in $\mathbb{R}^n$ such that $F_1 \supseteq F_2 \supseteq \dots \supseteq F_n \supseteq \dots$, Then $\bigcap_{i \in \mathbb{N}} F_i \neq \emptyset$.
9. A set $A \subset \mathbb{R}^n$ is said to be dense in $B \subset \mathbb{R}^n$ if $\bar{A} \supset B$. If A is dense in $\mathbb{R}^n$ it is simply called a dense set.
10. If $\{E_n : n \in \mathbb{N}\}$ is a collection of dense open set, then $\bigcap_{n \in \mathbb{N}} E_n$ is nonempty.
11. The set $\mathbb{Q}$ of rational numbers is not the intersection of a countable collection of open sets.
12. The set of all irrational numbers $\mathbb{Q}^c$ is not the union of a countable collection of closed sets in $\mathbb{R}$.
13. If $f: A \rightarrow \mathbb{R}^m, A \subset \mathbb{R}^n$ is continuous and A is compact then $f(A)$ is compact.
14. **Extremum Value Theorem:** Let $A \subset \mathbb{R}^n$ be compact. If $F: A \rightarrow \mathbb{R}$ is continuous then f takes on a maximum and minimum value on A .
15. Let $A \subset \mathbb{R}^n$ be compact and if $f: A \rightarrow \mathbb{R}^m$ is continuous and one to one then the inverse function is continuous.
16. Let $A \subset \mathbb{R}^n$ be compact and if $f: A \rightarrow \mathbb{R}^m$ is continuous then f is uniformly continuous.
17. Let $A \subset \mathbb{R}^n$ be compact and B be closed subset of A then B be closed and compact subset of A .
18. Let C be closed and K be compact in $\mathbb{R}^n$ then $K + C$ is closed.

19. Let A, B be compact subsets of $\mathbb{R}^n$ then their sum $A+B$ is compact.
20. Let $A \subseteq \mathbb{R}^n$ and U, V open subsets of $\mathbb{R}^n$ then $U \& V$ are said to be a separation of A if
 - (i) $U \cap A \neq \phi$
 - (ii) $V \cap A \neq \phi$
 - (iii) $(U \cap A) \cap (V \cap A) = \phi$
 - (iv) $(U \cap A) \cup (V \cap A) = A$
21. A subset A of $\mathbb{R}^n$ is said to be a disconnected set if it has a separation U and V . A is called connected if it is not disconnected.
22. A set $C \subseteq \mathbb{R}$ is connected iff it is an interval.
23. Let $A \subseteq \mathbb{R}^n$ be a connected set and if $f: A \rightarrow \mathbb{R}^m$ is continuous then $f(A)$ is connected.
24. In $\mathbb{R}^n$, the only subsets that are open and closed are ϕ and $\mathbb{R}^n$.
25. Let $\{A_i : i \in I\}$, where I is index set, be a collection of connected subsets of a (metric) space X with the property that for all $i, j \in I$ we have $A_i \cap A_j \neq \phi$, then $A = \bigcup_{i \in I} A_i$ is connected.
26. Let A be a countable subset of $\mathbb{R}^n$, $n > 1$, then $\mathbb{R}^n - A$ is connected.
27. Let X and Y be connected (metric) spaces then the product space $X \times Y$ is connected.

ASSIGNMENT SHEET - 5

- If $f(x) = 1/(1-x)$, $g(x) = f[f(x)]$ and $h(x) = f[g(x)]$, then what is $f(x)g(x)h(x)$ equal to?
(a) ~~-1~~
(b) 0
(c) 1
(d) 2
- What is the value of $\lim_{n \rightarrow \infty} \frac{2^{n+1} + 3^{n+1}}{2^n + 3^n}$?
(a) 1/3
(b) 1
(c) ~~3~~
(d) ∞
- What is the value of $\lim_{x \rightarrow 0} \frac{e^x - x - 1}{x^2}$?
(a) 0
(b) ~~1/2~~
(c) 2
(d) e
- If D is the set of all real x such that $f(x) = 1 - e^{\left(\frac{1}{x}\right) - 1}$ is positive, then what is D equal to?
(a) $(-\infty, 0) \cup (1, \infty)$
(b) $(-\infty, 0) \cup [1, \infty)$
(c) ~~(1, \infty)~~
(d) $(-\infty, 1)$
- If $\lim_{x \rightarrow 0} \frac{(x + 3 \sin x - x^3 - k \sinh x)}{1 - \cos x + x^2 - 3x^3}$ exists then what is the value of k ?
(a) ~~-1~~
(b) 2
(c) ~~3~~
(d) 4
- What is the value of $\lim_{x \rightarrow y} \frac{x^y - y^x}{x^x - y^y}$?
(a) $\frac{1 + \ln y}{1 - \ln y}$
(b) $\frac{1 - \ln y}{1 + \ln y}$
(c) $\frac{-1 + \ln y}{1 + \ln y}$
(d) $\frac{-1 - \ln y}{1 - \ln y}$
- Assertion (A): $\lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0$
Reason (R): $\lim_{x \rightarrow \infty} \frac{\sin x}{x} = \lim_{x \rightarrow \infty} \sin x \lim_{x \rightarrow \infty} \frac{1}{x}$
(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true
- What is the value of $\lim_{x \rightarrow \infty} \{\sin(1/x) + \cos(1/x)\}$?
(a) 0
(b) $\frac{1}{e}$
(c) e
(d) ~~1~~
- Which one of the following functions is not well defined?
(a) $\sqrt{1 + \sin x}$
(b) $\sqrt{5 \sec^2 x - 4}$
(c) $\sqrt{3 - \cos^2 x}$
(d) $\sqrt{x^4 + x^2 + \frac{1}{100}}$

10. Which one of the following graphs is the correct graph of the function $y(x) = x \cdot \ln(x)$?



11. What are the value of a & b respectively,

$$\text{if } \lim_{x \rightarrow 0} \frac{\sin ax - \ln(e^x \cos x)}{x \sin bx} = \frac{1}{2}$$

- (a.) 1, 1
(b.) 1, $\frac{1}{2}$
(c.) -1, 1
(d.) -1, $\frac{1}{2}$

12. **Assertion (A):** e^x cannot be expressed as sum of even and odd functions.

Reason (R): e^x is neither even nor odd function

- (a.) Both A and R are individually true and R is the correct explanation of A
(b.) Both A and R are individually true but R is not the correct explanation of A
(c.) A is true but R is false
(d.) A is false but R is true

13. Given the following limits:

1. $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$
2. $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right) = 1$
3. $\lim_{x \rightarrow 0} \frac{\log(x+1)}{x} = 1$

Which of these are correct?

- (a.) 1, 2 and 3
(b.) 1 and 2
(c.) 1 and 3
(d.) 2 and 3

14. If $f(x) = \frac{x-5}{x+5}$, $x \neq -5$ then the domain of

$f^{-1}(x)$, is

- (a.) $\mathbb{R}$
(b.) $\mathbb{R} - \{1\}$
(c.) $(-\infty, 1)$
(d.) $(1, \infty)$

15. $\lim_{x \rightarrow \infty} (4^x + 5^x)^{\frac{1}{x}}$ equals to

- (a.) 4
(b.) 5
(c.) e
(d.) 5e

16. $\lim_{x \rightarrow 0} \frac{e^x + e^{-x} - 2}{x^2}$ is equal to

- (a.) 1
(b.) -1
(c.) 1/2
(d.) -1/2

17. Consider a mapping f from the set of natural numbers N to the set of integers Z defined by

$$f(n) = \begin{cases} \frac{n-1}{2}, & \text{when } n \text{ is odd} \\ -\frac{n}{2}, & \text{when } n \text{ is even} \end{cases}$$

Then $f: N \rightarrow Z$ is:

- (a.) One - one but not onto
(b.) Onto but not one - one
(c.) Both one-one and onto
(d.) Neither one-one nor onto

18. If α, β are the roots of the equation $ax^2 + bx + c = 0$, then

$\lim_{x \rightarrow \alpha} \frac{1 - \cos(ax^2 + bx + c)}{(x - \alpha)^2}$ is equal to

- (a.) 0
 (b.) $\frac{a}{2}(\alpha - \beta)^2$
 (c.) $-\frac{a}{2}(\alpha - \beta)^2$
 (d.) $\frac{a^2}{2}(\alpha - \beta)^2$

19. If $\lim_{x \rightarrow 0} \frac{x(1 - \cos x) - ax^2 \sin x}{x^5}$ exists and is finite, then the value of a must be

- (a.) 1
 (b.) $\frac{1}{2}$
 (c.) $\frac{1}{3}$
 (d.) $\frac{1}{4}$

20. If $f(x) = 2x + 1$, then $f\left(\frac{1}{x}\right)$ is

- (a.) $\frac{1}{2x+1}$
 (b.) $\frac{2x+1}{x}$
 (c.) $\frac{1}{x}$
 (d.) $\frac{2}{x} + 1$

21. If $f(x) = \begin{cases} x^2 & 0 \leq x \leq \frac{1}{2} \\ x & \frac{1}{2} < x \leq 1 \end{cases}$ then

$\lim_{x \rightarrow \frac{1}{2}} f(x)$

- (a.) has the value $\frac{1}{2}$
 (b.) has the value 1
 (c.) has the value 2
 (d.) does not exist

22. If $\lim_{x \rightarrow 0} \frac{\sin 2x + a \sin x}{x^3} = b$, where

$\lim_{x \rightarrow 0} \frac{\sin 2x + a \sin x}{x^3}$ is finite, then the values of a and b respectively will be

- (a.) -2, -1
 (b.) 2, 1
 (c.) -2, 1
 (d.) 2, -1

23. $\lim_{x \rightarrow \infty} x \sin\left(\frac{1}{x}\right)$ equals

- (a.) 1
 (b.) 0
 (c.) -1
 (d.) $-\infty$

24. Consider the following statements: if

$$f(x) = \begin{cases} x & 0 \leq x < 1 \\ 3 - x & 1 \leq x \leq 2 \end{cases} \text{ then}$$

1. $\lim_{x \rightarrow 1^-} f(x) = 1$
 2. $\lim_{x \rightarrow 1^+} f(x) = 2$
 3. $\lim_{x \rightarrow 1} f(x) = 2$

Of these statements

- (a.) 1 and 3 are correct
 (b.) 1, 2 and 3 are correct
 (c.) 1 and 2 are correct
 (d.) 2 alone is correct

25. $\lim_{x \rightarrow 0} \frac{\sin(ax)}{\sin(bx)}$ ($a \neq 0, b \neq 0$) is

- (a.) $\frac{a}{b}$
 (b.) $\frac{b}{a}$
 (c.) $a.b$
 (d.) does not exist

26. $\lim_{x \rightarrow \infty} \frac{ax + b}{cx}$ is

- (a.) a/c
 (b.) ∞
 (c.) B
 (d.) none of the above

27. The correct value of $\lim_{x \rightarrow 0} \frac{x}{\sqrt{1-\cos x}}$
- does not exist
 - is $\sqrt{2}$
 - is $-\sqrt{2}$
 - is $1/\sqrt{2}$
28. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{\frac{\pi}{2} - x}$ is
- 1
 - 0
 - ∞
 - does not exist
29. If X and Y are two non-empty finite sets and $f: X \rightarrow Y$ and $g: Y \rightarrow X$ are mappings such that $g \circ f: X \rightarrow X$ is a surjective (i.e., onto) map, then
- f must be one-to-one
 - f must be onto
 - g must be one-to-one
 - X and Y must have the same number of elements.
30. Let $P(x)$ be a non-constant polynomial such that $P(n) = P(-n)$ for all $n \in \mathbb{N}$. Then $P'(0)$
- Equals 1
 - Equals 0
 - Equals -1
 - Can not be determined from the given data.
31. If $\lim_{x \rightarrow 0} \left(\frac{1+cx}{1-cx} \right)^{\frac{1}{x}} = 4$, then $\lim_{x \rightarrow 0} \left(\frac{1+2cx}{1-2cx} \right)^{\frac{1}{x}}$ is
- 2
 - 4
 - 16
 - 64
32. Let $f(x) = e^{(e^x)}$ and define $g(x) = f(x+1) - f(x)$. Then, as $x \rightarrow \infty$, the function $g(x)$ converges to
- 0
 - 1
 - e
 - e^e
33. Consider the functions $f, g: \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(n) = 3n+2$ and $g(n) = n^2 - 5$
- Neither f and g is a one-to-one function
 - The function f is one-to-one, but not g
 - The function g is one-to-one, but not f
 - Both f and g are one-to-one functions
34. Let T be the graph of the function
- $$f(x) = \begin{cases} 1+x, & -1 \leq x \leq 0; \\ 1-x, & 0 \leq x \leq 1 \end{cases}$$
- Then the reflection of T in the line $y=0$ is given by the graph of $g(x)$ where
- $g(x) = \begin{cases} -1-x, & -1 \leq x \leq 0 \\ -1+x, & 0 \leq x \leq 1 \end{cases}$
 - $g(x) = \begin{cases} -1+x, & -1 \leq x \leq 0 \\ -1-x, & 0 \leq x \leq 1 \end{cases}$
 - $g(x) = \begin{cases} -1-x, & -1 \leq x \leq 0 \\ 1-x, & 0 \leq x \leq 1 \end{cases}$
 - $g(x) = \begin{cases} 1-x, & -1 \leq x \leq 0 \\ -1-x, & 0 \leq x \leq 1 \end{cases}$
35. Let a, b, c, d be rational number with $ad - bc \neq 0$. then the function $f: \mathbb{R} \setminus \mathbb{Q} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{ax+b}{cx+d}$ is
- onto but not one-one.
 - one-one but not onto.
 - neither one-one nor onto.
 - both one-one and onto.
36. Let $f: (0, \infty) \rightarrow \mathbb{R}$ be the function defined by $f(x) = \frac{e^x}{x^x}$. Then the $\lim_{x \rightarrow \infty} f(x)$
- does not exist.
 - exists and is 0
 - exists and is 1
 - exists and is e

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- Let $f: X \rightarrow X$ such that $f(f(x)) = x$ for all $x \in X$. Then
 - f is one-to-one and onto.
 - f is one-to-one but not onto.
 - f is onto but not one-to-one.
 - f need not be either one-to-one or onto.
- A polynomial of odd degree with real coefficients must have
 - at least one real root.
 - no real root.
 - only real roots.
 - at least one root which is not real.
- Consider the following sets of functions on $\mathbb{R}$, W = The set of constant functions on $\mathbb{R}$, X = The set of polynomial functions on $\mathbb{R}$, Y = The set of continuous functions on $\mathbb{R}$, Z = The set of all functions on $\mathbb{R}$ Which of these sets has the same cardinality as that of $\mathbb{R}$?
 - Only W
 - Only W and X
 - Only W , X and Z
 - Only W , X , Y
- If $f: [0, 1] \rightarrow (0, 1)$ is a continuous mapping then which of the following is NOT true?
 - $F \subseteq [0, 1]$ is a closed set implies $f(F)$ is closed in $\mathbb{R}$.
 - If $f(0) < f(1)$ then $f([0, 1])$ must be equal to $[f(0), f(1)]$.
 - There must exist $x \in (0, 1)$ such that $f(x) = x$.
 - $f: ([0, 1]) \neq (0, 1)$.
- In which of the following cases, there is no continuous function f from the set S onto the set T ?
 - $S = [0, 1], T = \mathbb{R}$
 - $S = (0, 1), T = \mathbb{R}$
 - $S = (0, 1), T = (0, 1]$
 - $S = \mathbb{R}, T = (0, 1)$
- Let $A_n \subseteq \mathbb{R}$ for $n \geq 1$, and $\chi_n: \mathbb{R} \rightarrow \{0, 1\}$ be the function $\chi_n(x) = \begin{cases} 0 & \text{if } x \notin A_n \\ 1 & \text{if } x \in A_n \end{cases}$, let $g(x) = \limsup_{n \rightarrow \infty} \chi_n(x)$ and $h(x) = \liminf_{n \rightarrow \infty} \chi_n(x)$
 - If $g(x) = h(x) = 1$, then there exists m such that for all $n \geq m$ we have $x \in A_n$.
 - If $g(x) = 1$ and $h(x) = 0$, then there exist m such that for all $n \geq m$ we have $x \in A_n$.
 - If $g(x) = 1$ and $h(x) = 0$ then there exists a sequence $n_1, n_2, \dots$ of distinct integers such that $x \in A_{n_k}$ for all $k \geq 1$.
 - If $g(x) = h(x) = 0$ then there exists m such that for all $n \geq m$ we have $x \notin A_n$.
- Let f be a monotonically non-decreasing real-valued function on $\mathbb{R}$. Then
 - $\lim_{x \rightarrow a} f(x)$ exists at each point a
 - If $a < b$, then $\lim_{x \rightarrow a^+} f(x) \leq \lim_{x \rightarrow b^-} f(x)$
 - f is an unbounded function
 - The function $g(x) = e^{-f(x)}$ is a bounded function
- Which of the following statement is true?
 - $\lim_{x \rightarrow \infty} \frac{\log x}{x^{1/2}} = 0$ and $\lim_{x \rightarrow \infty} \frac{\log x}{x} = \infty$
 - $\lim_{x \rightarrow \infty} \frac{\log x}{x^{1/2}} = \infty$ and $\lim_{x \rightarrow \infty} \frac{\log x}{x} = 0$
 - $\lim_{x \rightarrow \infty} \frac{\log x}{x^{1/2}} = 0$ and $\lim_{x \rightarrow \infty} \frac{\log x}{x} = 0$
 - $\lim_{x \rightarrow \infty} \frac{\log x}{x^{1/2}} = 0$ but $\lim_{x \rightarrow \infty} \frac{\log x}{x}$ does not
- Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = [x^2]$. The points of discontinuity of f are
 - Only the integral points
 - All rational numbers
 - $\{\pm\sqrt{n} : n \text{ is a non-negative integer}\}$
 - All real numbers

10. A monotonic function
- Is always continuous
 - Is continuous only if it intermediate value property
 - Can be nowhere continuous
 - Can be discontinuous at finitely many points
11. Which one of the following functions has exactly two points of discontinuity?
- $f(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}$
 - $f(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$
 - $f(x) = \begin{cases} x & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$
 - $f(x) = \begin{cases} 1 & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$
12. Let $f: [a, b] \rightarrow \mathbb{R}$ be a continuous function and let $f(a) < f(b)$. Then by intermediate value theorem
- $f([a, b]) = [f(a), f(b)]$
 - $f([a, b]) \supseteq [f(a), f(b)]$
 - $f([a, b]) \subseteq [f(a), f(b)]$
 - $f([a, b]) \neq [f(a), f(b)]$
13. Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \begin{cases} x^4, & \text{if } x \text{ is rational,} \\ 2x^2 - 1, & \text{if } x \text{ is irrational} \end{cases}$. Let S be the set of points where f is continuous. Then
- $S = \{1\}$
 - $S = \{-1\}$
 - $S = \{-1, 1\}$
 - $S = \emptyset$
14. Let $f: [0, 1] \rightarrow \mathbb{R}$ be the continuous function defined by $f(x) = \frac{(x-1)(x-2)}{(x-3)(x-4)}$. Then the maximal subset of $\mathbb{R}$ on which f has a continuous extension is
- $(-\infty, 3)$
 - $(-\infty, 3) \cup (4, \infty)$
 - $\mathbb{R} \setminus \{3, 4\}$
 - $\mathbb{R}$
15. The continuous function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = (x^2 + 1)^{2003}$ is
- Onto but not one-one
 - One-one but not onto
 - Both one-one and onto
 - Neither one-one nor onto
16. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. If $f(\mathbb{Q}) \subseteq \mathbb{N}$, then
- $f(\mathbb{R}) = \mathbb{N}$
 - $f(\mathbb{R}) \subseteq \mathbb{N}$ but f need not be constant
 - f is unbounded
 - f is a constant function
17. Let $f: [a, b] \rightarrow \mathbb{R}$ be a monotonic function. Then
- f is continuous
 - f is discontinuous at at most two points
 - f is discontinuous at at finitely many points
 - f is discontinuous at at most countable points
18. Let $f: [0, 10] \rightarrow [0, 10]$ be a continuous mapping. Then
- f need not have any fixed point
 - f has at least 10 fixed points
 - f has at least 9 fixed points
 - f has at least one fixed point

19. Let $S = [0, 1] \cup [2, 3]$ and let $f: S \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} 2x & \text{if } x \in [0, 1], \\ 8-2x & \text{if } x \in [2, 3]. \end{cases}$ If $T = \{f(x) : x \in S\}$, then the inverse function $f^{-1}: T \rightarrow S$
- does NOT exist
 - exists and is continuous
 - exists and is NOT continuous
 - exists and is monotonic
20. Let f be a continuous function from $[0, 4]$ to $[3, 9]$. Then
- There must be an x such that $f(x) = 4$
 - There must be an x such that $3f(x) = 2x + 6$
 - There must be an x such that $2f(x) = 3x + 6$
 - There must be an x such that $f(x) = x$
21. Let f_1 and f_2 be two real valued functions defined on the real line. Define two functions g and h by $g(x) = \max\{f_1(x), f_2(x)\}$ and $h(x) = \min\{f_1(x), f_2(x)\}$. Then $g(x^2) + h(x)^2 + 3g(x)h(x) = f_1(x)^2 + f_2(x)^2 + 3f_1(x)f_2(x)$ holds for all $x \in \mathbb{R}$
- Always
 - Only if $f_1(x) = f_2(x)$ for all x in $\mathbb{R}$
 - Only f_1 and f_2 are both positive functions or both negative functions
 - Only if at least one of the functions f_1 and f_2 is identically zero.
22. If X and Y are two non-empty finite sets and $f: X \rightarrow Y$ and $g: Y \rightarrow X$ are mappings such that $g \circ f: X \rightarrow X$ is a surjective (i.e., onto) map, then
- f must be one-to-one
 - f must be onto
 - g must be one-to-one
 - X and Y must have the same number of elements.
23. Let X and Y be two non-empty sets and let $f: X \rightarrow Y$, $g: Y \rightarrow X$ be two mappings. If both f and g are injective (i.e., one-to-one) then
- X and Y must be infinite sets.
 - $g = f^{-1}$ always.
 - One of $f \circ g: Y \rightarrow Y$ and $g \circ f: X \rightarrow X$ is always bijective (one-to-one and onto).
 - There exists a bijective mapping $h: X \rightarrow Y$
24. Define a function f on the real line by $f(x) = \begin{cases} x - [x] - \frac{1}{2} & \text{if } x \text{ is not an integer,} \\ 0 & \text{if } x \text{ is an integer} \end{cases}$ Then which of the following is true:
- f is periodic with period 1, i.e., $f(x+1) = f(x)$ for all x
 - f is continuous
 - f is one-to-one
 - $\lim_{x \rightarrow a} f(x)$ exists for all $a \in \mathbb{R}$.
25. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x+y) = f(x)f(y)$ for all $x, y \in \mathbb{R}$ and $f(x) = 1 + xg(x)$ where $\lim_{x \rightarrow 0} g(x) = 1$. Then the function $f(x)$ is
- e^x
 - 2^x
 - A non-constant polynomial
 - Equal to 1 for all $x \in \mathbb{R}$.
26. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function satisfying $f \circ f = f$. Then
- f must be constant
 - $f(x) = x$ for all x in the range of f
 - f must be a non-constant polynomial
 - There is no such function
27. Let $f: [0, 1] \rightarrow [0, 1]$ be continuous and $f(0) = 0, f(1) = 1$. Then, f is necessarily
- injective, but not surjective.
 - surjective, but not injective.
 - bijective.

- (d.) surjective.
28. Let p be a polynomial of degree $2n+1$ with real coefficients. We say that a real number a is a fixed point of p if $p(a)=a$. Then, p has
- exactly $2n+1$ fixed points
 - at least one fixed point
 - at most one fixed point
 - n fixed points
29. Define the function $f: \mathbb{R} \rightarrow \mathbb{R}$ by
- $$f(x) = \begin{cases} ax^2 + b, & \text{if } x \leq 1 \\ cx + 1, & \text{if } x > 1 \end{cases}$$
- We want to find appropriate values of a, b and c such that
- f is increasing in the interval $(0, \infty)$;
 - f' is continuous on $\mathbb{R}$.
- Which of the following is the correct statement about the values of (a, b, c) for which both conditions (1) and (2) are satisfied?
- $(3, 2, 6)$ is the only possible value
 - There are finitely many values of (a, b, c)
 - $(-2, -3, -4)$ is one of the values
 - There are infinitely many values of (a, b, c)
30. The limit $\lim_{x \rightarrow 0} \frac{3^x - 2^x}{4^x - 3^x}$ equals
- $\frac{3}{4}$
 - $\log_{4/3} \left(\frac{3}{2} \right)$
 - $\log_e \left(\frac{3}{2} \right)$
 - $\frac{2}{3}$
31. The equation $x^2 = x \sin x + \cos x$ is true for
- no real value of x
 - exactly one real value of x
 - exactly two real values of x
 - infinitely many real values of x
32. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous with $f(0) = f(1) = 0$. Which of the following is not possible?
- $f([0, 1]) = \{0\}$
 - $f([0, 1]) = [0, 1]$
 - $f([0, 1]) = [0, 1]$
 - $f([0, 1]) = \left[-\frac{1}{2}, \frac{1}{2}\right]$
33. Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by
- $$f(x) = \begin{cases} 2x, & \text{if } x \text{ is rational} \\ 3-x, & \text{if } x \text{ is irrational} \end{cases}$$
- Then f is continuous
- only at $x = 1$.
 - only at $x = 2$.
 - only at $x = 0$.
 - at no point of $\mathbb{R}$.
34. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = |x+1| - |x|$, then the range of f is
- The whole of $\mathbb{R}$.
 - The closed interval $[-1, 1]$
 - The unbounded subset of $\mathbb{R}$.
 - None of the above.
35. How many zeroes are there for the function $x^2 - 5|x| + 6$?
- 3
 - 4
 - 6
 - None of these
36. Let $f(x) = \cos^{-1} x$. Then f is one-one and onto if the domain and range are specified respectively as
- $[-1, 1]$ and $(-\infty, \infty)$
 - $[-1, 1]$ and $(0, 2\pi)$
 - $[-1, 1]$ and $(0, \pi/2)$
 - $[-1, 1]$ and $[0, \pi]$

37. A bijection is a map that is both one-one and onto. It is given that $f: \mathbb{R} \rightarrow \mathbb{R}$ is not a bijection. Which of the following must be true?
- If f is not one-one, then f is not onto.
 - If f is not onto, then f is not one-one.
 - f is neither one-one nor onto.
 - If f is one-one, then f is not onto.
38. Define $f: \mathbb{R} \rightarrow \mathbb{R}$ as follows:

$$f(x) = \begin{cases} 1, & \text{if } x \text{ is rational} \\ \frac{\sin x}{x}, & \text{if } x \text{ is irrational} \end{cases}$$
 Then
- f is continuous everywhere
 - f is continuous only at $x = 0$.
 - f is continuous all rational points.
 - f is continuous at all irrational points.
39. Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be continuous functions whose graphs do not intersect. Then for which function below the graph lies entirely on one side of the X-axis
- f
 - $g + f$
 - $g - f$
 - $g.f$
40. $f(x) = e^x - e^{-x}$, $g(x) = e^x + e^{-x}$. Then
- Both f and g are even functions
 - Both f and g are odd functions
 - f is odd, g is even
 - f is even, g is odd
41. Let X be a set and $f, g: X \rightarrow X$ be functions. We can say that $f \circ g$ is bijective if
- at least one of f, g is bijective
 - both f and g are bijective
 - f is 1-1 and g is onto
 - f is onto and g is 1-1
42. Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be functions. We can conclude that $h(x) \leq f(x) \forall x \in \mathbb{R}$ if we define $h: \mathbb{R} \rightarrow \mathbb{R}$ as
- $\min\{g(x), f(x) + g(x)\}$
 - $\min\{f(x), f(x) + g(x)\}$
 - $\max\{g(x), f(x) + g(x)\}$
 - $\max\{f(x), f(x) + g(x)\}$
43. Let X be a non-empty set, $f: X \rightarrow X$ be a function and let $A, B \subset X$. Then the identity $f(A \cap B) = f(A) \cap f(B)$ is
- Always holds
 - holds if f is 1-1
 - holds if f is onto
 - holds if $A \cup B = X$
44. The range of the function $f(x) = \frac{x}{\sqrt{x^2 + 1}}$, $x \in \mathbb{R}$ is
- $[-1, 1]$
 - $[-1, 1)$
 - $(-1, 1]$
 - None of these

ASSIGNMENT SHEET - 7

1. The value of $\lim_{x \rightarrow 0} \frac{1}{x} \int_x^{2x} e^{-t^2} dt$
 - (a.) does not exist.
 - (b.) is infinite.
 - (c.) exists and equals 1.
 - (d.) exists and equals 0.
2. Let f be a bounded function on $\mathbb{R}$ and $a \in \mathbb{R}$. For $\delta > 0$, let $\omega(a, \delta) = \sup |f(x) - f(a)|$, $x \in [a - \delta, a + \delta]$. Then
 - (a.) $\omega(a, \delta_1) \leq \omega(a, \delta_2)$ if $\delta_1 \leq \delta_2$.
 - (b.) $\lim_{\delta \rightarrow 0+} \omega(a, \delta) = 0$ for all $a \in \mathbb{R}$.
 - (c.) $\lim_{\delta \rightarrow 0+} \omega(a, \delta)$ need not exist.
 - (d.) $\lim_{\delta \rightarrow 0+} \omega(a, \delta) = 0$ if and only if f is continuous at a .
3. Let f be a continuously differentiable function on $\mathbb{R}$ suppose that $L = \lim_{x \rightarrow \infty} (f(x) + f'(x))$ exists. If $0 < L < \infty$ then which of the following statements is/are correct?
 - (a.) If $\lim_{x \rightarrow \infty} f'(x)$ exists, then it is 0
 - (b.) If $\lim_{x \rightarrow \infty} f(x)$ exists it is L
 - (c.) If $\lim_{x \rightarrow \infty} f'(x)$ exists then $\lim_{x \rightarrow \infty} f(x) = 0$
 - (d.) If $\lim_{x \rightarrow \infty} f(x)$ exists then $\lim_{x \rightarrow \infty} f'(x) = L$
4. Let $A \subseteq \mathbb{R}$ and $f: A \rightarrow \mathbb{R}$ be given by $f(x) = x^2$. Then f is uniformly continuous if
 - (a.) A is a bounded subset of $\mathbb{R}$
 - (b.) A is a dense subset of $\mathbb{R}$
 - (c.) A is an unbounded and connected subset of $\mathbb{R}$
 - (d.) A is an unbounded and open subset of $\mathbb{R}$
5. If $f: [0, 1] \rightarrow (0, 1)$ is a continuous mapping then which of the following is NOT true?
 - (a.) $F \subseteq [0, 1]$ is a closed set implies $f(F)$ is closed in $\mathbb{R}$.
 - (b.) If $f(0) < f(1)$ then $f([0, 1])$ must be equal to $[f(0), f(1)]$.
 - (c.) There must exist $x \in (0, 1)$ such that $f(x) = x$.
 - (d.) $f: ([0, 1]) \neq (0, 1)$.
6. The function $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = e^{|x|+x^2} + |x^2 - 1|$ Which of the following is true about the function f ?
 - (a.) It is not differentiable exactly at three points of $\mathbb{R}$.
 - (b.) It is not differentiable at $x = 0$.
 - (c.) It is differentiable at $x = 2$.
 - (d.) It is not differentiable at $x = 1$ and $x = -1$.
7. The function $f(x) = a_0 + a_1|x| + a_2|x|^2 + a_3|x|^3$ is differentiable at $x = 0$
 - (a.) For no values of a_0, a_1, a_2, a_3
 - (b.) For any value of a_0, a_1, a_2, a_3
 - (c.) Only if $a_1 = 0$
 - (d.) Only if both $a_1 = 0$ and $a_3 = 0$.
8. Suppose $f: \mathbb{R} \rightarrow \mathbb{R}$ is a function that satisfies $|f(x) - f(y)| \leq |x - y|^\beta$, $\beta > 0$. Which of the following is correct?
 - (a.) If $\beta = 1$ then f is differentiable
 - (b.) If $\beta > 0$ then f is uniformly continuous
 - (c.) If $\beta > 1$ then f is a constant function
 - (d.) f must be a polynomial

9. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function with period $\rho > 0$. Then $g(x) = \int_x^{x+\rho} f(t) dt$ is a
- Constant function
 - Continuous function
 - Continuous function but not differentiable
 - Neither continuous nor differentiable
10. Let A be the set of rational numbers in the open interval $(0, 7)$ and $f: A \rightarrow \mathbb{R}$ be a uniformly continuous function. Which of the following are true?
- f is bounded
 - f is necessarily a constant function
 - f is differentiable on $(0, 7)$
 - f is differentiable at all the rational points in $(0, 7)$
11. Let f be a twice differentiable function on $\mathbb{R}$. Given that $f''(x) > 0$ for all $x \in \mathbb{R}$,
- $f(x) = 0$ has exactly two solution on $\mathbb{R}$
 - $f(x) = 0$ has a positive solution if $f(0) = 0$ and $f'(0) = 0$
 - $f(x) = 0$ has no positive solution if $f(0) = 0$ and $f'(0) > 0$
 - $f(x) = 0$ has no positive solution if $f(0) = 0$ and $f'(0) < 0$
12. Suppose $f: \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function. Then which of the following statements are necessarily true?
- If $f'(x) \leq r < 1$ for all $x \in \mathbb{R}$, then f has at least one fixed point.
 - If f has a unique fixed point, then $f'(x) \leq r < 1$ for all $x \in \mathbb{R}$
 - If f has a unique fixed point, then $f'(x) \geq r > -1$ for all $x \in \mathbb{R}$
 - None of these
13. Let $f(x) = \sin x - x + \frac{x^3}{3!}$ and $g(x) = \cos x - 1 + \frac{x^2}{2!}$ for $x \in \mathbb{R}$. Which of the following statements are correct?
- $f(x) \geq 0$ for all $x > 0$
 - g is an increasing function on $[0, \infty)$
 - g is an decreasing function on $[0, \infty)$
 - f is a decreasing function on $[0, \infty)$
14. Let $I = [0, 1] \subset \mathbb{R}$. For $x \in \mathbb{R}$, let $\phi(x) = \text{dist}(x, I) = \inf \{|x - y| : y \in I\}$. Then
- $\phi(x)$ is discontinuous somewhere on $\mathbb{R}$
 - $\phi(x)$ is continuous on $\mathbb{R}$ but not continuously differentiable exactly at $x = 0$
 - $\phi(x)$ is continuous on $\mathbb{R}$ but not continuously differentiable exactly at $x = 0$ and $x = 1$
 - $\phi(x)$ is differentiable on $\mathbb{R}$.
15. Consider the function $f(x) = \cos(|x - 5|) + \sin(|x - 3|) + |x + 10|^3 - (|x| + 4)^2$. At which of the following point is f not differentiable?
- $x = 5$
 - $x = 3$
 - $x = -10$
 - $x = 0$
16. Define $f: \mathbb{R} \rightarrow \mathbb{R}$ by $f(x) = \begin{cases} x^2 & \text{if } x < 0, \\ 2x + x^2 & \text{if } x \geq 0 \end{cases}$. Then which of the following statements are correct?
- $f''(x) = 2$ for all $x \in \mathbb{R}$
 - $f''(0)$ does not exist
 - $f'(x)$ exists for each $x \neq 0$
 - $f'(0)$ does not exist

17. Consider the function $f(x) = |\cos x| + |\sin(2-x)|$. At which of the following points is f not differentiable?
- $\{(2n+1)\frac{\pi}{2} : n \in \mathbb{Z}\}$
 - $\{n\pi : n \in \mathbb{Z}\}$
 - $\{n\pi + 2 : n \in \mathbb{Z}\}$
 - $\{\frac{n\pi}{2} : n \in \mathbb{Z}\}$
18. Let $F = \{f : \mathbb{R} \rightarrow \mathbb{R} : |f(x) - f(y)| \leq K|x - y|^\alpha\}$. For all $x, y \in \mathbb{R}$ and for some $\alpha > 0$ and some $K > 0$. Which of the following is/are true?
- Every $f \in F$ is continuous
 - Every $f \in F$ is uniformly continuous
 - Every differentiable function f is in F
 - Every $f \in F$ is differentiable
19. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^2 \sin \frac{1}{x^2}, x \neq 0; f(0) = 0$
- f is differentiable everywhere with continuous derivative
 - f is differentiable everywhere other than the point 1
 - f is nowhere differentiable
 - f is differentiable but its derivative is not continuous
20. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = [x^2]$. The points of discontinuity of f are
- Only the integral points
 - All rational numbers
 - $\{\pm\sqrt{n} : n \text{ is a non-negative integer}\}$
 - All real numbers
21. The function $f(x) = |x| + 3$ is
- Continuous as well as differentiable on $\mathbb{R}$
 - Continuous on $\mathbb{R}$ but not differentiable anywhere on $\mathbb{R}$
 - Continuous on $\mathbb{R}$ and differentiable at all but finitely many points on $\mathbb{R}$
 - Non continuous on $\mathbb{R}$
22. Let $f(x) = |\sin \pi x|, x \in \mathbb{R}$. Then
- f is continuous nowhere
 - f is continuous everywhere and differentiable nowhere
 - f is continuous everywhere and differentiable everywhere except at integral value of x
 - f is differentiable everywhere
23. The function $f(x) = 1 - |1 - x|$ on $\mathbb{R}$ is
- Not continuous
 - Continuous but not differentiable
 - Differentiable but not continuous
 - Differentiable at only one point
24. $f : [0, 1] \rightarrow \mathbb{R}$ is a function. Which of the following is possible?
- f is differentiable but not continuous
 - f is unbounded and continuous
 - f is differentiable but its derivative is not continuous
 - f is continuous but not integrable
25. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable. Which of these will follow from mean value theorem?
- For all a, b in $\mathbb{R}$, if $a < c < b$ then $\frac{f(b) - f(a)}{b - a} = f'(c)$
 - For some a, b in $\mathbb{R}$ and for all c in (a, b) , $\frac{f(b) - f(a)}{b - a} = f'(c)$
 - For all a, b in $\mathbb{R}$, there is some c in (a, b) such that $\frac{f(b) - f(a)}{b - a} = f'(c)$
 - For all a, b in $\mathbb{R}$, there is a unique element c in (a, b) then the $\frac{f(b) - f(a)}{b - a} = f'(c)$
26. The function $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$ is
- Differentiable at 0 but not continuous
 - Has 2nd derivative at the origin
 - Continuous at the origin but not differentiable
 - Neither continuous nor differentiable at the origin

27. The set of points where $f(x) = |\sin x|$ is not differentiable is
- Empty
 - $\{0\}$
 - $\{k\pi : k \in \mathbb{Z}\}$
 - $\left\{\frac{k\pi}{2} : k \in \mathbb{Z}\right\}$
28. Let $f : (0, 2) \rightarrow \mathbb{R}$ be defined by
- $$f(x) = \begin{cases} x^2 & \text{If } x \text{ is rational} \\ 2x-1 & \text{If } x \text{ is irrational} \end{cases} \quad \text{Then}$$
- f is differentiable exactly at one point
 - f is differentiable exactly at two points
 - f is not differentiable at any point in $(0, 2)$
 - f is differentiable at every point in $(0, 2)$
29. Let $X = (0, 1) \cup (2, 3)$ be an open set in $\mathbb{R}$. Let f be a continuous function on X such that the derivative $f'(x) = 0$ for all x . Then the range of f has
- Uncountable number of points
 - Countably infinite number of points
 - At most 2 points
 - At most 1 point
30. If $f(x)$ is real valued function defined on $[0, \infty]$ such that $f(0) = 0$ and $f''(x) > 0$ for all x , then the function $h(x) = \frac{f(x)}{x}$ is
- Increasing in $[0, \infty]$
 - Decreasing in $[0, 1]$
 - Increasing in $[0, 1]$ and decreasing in $[1, \infty]$
 - Decreasing in $[0, 1]$ and increasing in $[1, \infty]$

ASSIGNMENT SHEET - 8

- Let a be a positive real number. Which of the following integrals are convergent?
 - $\int_0^a \frac{1}{x^4} dx$.
 - $\int_0^a \frac{1}{\sqrt{x}} dx$.
 - $\int_4^\infty \frac{1}{x \log_e x} dx$.
 - $\int_5^\infty \frac{1}{x(\log x)^2} dx$.
- Let k be a positive integer. The radius of convergence of the series $\sum_{n=0}^\infty \frac{(n!)^k}{(kn)!} z^n$ is
 - k
 - k^{-k}
 - k^k
 - ∞
- Let $\{a_n : n \geq 1\}$ be a sequence of real numbers such that $\sum_{n=1}^\infty a_n$ is convergent and $\sum_{n=1}^\infty |a_n|$ is divergent. Let R be the radius of convergence of the power series $\sum_{n=1}^\infty a_n x^n$. then we can conclude that
 - $0 < R < 1$
 - $R = 1$
 - $1 < R < \infty$
 - $R = \infty$
- Let f be a monotonically increasing function from $[0,1]$ into $[0,1]$. Which of the following statements is/are true?
 - f must be continuous at all but finitely many points in $[0,1]$
 - f must be continuous at all but countably many points in $[0,1]$
 - f must be Riemann integrable
 - f must be Lebesgue integrable
- Let α, p be real numbers and $\alpha > 1$
 - If $p > 1$ then $\int_{-\infty}^\infty \frac{1}{|x|^{p\alpha}} dx < \infty$
 - If $p > \frac{1}{\alpha}$ then $\int_{-\infty}^\infty \frac{1}{|x|^{p\alpha}} dx < \infty$
 - If $p < \frac{1}{\alpha}$ then $\int_{-\infty}^\infty \frac{1}{|x|^{p\alpha}} dx < \infty$
 - For any $p \in \mathbb{R}$ we have $\int_{-\infty}^\infty \frac{1}{|x|^{p\alpha}} dx = \infty$
- Let $L = \int_0^1 \frac{dx}{1+x^8}$. Then
 - $L < 1$
 - $L > 1$
 - $L < \frac{\pi}{4}$
 - $L > \frac{\pi}{4}$
- Let, f, g and h be bounded functions on the closed interval $[a, b]$, such that $f(x) \leq g(x) \leq h(x)$ for all $x \in [a, b]$. Let $P = \{a = a_0 < a_1 < a_2 < \dots < a_n = b\}$ be a partition of $[a, b]$. We denote by $U(f, P)$ and $L(f, P)$, the upper and lower Riemann sums of f with respect to the partition P and similarly for g and h . Which of the following statements is necessarily true?
 - If $U(h, P) - U(f, P) < 1$ then $U(g, P) - L(g, P) < 1$
 - If $L(h, P) - L(f, P) < 1$ then $U(g, P) - L(g, P) < 1$
 - If $U(h, P) - L(f, P) < 1$ then $U(g, P) - L(g, P) < 1$
 - If $L(h, P) - U(f, P) < 1$ then $U(g, P) - L(g, P) < 1$

8. The power series $\sum_{n=0}^{\infty} \frac{[2+(-1)^n]^n}{3^n} x^n$ converges
- Only for $x=0$
 - For all $x \in \mathbb{R}$
 - Only for $-1 < x < 1$
 - Only for $-1 < x \leq 1$
9. Area enclosed between x -axis from a to b and the curve $f(x)$ is finite when
- $a=0, b=\infty, f(x)=e^{-5x^5}$
 - $a=-\infty, b=\infty, f(x)=e^{-5x^5}$
 - $a=-7, b=\infty, f(x)=\frac{1}{x^4}$
 - $a=-7, b=7, f(x)=\frac{1}{x^4}$
10. Let f be a continuously differentiable real-valued function on $[a, b]$ such that $|f'(x)| \leq K$ for all $x \in [a, b]$. For a partition $P = \{a = a_0 < a_1 < \dots < a_n = b\}$, let $U(f, P)$ and $L(f, P)$ denote the upper and lower Riemann sums of f with respect to P . Then
- $|L(f, P) - U(f, P)| \leq K(b-a)$
 - $U(f, P) - L(f, P) \leq K(b-a)$
 - $U(f, P) - L(f, P) \leq K\|P\|$, where $\|P\| = \max_{0 \leq i \leq n-1} (a_{i+1} - a_i)$ is the norm of the partition
 - $U(f, P) - L(f, P) \leq K\|P\|(b-a)$
11. Which of the conditions below imply that a function $f: [0, 1] \rightarrow \mathbb{R}$ is necessarily of bounded variation?
- f is a monotone function on $[0, 1]$
 - f is a continuous and monotone function on $[0, 1]$
 - f has a derivative at each $x \in (0, 1)$
 - f has a bounded derivative on the interval $(0, 1)$.
12. Define $f: [0, 1] \rightarrow [0, 1]$ by $f(x) = \frac{2^k - 1}{2^k}$ for $f \in \left[\frac{2^{k-1} - 1}{2^{k-1}}, \frac{2^k - 1}{2^k}\right], k \geq 1$. Then f is a Riemann-integrable function such that
- $\int_0^1 f(x) dx = \frac{2}{3}$
 - $\frac{1}{2} < \int_0^1 f(x) dx < \frac{2}{3}$
 - $\int_0^1 f(x) dx = 1$
 - $\frac{2}{3} < \int_0^1 f(x) dx < 1$
13. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that $\int_0^{\infty} f(x) dx$ exists. Which of the following statements are correct?
- If $\lim_{x \rightarrow \infty} f(x)$ exists, then $\lim_{x \rightarrow \infty} f(x) = 0$
 - The limit $\lim_{x \rightarrow \infty} f(x)$ must exist and is zero
 - In case f is a non-negative function, $\lim_{x \rightarrow \infty} f(x)$ must exist and is zero
 - In case f is a differentiable function $\lim_{x \rightarrow \infty} f'(x)$ must exist and is zero
14. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that f' is bounded. Given a closed and bounded interval $[a, b]$, and a partition $P = \{a_0 = a < a_1 < \dots < a_n = b\}$ of $[a, b]$, let $M(f, P)$ and $m(f, P)$ denote, respectively, the upper Riemann sum and the lower Riemann sum of f with respect to P . Then
- $\left| M(f, P) - \int_a^b f(x) dx \right| \leq (b-a) \sup \{ |f(x)| : x \in [a, b] \}$
 - $\left| m(f, P) - \int_a^b f(x) dx \right| \leq (b-a) \inf \{ |f(x)| : x \in [a, b] \}$
 - $\left| M(f, P) - \int_a^b f(x) dx \right| \leq (b-a)^2 \sup \{ |f'(x)| : x \in [a, b] \}$
 - $\left| m(f, P) - \int_a^b f(x) dx \right| \leq (b-a)^2 \inf \{ |f'(x)| : x \in [a, b] \}$

15. Let $f: [0, 1] \rightarrow \mathbb{R}$ be continuous such that $f(t) \geq 0$ for all t in $[0, 1]$. Define $g(x) = \int_0^x f(t) dt$ then
- g is monotone and bounded
 - g is monotone, but not bounded
 - g is bounded, but not monotone
 - g is neither monotone nor bounded
16. Let f be a continuous function on $[0, 1]$ with $f(0) = 1$. Let $G(a) = \frac{1}{a} \int_0^a f(x) dx$
- $\lim_{a \rightarrow 0} G(a) = \frac{1}{2}$
 - $\lim_{a \rightarrow 0} G(a) = 1$
 - $\lim_{a \rightarrow 0} G(a) = 0$
 - The limit $\lim_{a \rightarrow 0} G(a)$ does not exist
17. Match the following
- A. $\int_1^2 \frac{x}{\sqrt{x-1}} dx$ B. $\int_0^\pi \frac{dx}{1+\cos x}$
- Divergent & diverges to $+\infty$
 - Convergent & value is $\frac{4}{3}$
 - Divergent & diverges to $-\infty$
 - Convergent & value is $\frac{8}{3}$
- A-1, B-2
 - A-3, B-4
 - A-4, B-1
 - A-2, B-3
18. Suppose f is an increasing real valued function on $[0, \infty)$ with $f(x) > 0 \forall x$ and let $g(x) = \frac{1}{x} \int_0^x f(u) du$; $0 < x < \infty$. Then which of the following are true:
- $g(x) \leq f(x)$ for all $x \in (0, \infty)$
 - $xg(x) \leq f(x)$ for all $x \in (0, \infty)$
 - $xg(x) \geq f(0)$ $x \in (0, \infty)$
 - $yg(y) - xg(x) \leq (y-x)f(y)$ for all $x < y$
19. Let $f: [0, 1] \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} x \cos(\pi/(2x)) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$ then
- f is continuous on $[0, 1]$
 - f is of bounded variation on $[0, 1]$
 - f is differentiable $[0, 1]$ and its derivative f' is bounded on $(0, 1)$
 - f is Riemann integrable on $[0, 1]$
20. Let $f: [0, \infty) \rightarrow \mathbb{R}$, and $g: [0, \infty) \rightarrow \mathbb{R}$ be continuous functions satisfying $\int_0^x t^2 dt = x^3(1+x)^2$ and $\int_0^{x^2(1+x)} g(t) dt = x$ for all $x \in [0, \infty)$. Then $f(2) + g(2)$ is equal to
- 0
 - 5
 - 6
 - 11
21. Find the value of p for which the integral $\int_0^1 \frac{x^{p-1}}{1+x} dx$ is convergent.
- Convergent if $p > 1$
 - Convergent if $p = 1$
 - Convergent if $p = 0$
 - Convergent iff $p > 0$
22. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Which of the following is always true?
- $f^{-1}(U)$ is open for all open sets $U \subseteq \mathbb{R}$
 - $f^{-1}(C)$ is closed for all closed sets $C \subseteq \mathbb{R}$
 - $f^{-1}(K)$ is compact for all compact sets $K \subseteq \mathbb{R}$
 - $f^{-1}(G)$ is connected for all connected sets $G \subseteq \mathbb{R}$
23. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be differentiable with $0 < f'(x) < 1$ for all x . Then
- f is increasing and f is bounded.
 - f is increasing and f is Riemann integrable on i .
 - f is increasing and f is uniformly continuous.
 - f is of bounded variation.

24. c is a fixed point in $[0, 1]$. A function $f: [0, 1] \rightarrow \mathbb{R}$ is defined by

$$f(x) = \begin{cases} c & \text{if } 0 \leq x \leq c \\ 2c & \text{if } c < x \leq 1 \end{cases}$$

It is given that the Riemann integral $\int_0^1 f(x) dx$ is $\frac{7}{16}$. Then the value of c is

(a.) $\frac{1}{2}$

(b.) $\frac{1}{3}$

(c.) $\frac{1}{4}$

(d.) $\frac{1}{5}$

25. The improper integral $\int_1^{\infty} \frac{x}{1+x^2} \sin x dx$ is

(a.) Divergent

(b.) Convergent

(c.) Oscillatory

(d.) Diverges to $-\infty$

26. Let $f(x) = x^3 - \sin x$, $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. Then f is

(a.) Bounded and of bounded variation

(b.) Unbounded and of bounded variation

(c.) Unbounded but not of bounded variation

(d.) Bounded but not of bounded variation

27. If $f: [1, 2] \rightarrow \mathbb{R}$ is a non-negative Riemann-integrable function such that $\int_1^2 \frac{f(x)}{\sqrt{x}} dx = k \int_1^2 f(x) dx \neq 0$, then k belongs to the interval

(a.) $\left[0, \frac{1}{3}\right]$

(b.) $\left[\frac{1}{3}, \frac{2}{3}\right]$

(c.) $\left[\frac{2}{3}, 1\right]$

(d.) $\left[1, \frac{4}{3}\right]$

28. The improper integral $\int_0^4 \frac{dx}{x(4-x)}$ is

(a.) Convergent & value in 6

(b.) Divergent & diverges to $+\infty$

(c.) Convergent & value is 4

(a.) None of these

29. Consider the power series $\sum_{n=1}^{\infty} \frac{x^n}{\sqrt{n}}$ and

$$\sum_{n=1}^{\infty} \frac{x^n}{n}$$

Then

(a.) both converge on $(-1, 1]$

(b.) both converge on $[-1, 1)$

(c.) exactly one of them converges on $(-1, 1]$

(d.) none of them converges on $[-1, 1)$

30. Which one of the following does NOT hold for all continuous functions $f: [-\pi, \pi] \rightarrow \mathbb{R}$?

(a.) If $f(-t) = f(t)$ for each $t \in [-\pi, \pi]$, then

$$\int_{-\pi}^{\pi} f(t) dt = 2 \int_0^{\pi} f(t) dt$$

(b.) If $f(-t) = -f(t)$ for each $t \in [-\pi, \pi]$, then

$$\int_{-\pi}^{\pi} f(t) dt = 0$$

(c.) $\int_{-\pi}^{\pi} f(t) dt = - \int_{-\pi}^{\pi} f(t) dt$

(d.) There is an α with $-\pi < \alpha < \pi$ such that

$$\int_{-\pi}^{\pi} f(t) dt = 2\pi f(\alpha)$$

31. For a positive real number p , let $(f_n: n = 1, 2, \dots)$ be a sequence of functions defined on

$$[0, 1] \text{ by } f_n(x) = \begin{cases} n^{p+1}x, & \text{if } 0 \leq x < \frac{1}{n} \\ \frac{1}{x^p}, & \text{if } \frac{1}{n} \leq x \leq 1 \end{cases} \text{ Let}$$

$f(x) = \lim_{n \rightarrow \infty} f_n(x), x \in [0, 1]$. Then, on $[0, 1]$

- (a.) f is Riemann integrable
- (b.) the improper integral $\int_0^1 f(x) dx$ converges for $p \geq 1$
- (c.) the improper integral $\int_0^1 f(x) dx$ converges for $p < 1$
- (d.) f_n converges uniformly

32. Let $f: [0, 1] \rightarrow \mathbb{R}$ and $g: [0, 1] \rightarrow \mathbb{R}$ be two functions defined by

$$f(x) = \begin{cases} \frac{1}{n} & \text{If } x = \frac{1}{n}, n \in \mathbb{N} \\ 0 & \text{Otherwise} \end{cases} \text{ and}$$

$$g(x) = \begin{cases} n & \text{If } x = \frac{1}{n}, n \in \mathbb{N} \\ 0 & \text{Otherwise} \end{cases}$$

- (a.) Both f and g are Riemann integrable
- (b.) f is Riemann integrable but g is not
- (c.) g is Riemann integrable but f is not
- (d.) Neither f nor g is Riemann integrable

ASSIGNMENT SHEET - 9

1. For $n \geq 1$, let $g_n(x) = \sin^2\left(x + \frac{1}{n}\right)$, $x \in [0, \infty)$

and $f_n(x) = \int_0^x g_n(t) dt$. Then

- $\{f_n\}$ converges pointwise to a function f on $[0, \infty)$, but does not converge uniformly on $[0, \infty)$.
- $\{f_n\}$ does not converge to any function on $[0, \infty)$.
- $\{f_n\}$ converges uniformly on $[0, 1)$.
- $\{f_n\}$ converges uniformly on $[0, \infty)$.

2. Let $\{b_n\}$ and $\{c_n\}$ be sequence of real numbers then a necessary and sufficient condition for the sequence of polynomials $f_n(x) = b_n x + c_n x^2$ to converge uniformly to 0 on the real line is

- $\lim_{n \rightarrow \infty} b_n = 0$ and $\lim_{n \rightarrow \infty} c_n = 0$
- $\sum_{n=1}^{\infty} |b_n| < \infty$ and $\sum_{n=1}^{\infty} |c_n| < \infty$
- There exists a positive integer N such that $b_n = 0$ and $c_n = 0$ for all $n > N$
- $\lim_{n \rightarrow \infty} c_n = 0$

3. Let $\{f_n\}$ be a sequence of continuous functions on R .

- If $\{f_n\}$ converges to f pointwise on R then $\lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} f_n(x) dx = \int_{-\infty}^{\infty} f(x) dx$
- If $\{f_n\}$ converges to f uniformly on R then $\lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} f_n(x) dx = \int_{-\infty}^{\infty} f(x) dx$
- If $\{f_n\}$ converges to f uniformly on R then f is continuous on R
- There exists a sequence of continuous functions $\{f_n\}$ on R , such that $\{f_n\}$ converges to f uniformly on R , but $\lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} f_n(x) dx \neq \int_{-\infty}^{\infty} f(x) dx$

4. Let $P_n(x) = a_n x^2 + b_n x + c_n$ be a sequence of quadratic polynomials where $a_n, b_n, c_n \in R$ for all $n \geq 1$. Let $\lambda_0, \lambda_1, \lambda_2$ be distinct real numbers such that $\lim_{n \rightarrow \infty} P_n(\lambda_0) = A_0$, $\lim_{n \rightarrow \infty} P_n(\lambda_1) = A_1$ and $\lim_{n \rightarrow \infty} P_n(\lambda_2) = A_2$. Then

- $\lim_{n \rightarrow \infty} P_n(x)$ exists for all $x \in R$
- $\lim_{n \rightarrow \infty} P_n'(x)$ exists for all $x \in R$
- $\lim_{n \rightarrow \infty} P_n\left(\frac{\lambda_0, \lambda_1, \lambda_2}{3}\right)$ does not exist
- $\lim_{n \rightarrow \infty} P_n'\left(\frac{\lambda_0, \lambda_1, \lambda_2}{3}\right)$ does not exist

5. Let $f_n : [1, 2] \rightarrow [0, 1]$ be given by $f_n(x) = (2-x)^n$ for all non-negative integers n . Let $f(x) = \lim_{n \rightarrow \infty} f_n(x)$ for $1 \leq x \leq 2$. Then which of the following is true?

- f is a continuous function on $[1, 2]$.
- f_n converges uniformly to f on $[1, 2]$ as $n \rightarrow \infty$.
- $\lim_{n \rightarrow \infty} \int_1^2 f_n(x) dx = \int_1^2 f(x) dx$.
- for any $a \in (1, 2)$ we have $\lim_{n \rightarrow \infty} f_n'(a) \neq f'(a)$.

6. Let $f : R \rightarrow [0, \infty)$ be a non-negative real valued continuous function. Let

$$\phi_n(x) = \begin{cases} n & \text{if } f(x) \geq n \\ 0 & \text{if } f(x) < n \end{cases}$$

$$\phi_{n,k}(x) = \begin{cases} \frac{k}{2^n} & \text{if } f(x) \in \left[\frac{k}{2^n}, \frac{k+1}{2^n}\right] \\ 0 & \text{if } f(x) \notin \left[\frac{k}{2^n}, \frac{k+1}{2^n}\right] \end{cases}, \text{ And}$$

$g_n(x) = \phi_n(x) + \sum_{k=0}^{n2^n-1} \phi_{n,k}(x)$. As $n \uparrow \infty$, which of the following are true?

- $g_n(x) \uparrow f(x)$ for every $x \in R$
- Given any $C > 0$, $g_n(x) \uparrow f(x)$ uniformly on the set $\{x : f(x) < C\}$
- $g_n(x) \uparrow f(x)$ uniformly for $x \in R$
- Given any $C > 0$, $g_n(x) \uparrow f(x)$ uniformly on the set $\{x : f(x) \geq C\}$

7. Let $\{f_n\}$ be a sequence of continuous real-valued functions defined on $[0, \infty)$. Suppose $f_n(x) \rightarrow f(x)$ for all $x \in [0, \infty)$ and that f is integrable. Then
- $\int_0^\infty f_n(x) dx \rightarrow \int_0^\infty f(x) dx$ as $n \rightarrow \infty$
 - If $f_n \rightarrow f$ uniformly on $[0, \infty)$, then $\int_0^1 f_n(x) dx \rightarrow \int_0^1 f(x) dx$
 - If $f_n \rightarrow f$ uniformly on $[0, \infty)$, then $\int_0^\infty f_n(x) dx \rightarrow \int_0^\infty f(x) dx$
 - If $\int_0^1 |f_n(x) - f(x)| dx \rightarrow 0$, then $f_n \rightarrow f$ uniformly on $[0, 1]$.
8. Let $f_n(x) = \begin{cases} 1-nx & \text{for } x \in [0, \frac{1}{n}] \\ 0 & \text{for } x \in [\frac{1}{n}, 1] \end{cases}$ Then
- $\lim_{n \rightarrow \infty} f_n(x)$ defines a continuous function on $[0, 1]$
 - $\{f_n\}$ converges uniformly on $[0, 1]$
 - $\lim_{n \rightarrow \infty} f_n(x) = 0$ for all $x \in [0, 1]$
 - $\lim_{n \rightarrow \infty} f_n(x)$ exists for all $x \in [0, 1]$
9. Let $f_n(x) = x^{1/n}$ for $n \in [0, 1]$. Then
- $\lim_{n \rightarrow \infty} f_n(x)$ exists for all $x \in [0, 1]$
 - $\lim_{n \rightarrow \infty} f_n(x)$ defines a continuous function on $[0, 1]$
 - $\{f_n\}$ converges uniformly on $[0, 1]$
 - $\lim_{n \rightarrow \infty} f_n(x) = 0$ for all $x \in [0, 1]$.
10. Let $\{f_n\}$ be a sequence of integrable functions defined on an interval $[a, b]$. Then
- If $f_n(x) \rightarrow 0$ a.e., then $\int_a^b f_n(x) dx \rightarrow 0$
 - If $\int_a^b f_n(x) dx \rightarrow 0$, then $f_n(x) \rightarrow 0$ a.e.
 - If $f_n(x) \rightarrow 0$ a.e. and each f_n is a bounded function, then $\int_a^b f_n(x) dx \rightarrow 0$
 - If $f_n(x) \rightarrow 0$ a.e. and the f_n 's are uniformly bounded, then $\int_a^b f_n(x) dx \rightarrow 0$
11. Let $f(x) = \sum_{n=1}^{\infty} \frac{\sin(nx)}{n^2}$. Then
- $\lim_{x \rightarrow 0} f(x) = 0$
 - $\lim_{x \rightarrow 0} f(x) = 1$
 - $\lim_{x \rightarrow 0} f(x) = \frac{\pi^2}{6}$
 - $\lim_{x \rightarrow 0} f(x)$ does not exist
12. The series $\sum_{m=1}^{\infty} x^{\ln m}$, $x > 0$, is convergent on the interval
- $(0, 1/e)$
 - $(1/e, e)$
 - $(0, e)$
 - $(1, e)$
13. Let $f_n(x) = \frac{x}{\{(n-1)x+1\}\{nx+1\}}$ and $s_n(x) = \sum_{j=1}^n f_j(x)$ for $x \in [0, 1]$. The sequence $\{s_n\}$
- Converges uniformly on $[0, 1]$
 - Converges pointwise on $[0, 1]$ but not uniformly
 - Converges pointwise for $x=0$ but not for $x \in (0, 1]$
 - Does not converge for $x \in [0, 1]$
14. Let $\{f_n\}$ be a sequence of real valued differentiable function on $[a, b]$ such that $f_n(x) \rightarrow f(x)$ as $n \rightarrow \infty$ for every $x \in [a, b]$ and for some Riemann-integrable function $f: [a, b] \rightarrow \mathbb{R}$. Consider the statements:
- $\{f_n\}$ converges uniformly
 - $\{f_n'\}$ converges uniformly
 - $\int_a^b f_n(x) dx \rightarrow \int_a^b f(x) dx$
 - f is differentiable
- Then which one of the following need NOT be true
- P_1 implies P_3
 - P_2 implies P_1
 - P_2 implies P_4
 - P_3 implies P_1

15. Let $f_n(x) = \frac{x^n}{1+x}$ and $g_n(x) = \frac{x^n}{1+nx}$ for $x \in \mathbb{N}$. Then on the interval $[0, 1]$,

(a.) both $\{f_n\}$ and $\{g_n\}$ converges uniformly

(b.) neither $\{f_n\}$ nor $\{g_n\}$ converges uniformly

(c.) $\{f_n\}$ converges uniformly but $\{g_n\}$ does not converge uniformly

(d.) $\{g_n\}$ converges uniformly but $\{f_n\}$ does not converges uniformly

16. Which of the following sequence $\{f_n\}_{n=1}^{\infty}$ of functions does NOT converge uniformly on $[0, 1]$?

(a.) $f_n(x) = \frac{e^{-x}}{n}$

(b.) $f_n(x) = (1-x)^n$

(c.) $f_n(x) = \frac{x^2 + nx}{n}$

(d.) $f_n(x) = \frac{\sin(nx+n)}{n}$

17. Which one of the following statements holds?

(a.) The series $\sum_{n=0}^{\infty} x^n$ converges for each $x \in [-1, 1]$

(b.) The series $\sum_{n=0}^{\infty} x^n$ converges uniformly in $(-1, 1)$

(c.) The series $\sum_{n=1}^{\infty} \frac{x^n}{n^2}$ converges for each $x \in [-1, 1]$

(d.) The series $\sum_{n=1}^{\infty} \frac{x^n}{n^2}$ converges uniformly in $(-1, 1)$

18. Let $\sigma > 1$ and $g(x) = \sum_{n=1}^{\infty} \frac{1}{n^x}$, $\sigma \leq x < \infty$.

Then $g(x)$ is

(a.) Not continuous

(b.) Continuous but not differentiable

(c.) Differentiable but not continuously differentiable

(d.) Continuously differentiable

19. Consider two sequences $\{f_n\}$ and $\{g_n\}$ of functions where $f_n: [0, 1] \rightarrow \mathbb{R}$ and $g_n: \mathbb{R} \rightarrow \mathbb{R}$ are defined by

$f_n(x) = x^n$ and

$g_n(x) = \begin{cases} \cos(x-n)\pi/2 & \text{If } x \in [n-1, n+1] \\ 0 & \text{otherwise} \end{cases}$

Then

(a.) Neither $\{f_n\}$ nor $\{g_n\}$ is uniformly convergent

(b.) $\{f_n\}$ is not uniformly convergent but $\{g_n\}$ is

(c.) $\{g_n\}$ is not uniformly convergent but $\{f_n\}$

(d.) Both $\{f_n\}$ and $\{g_n\}$ are uniformly convergent

20. Let $f_x(x) = \frac{\sin x}{\sqrt{n}}$, $n = 1, 2, \dots$ and $x \in [-1, 1]$. Then as $n \rightarrow \infty$,

(a.) $\{f_n(x)\}_{n=1}^{\infty}$ does not converge uniformly in $[-1, 1]$

(b.) $\lim_{n \rightarrow \infty} \int_{-1}^1 f_n(x) dx \neq 0$

(c.) $\{f'_n(x)\}$ does not converge uniformly in $[-1, 1]$

(d.) $f_n(x)$, $n = 1, 2$ is not uniformly continuous in $[-1, 1]$

ASSIGNMENT SHEET - 10

- Let $A = \{(x, y) \in \mathbb{R}^2 : x + y \neq -1\}$. Define $f : A \rightarrow \mathbb{R}^2$ by

$$f(x, y) = \left(\frac{x}{1+x+y}, \frac{y}{1+x+y} \right).$$
 Then,
 (a.) The Jacobian matrix of f does not vanish on A
 (b.) f is infinitely differentiable on A
 (c.) f is injective on A
 (d.) $f(A) = \mathbb{R}^2$
- Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by

$$f(x, y) = (x + 2y + y^2 + |xy|, 2x + y + x^2 + |xy|)$$
 for $(x, y) \in \mathbb{R}^2$. Then
 (a.) f is discontinuous at $(0, 0)$
 (b.) f is continuous at $(0, 0)$ but not differentiable at $(0, 0)$
 (c.) f is differentiable at $(0, 0)$
 (d.) f is differentiable at $(0, 0)$ and the derivative $Df(0, 0)$ is invertible
- Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by $f(x, y) = (x + y, xy)$. Then
 (a.) f is not differentiable at the point $(0, 0)$.
 (b.) The derivative of f is invertible except on the set $\{(x, y) \in \mathbb{R}^2 : x = y\}$.
 (c.) The inverse image of each point in $\mathbb{R}^2$ under f has at most two elements.
 (d.) f is surjective.
- Let $f : \mathbb{R}^m \rightarrow \mathbb{R}^m$ be a differential function. Let $Df(x)$ be the derivative of f at $x \in \mathbb{R}^m$. Which of the following is/are correct?
 (a.) $Df(x)(u) = 0$ for all u in $\mathbb{R}^m$.
 (b.) $Df(0)(u) = 0$ for all u in $\mathbb{R}^m$ and some $x \in \mathbb{R}^m$ only if f is a constant.
 (c.) $Df(0)(u) = 0$ for all $u \in \mathbb{R}^m$ and all $x \in \mathbb{R}^m$ only if f is a constant.
 (d.) If f is not a constant function, then $Df(x)$ is a one to one function for some $x \in \mathbb{R}^m$
- If $f : S \rightarrow S$ is a function, then we denote by f^k , the function $f \circ f \circ \dots \circ f$ (k times). Let f_1 and f_2 be two functions defined on $\mathbb{R}^2$ as follows:

$$f_1(x, y) = (x + 1, y + 3), f_2(x, y) = (x - 3, y - 2).$$
 Then
 (a.) For any positive integer k , there exists a unique $(a, b) \in \mathbb{R}^2$, such that $f_1^k(0, 0) = f_2^k(a, b)$.
 (b.) For any real number a and any positive integer, there is at most one solution y for $f_1^k(0, 0) = f_2^k(a, y)$
 (c.) There exists $(a, b) \in \mathbb{R}^2$ such that $f_1^k(a, b) \neq f_2^k(x, y)$ for any $(x, y) \in \mathbb{R}^2$ and any positive integer k .
 f_1 is linear transformation.
- Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be the map $f(x_1, \dots, x_n) = a_1 x_1 + \dots + a_n x_n$, where $a = (a_1, \dots, a_n)$ is a fixed non-zero vector. Let $Df(0)$ denote the derivative of f at 0. Which of the following are correct?
 (a.) $(Df)(0)$ is a linear map from $\mathbb{R}^n$ to $\mathbb{R}$
 (b.) $[(Df)(0)](a) = \|a\|^2$
 (c.) $[(Df)(0)](a) = 0$
 (d.) $[(Df)(0)](b) = a_1 b_1 + \dots + a_n b_n$ for $b = (b_1, \dots, b_n)$
- Let $f(x, y) = \sqrt{|xy|}$. Then
 (a.) f_x and f_y do not exist at $(0, 0)$
 (b.) $f_x(0, 0) = 1$
 (c.) $f_y(0, 0) = 0$
 (d.) f is differentiable at $(0, 0)$

8. Let $F_1, F_2 : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the functions

$$F_1(x_1, x_2) = \frac{-x_2}{x_1^2 + x_2^2} \text{ and } F_2(x_1, x_2) = \frac{x_1}{x_1^2 + x_2^2}.$$

Which of the following are correct?

(a.) $\frac{\partial F_1}{\partial x_2} = \frac{\partial F_2}{\partial x_1}$

- (b.) There exists a function

$$f : \mathbb{R}^2 - \{(0, 0)\} \rightarrow \mathbb{R} \text{ such that } \frac{\partial f}{\partial x_1} = F_1$$

$$\text{and } \frac{\partial f}{\partial x_2} = F_2$$

- (c.) There exists no function

$$f : \mathbb{R}^2 - \{(0, 0)\} \rightarrow \mathbb{R} \text{ such that } \frac{\partial f}{\partial x_1} = F_1$$

$$\text{and } \frac{\partial f}{\partial x_2} = F_2$$

- (d.) There exists a function $f : D \rightarrow \mathbb{R}$ where D is the open disc of radius 1 centred at

$$(2, 0), \text{ which satisfies } \frac{\partial f}{\partial x_1} = F_1 \text{ and}$$

$$\frac{\partial f}{\partial x_2} = F_2 \text{ on } D.$$

9. Let $z = x + iy$ and $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the function

$$f(x, y) = f(z) = z^2 = (x^2 - y^2, 2xy) \in \mathbb{R}^2. \text{ Let}$$

$(Df)(a)$ denote the derivative of f at a .

Which of the following are true?

- (a.) $(Df)(a)h = 2ah$, where $a = a_1 + ia_2$ and $h = h_1 + ih_2$

(b.) $(Df)(a) = 2 \begin{pmatrix} a_1 & -a_2 \\ a_2 & a_1 \end{pmatrix}, a = (a_1, a_2) \in \mathbb{R}^2$

- (c.) f is one to one on $\mathbb{R}^2$

- (d.) For any $a \in \mathbb{R}^2 - \{(0, 0)\}$, f is one to one on some neighbourhood of a .

10. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be the function defined by $f(x) = x\|x\|^2$ for $x \in \mathbb{R}^n$. Which of the following statements are correct?

(a.) $(Df)(0) = 0$

(b.) $(Df)(x) = 0$ for all $x \in \mathbb{R}^n$

- (c.) f is one-one

- (d.) f has an inverse

11. Let $f : A \cup E \rightarrow \mathbb{R}^2$ be differentiable, where

$$A = \left\{ (x, y) \in \mathbb{R}^2 : \frac{1}{2} < x^2 + y^2 < 1 \right\} \text{ and}$$

$$E = \left\{ (x, y) \in \mathbb{R}^2 : (x-2)^2 + (y-2)^2 < \frac{1}{2} \right\}. \text{ Let}$$

Df be the derivative of the function f .

Which of the following are necessarily correct?

- (a.) If $(Df)(x, y) = 0$ for all $(x, y) \in A \cup E$, then f is constant

- (b.) If $(Df)(x, y) = 0$ for all $(x, y) \in A$, then f is constant on A

- (c.) If $(Df)(x, y) = 0$ for all $(x, y) \in E$, then f is constant on E

- (d.) If $(Df)(x, y) = 0$ for all $(x, y) \in A \cup E$, then for some $(x_0, y_0), (x_1, y_1) \in \mathbb{R}^2$, $f(x, y) = (x_0, y_0)$ for all $(x, y) \in A$ and $f(x, y) = (x_1, y_1)$ for all $(x, y) \in E$.

12.

Let $L : \mathbb{R}^n \rightarrow \mathbb{R}$ be the function $L(x) = \langle x, y \rangle$, where $\langle \cdot, \cdot \rangle$ is some inner product on $\mathbb{R}^n$ and y is a fixed vector in $\mathbb{R}^n$. Further denote by DL , the derivative of L . Which of the following are necessarily correct?

(a.) $DL(u) = DL(v)$ for all $u, v \in \mathbb{R}^n$

(b.) $DL(0, 0, \dots, 0) = L$

(c.) $DL(x) = \|x\|^2$ for all $x \in \mathbb{R}^n$

(d.) $DL(1, 1, \dots, 1) = 0$.

13.

Let $f : [\pi, 2\pi] \rightarrow \mathbb{R}^2$ be the function $f(t) = (\cos t, \sin t)$. Which of the following are necessarily correct?

- (a.) There exists $t_0 \in [\pi, 2\pi]$ such that

$$f'(t_0) = \frac{1}{\pi}(f(2\pi) - f(\pi))$$

- (b.) There does not exist any $t_0 \in [\pi, 2\pi]$

$$\text{such that } f'(t_0) = \frac{1}{\pi}(f(2\pi) - f(\pi))$$

- (c.) There exists $t_0 \in [\pi, 2\pi]$ such that

$$\|f(2\pi) - f(\pi)\| \leq \pi \|f'(t_0)\|$$

- (d.) $f'(t) = (-\sin t, \cos t)$ for all $t \in [\pi, 2\pi]$.

14. Consider the map $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $f(x, y) = (7x + x^4, 3x + 4y + y^4)$. Then,

- (a.) f is discontinuous at $(0, 0)$.
- (b.) f is continuous at $(0, 0)$ and all directional derivatives exist at $(0, 0)$
- (c.) f is differentiable at $(0, 0)$ but the derivative $Df(0, 0)$ is not invertible.
- (d.) f is differentiable at $(0, 0)$ and the derivative $Df(0, 0)$ is invertible.

15. The map $L: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $L(x, y) = (x, -y)$ is

- (a.) Differentiable everywhere on $\mathbb{R}^2$
- (b.) Differentiable only at $(0, 0)$
- (c.) $DL(0, 0) = L$
- (d.) $DL(x, y) = L$ for all $(x, y) \in \mathbb{R}^2$

16. Consider the map $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $f(x, y) = (3x - 2y + x^2, 4x + 5y + y^2)$. Then

- (a.) f is discontinuous at $(0, 0)$
- (b.) f is continuous at $(0, 0)$ and all directional derivatives exists at $(0, 0)$
- (c.) f is differentiable at $(0, 0)$ but the derivative $Df(0, 0)$ is not invertible
- (d.) f is differentiable at $(0, 0)$ and the derivative $Df(0, 0)$ is invertible.

17. Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ and $g: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by $f(x, y) = |x| + |y|$ and $g(x, y) = |xy|$. Then

- (a.) f is differentiable at $(0, 0)$, but g is not differentiable at $(0, 0)$
- (b.) g is differentiable at $(0, 0)$, but f is not differentiable at $(0, 0)$
- (c.) Both f and g are differentiable at $(0, 0)$
- (d.) Both f and g are continuous at $(0, 0)$

18. Let

$$f(x, y) = \begin{cases} \frac{xy}{(x^2 + y^2)^{5/2}} [1 - \cos(x^2 + y^2)], & (x, y) \neq (0, 0) \\ k, & (x, y) = (0, 0) \end{cases}$$

Then the value of k for which $f(x, y)$ is continuous at $(0, 0)$ is

(a.) 0

(b.) $\frac{1}{2}$

(c.) 1

(d.) $\frac{3}{2}$

19. For $(x, y) \in \mathbb{R}^2$ $f(x, y) = \begin{cases} \frac{2xy}{x^2 + y^2} & \text{if } (x, y) \neq 0 \\ 0 & \text{if } (x, y) = 0 \end{cases}$

Then

- (a.) f_x and f_y exist at $(0, 0)$ f is continuous at $(0, 0)$
- (b.) f_x and f_y exist at $(0, 0)$ and f is discontinuous at $(0, 0)$
- (c.) f_x and f_y do not exist at $(0, 0)$, f is continuous at $(0, 0)$
- (d.) f_x and f_y do not exist at $(0, 0)$ and f is discontinuous at $(0, 0)$

20. Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} x^2 + y^2 & \text{If } x \text{ and } y \text{ are rational} \\ 0 & \text{Otherwise} \end{cases}$$

Then

- (a.) f is not continuous at $(0, 0)$
- (b.) f is continuous at $(0, 0)$ but not differentiable at $(0, 0)$
- (c.) f is differentiable only at $(0, 0)$
- (d.) f is differentiable every where

21. If

$$f(x, y) = \begin{cases} x^3 / (x^2 + y^2), & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

then at $(0, 0)$

- (a.) f_x, f_y do not exist
- (b.) f_x, f_y exist and are equal
- (c.) The directional derivative exists along any straight line
- (d.) f is differentiable

22. Let

$$f(x, y) = \frac{xy}{x^2 + y^2}, (x, y) \neq (0, 0); f(0, 0) = 0.$$

Then

- (a.) f is continuous at $(0, 0)$ and the partial derivatives f_x, f_y exist at every point of $\mathbb{R}^2$.
- (b.) f is discontinuous at $(0, 0)$ and f_x, f exist at every point of $\mathbb{R}^2$.
- (c.) f is discontinuous at $(0, 0)$ and f_x, f exist only at $(0, 0)$.
- (d.) None of the above